

v7 cloning campaign: summary of the results to 2026-09-15

2026-09-15

This is the short form of `REPORT_v7_2026-09-15` (the full report, 87 pages): the model and its equations, the generation of the training subsets, the pre-training methodology and results, and for the two optimization stages only the training losses, the trained enhancement factors and the DFS-units 3x3 figures with their per-cell tables. The figure and section numbers are those of the full report, so that a figure cited here can be found there with its neighbours; the figures the summary leaves out are cited as "of the full report". Every number is read from a file of the pulled runs or from a CSV the figure suite wrote beside the figure it belongs to.

Scope and inventory

The v7 campaign trains one network pair (an exchange network and a correlation network) per architecture and per training-subset size, under one protocol, on the cluster as separate runs that differ in their architecture axis. Each run carries its own pre-training (the cloning of the parent functional), a fidelity certificate that gates its training array, and the array itself, one cell per (architecture, subset size). The figures and tables are drawn once over every finished cell of every run (the runs merged into one view, `family_runs.yaml`); the architecture names are those of Sec. 1.5.

Run	Architectures	Cells in the figures / planned	Run directory
size	deep_3x16, deep_attn_3x16, deep_2x8, deep_attn_2x8	deep_3x16 11/11, deep_attn_3x16 10/11, deep_2x8 11/11, deep_attn_2x8 9/11	run_20260902T145245Z
families	deep0_3x16, deep0_attn_3x16, deep0_cusp_3x16	deep0_3x16 11/11, deep0_attn_3x16 8/11, deep0_cusp_3x16 0/11	run_20260902T145247Z
meta-GGA families	deep0_cusp_mgga_3x16, deep0_mgga_3x16, deep0_mgga_attn_3x16, deep0_rung35_mgga_3x16, deep0_rung35ms_mgga_3x16	0/55 (four certificates read FAIL, Sec. 3.6; the fifth fit was cancelled)	run_20260902T145250Z
25-cycle arm	deep_3x16, tag [25 cycles]	2/5 (7 and 12)	run_20260908T153856Z
dpyscf-parity arm	deep_3x16, tag [dpyscf parity]	2/5 (7 and 12)	run_20260908T153908Z

The subset-size axis is 1, 2, 3, 4, 5, 6, 7, 12, 15, 18 and 26 training points, drawn from the 26-point DFS training pool by the Jensen-Shannon subset selection of Sec. 2. Basis 6-311++G(3df,2pd), grid level 3, density fitting on, for every stage. The validation-best checkpoint is read for every optimization figure. The figures and tables hold 64 cells; the cells not among them are running or queued (deep_attn_3x16 at 26, stopped by the 48 h wall and relaunched from its resume checkpoint; deep_attn_2x8 at 26; deep0_attn_3x16 at 15, 18 and 26; deep0_cusp_3x16 at every size; the arms at 15 to 26), held (the meta-GGA group), or evaluated with held-out tables holding no reaction (deep_attn_2x8 at 18 and deep_3x16 [25 cycles] at 15: every benchmark reference was refused while the T1 backfill was rewriting the reference files, Sec. 5.8; both are to be re-evaluated in a later cluster session).

1. Model, descriptors and the functional form

Sources: xcquinox/pipeline/networks.py (the two networks), xcquinox/pipeline/descriptors.py and xcquinox/pipeline/metagga.py (the extra descriptors), xcquinox/net.py (the attention block), xcquinox/pipeline/config.py (the architecture registry). The coordinate convention is the DFS one (model.descriptor_coordinates: dfs in every v7 file), following Dick and Fernandez-Serra, PRB 104, L161109 (2021), whose equation numbers are quoted as "DFS Eq. n".

1.1 Density variables

With $n = n_\alpha + n_\beta$ the total density and $\sigma = |\nabla n|^2$,

$$k_F = (3\pi^2 n)^{1/3}, \quad s = \frac{|\nabla n|}{2k_F n}, \quad r_s = \left(\frac{3}{4\pi n}\right)^{1/3}, \quad \zeta = \frac{n_\alpha - n_\beta}{n}, \quad \phi(\zeta) = \frac{1}{2} \left[(1 + \zeta)^{4/3} + (1 - \zeta)^{4/3} \right].$$

s is the reduced density gradient (DFS Eq. 5), ϕ the spin-scaling factor of DFS Eq. 4 ($\phi = 1$ at $\zeta = 0$, $2^{1/3}$ at full polarization). The density is floored at the network's lower_rho_cutoff before any of these is formed.

1.2 Network coordinates (DFS coordinates)

The exchange network reads one coordinate at the GGA level and two at the meta-GGA level; the correlation network reads three and four. They are logarithmic compressions of the variables above (DFS Eqs. 7 to 10):

$$x_0 = \ln(n^{1/3} + 10^{-5}), \quad x_1 = \ln \phi(\zeta), \quad x_s = (1 - e^{-s^2}) \ln(s + 1), \quad x_\alpha = \ln \frac{\alpha + 1}{2}.$$

Exchange input: $[x_s, \dots]$, where s is that of the doubled spin channel $2n_\sigma$ (the exchange network is evaluated per spin channel, Sec. 1.4). Correlation input: $[x_0, x_1, x_s, \dots]$ on the total density. The ellipsis holds the extra descriptors of the architecture, in the registry's list order; on the meta-GGA rung the raw α column among them is replaced by x_α .

1.3 Extra descriptors

Iso-orbital indicator (meta-GGA rung, metagga), SCAN's α (Sun, Ruzsinszky and Perdew, PRL 115, 036402 (2015), Eq. 2; DFS Eq. 6):

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \quad \tau = \frac{1}{2} \sum_i |\nabla \psi_i|^2 = \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} \nabla \chi_\mu \cdot \nabla \chi_\nu, \quad \tau_W = \frac{|\nabla n|^2}{8n}, \quad \tau_{\text{unif}} = \frac{3}{10} (3\pi^2)^{2/3} n^{5/3}.$$

τ is a contraction of the live one-particle density matrix P against the AO gradients, so it is self-consistent and differentiable through the SCF without second derivatives. The bound $\alpha \geq 0$ (the

von Weizsacker inequality) is imposed by a smooth positive part rather than a hard clip, so the descriptor and its derivative stay continuous where a spin channel holds one orbital.

Nuclear cusp pair (cusp), two bounded features per grid point:

$$x_4 = \exp(-2Z_{\text{nearest}} r_{\text{min}}), \quad x_5 = \tanh\left(\frac{1}{5} \ln \sum_A \frac{Z_A}{r_A}\right).$$

x_4 is a proximity feature following the density-form cusp envelope e^{-2Zr} (Steiner, J. Chem. Phys. 39, 2365 (1963), the density counterpart of Kato’s wavefunction cusp); x_5 is the log-compressed bare-nuclear potential $-V_{\text{ext}}$, the log and the $1/5$ scale being the conditioning of this work. Both are geometric, so the synthetic pre-training mesh (Sec. 3.1) carries no values for them.

The remaining registry descriptors (density-matrix statistics, the rung-3.5 occupancies) are not on any v7 axis and are not described here.

1.4 The functional form

Each network is a multilayer perceptron with GELU activations, depth hidden layers of nodes units, a final linear layer (initialized at zero in the `deep_*` family, `zero_init_final_layer`), and an output squash. With g the MLP output at a grid point,

$$F_x = 1 + \mathcal{L}_{1.804}(G_x \cdot g_x), \quad F_c = 1 + \mathcal{L}_2(G_c \cdot g_c), \quad \mathcal{L}_\lambda(z) = \lambda \text{sigmoid}(z - \ln(\lambda - 1)) - 1.$$

$\mathcal{L}_\lambda(0) = 0$, so a zero MLP output returns $F = 1$ (the local density approximation), and F stays in $(0, \lambda)$ for every z . $\mathcal{L}_\lambda$ is the DFS I_a map (DFS Eq. 11) to machine precision. For exchange $\lambda = 1.804 = 1 + \kappa_{\text{PBE}}$, the PBE Lieb-Oxford-motivated ceiling (PBE, PRL 77, 3865 (1996), Eq. 14); DFS uses 1.174 (the tighter bound of Perdew, Ruzsinszky, Sun and Burke, J. Chem. Phys. 140, 18A533 (2014)), and the v7 meta-GGA architectures keep 1.804 as well (`config.resolved_xnet_lob_lim`), a recorded deviation. For correlation $\lambda = 2$ is the symmetric sigmoid of DFS Eq. 13; its purpose is $F_c \geq 0$, the upper value 2 is incidental.

G is the uniform-electron-gas gate that pins $F = 1$ where the gradient vanishes:

$$G = \tanh^2(s) \text{ (GGA rung)}, \quad G = x_s + \tanh^2(x_\alpha) \text{ (meta-GGA rung, DFS Eq. 12)},$$

evaluated on the raw s (and α), never on the compressed coordinate. On the meta-GGA rung the same gate form is used for both networks (DFS’s correlation gate applies $\tanh$ to x_s ; recorded deviation).

The exchange-correlation energy is assembled with the exchange spin-scaling relation of Oliver and Perdew (PRA 20, 397 (1979)) and the PW92 correlation baseline:

$$E_{xc} = \frac{1}{2} (E_x[2n_\alpha] + E_x[2n_\beta]) + E_c[n_\alpha, n_\beta], \quad E_x[n] = \int n \epsilon_x^{\text{LDA}}(n) F_x d^3r, \quad E_c = \int n \epsilon_c^{\text{PW92}}(r_s, \zeta) F_c d^3r,$$

so the exchange network sees each spin channel at its doubled density (its x_s is that of $2n_\sigma$), and the correlation network sees the total density with ζ through x_1 .

Attention variants. In the `_attn` architectures a self-attention block (Vaswani et al., 2017, Eqs. 1 and 2, pre-layer-norm as in Xiong et al., 2020) sits after the first hidden layer. The hidden vector of H units at a grid point is treated as h tokens of dimension $d_k = H/h$ and attention runs across the token axis,

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad x \leftarrow x + W^O \text{Concat}(\text{head}_1, \dots, \text{head}_h)(\text{LN}(x)).$$

It is a per-point re-mixing of hidden units, not a spatial or non-local operation.

1.5 Architectures on the v7 axes

The architectures lie on four axes: the size (hidden layers x units), the initialization of the last layer, attention, and the extra descriptors (which set the rung and the parent). The name states them: the family word is the initialization (deep for the Glorot start of every layer; deep0 for a last layer whose weight and bias are zeroed at construction, so that the untrained network is the LDA, $F = 1$, and pre-training starts from it), then the descriptor and attention tokens, then the size. deep_3x16 and deep0_3x16 are the same network (depth 3, width 16, the same inputs, the same parameter count), separated by the initialization alone, so the two columns are replicates of one architecture under two starts. Nothing is frozen and nothing is appended in either; after pre-training both are PBE clones that differ in their fitted weights.

Architecture	Start	Rung, parent	Layers x units	Heads	Extra descriptors
deep_2x8	Glorot	GGA, PBE	2 x 8	none	none
deep_attn_2x8	Glorot	GGA, PBE	2 x 8	2	none
deep_3x16	Glorot	GGA, PBE	3 x 16	none	none
deep_attn_3x16	Glorot	GGA, PBE	3 x 16	4	none
deep0_3x16	last layer zeroed	GGA, PBE	3 x 16	none	none
deep0_attn_3x16	last layer zeroed	GGA, PBE	3 x 16	4	none
deep0_cusp_3x16	last layer zeroed	GGA, PBE	3 x 16	none	cusp
deep0_mgga_3x16	last layer zeroed	meta-GGA, SCAN	3 x 16	none	α
deep0_mgga_attn_3x16	last layer zeroed	meta-GGA, SCAN	3 x 16	4	α
deep0_cusp_mgga_3x16	last layer zeroed	meta-GGA, SCAN	3 x 16	none	cusp, α
deep0_rung35_mgga_3x16	last layer zeroed	meta-GGA, SCAN	3 x 16	none	cusp, rung-3.5 occupancy, α
deep0_rung35ms_mgga_3x16	last layer zeroed	meta-GGA, SCAN	3 x 16	none	cusp, rung-3.5 multishell occupancy, α

The network inputs follow from the rung and the extra descriptors:

Architecture	Exchange input	Correlation input
deep_2x8, deep_attn_2x8, deep_3x16, deep_attn_3x16, deep0_3x16, deep0_attn_3x16	x_s	x_0, x_1, x_s
deep0_cusp_3x16	x_s, x_4, x_5	x_0, x_1, x_s, x_4, x_5
deep0_mgga_3x16, deep0_mgga_attn_3x16	x_s, x_α	x_0, x_1, x_s, x_α
deep0_cusp_mgga_3x16	x_s, x_4, x_5, x_α	$x_0, x_1, x_s, x_4, x_5, x_\alpha$
deep0_rung35_mgga_3x16, deep0_rung35ms_mgga_3x16	x_s, x_4, x_5, x_α and the occupancy columns (two; six in the multishell form)	$x_0, x_1, x_s, x_4, x_5, x_\alpha$ and the same occupancy columns

The deep0 networks also carry `dm_entropy_intensive` and `descriptor_log_transform`, both inert on this axis (no density-matrix descriptor; the DFS coordinates already compress s , and only the cusp column-1 logarithm acts). All v7 networks are unanchored (`model.parent_anchor: false`): the network output is the whole enhancement factor, and the parent functional enters only as the pre-training target. A cell of a run trained under another protocol carries the run’s tag on its name (deep_3x16 [25 cycles], deep_3x16 [dpyscf parity]; Sec. 6), so that the same architecture under two protocols sits on one axis.

2. Training-subset generation (step 0 of the pipeline)

Sources: `xcquinox/pipeline/dfs_pool.py` (the pool), `xcquinox/pipeline/training_points.py` (build_dfs_pool_points), `xcquinox/pipeline/subset_selection.py` (the descriptors, the histograms, the metric, the search), the selection ledger `notebooks/checkpoints_step7/alpha_on/subset_index_log.json`, the cached per-species descriptors `notebooks/checkpoints_step7/subset_descriptors/` and the cached reference histogram `notebooks/checkpoints_step7/alpha_on/dfs_pool_full_hist/reference.npz`. Every value quoted below was recomputed from those caches with the package’s own functions and agrees with the ledger to the last digit.

2.1 The pool

The training pool is the DFS training set (Dick and Fernandez-Serra, PRB 104, L161109 (2021), SI Sec. II): 21 atomization energies of G2/97 molecules, 3 BH76 reactions and 2 IP13 ionization potentials, 26 selectable points, with the H and Li atomic densities of the DFS set entering as the anchors of the points that contain those elements. The atomization references are the non-relativistic, zero-point-exclusive values of Haunschild and Klopper (J. Chem. Phys. 136, 164102 (2012)) with H₂O and C₂H₂ from W4-11 (Karton, Daon and Martin, CPL 510, 165 (2011)); the reaction energies are the GMTKN55 BH76RC values; the ionization potentials the NIST values. Each point carries the species its loss term needs: an atomization point its molecule plus the H and Li anchors it contains, a reaction point every reactant and product, an ionization point the neutral and the cation. The 26 points, in the index order of the ledger:

Index	Point	Kind	Species (name, charge, spin)
0	H2	atomization energy	H2 (0, 0), H (0, 1)
1	N2	atomization energy	N2 (0, 0)
2	FLi	atomization energy	FLi (0, 0), Li (0, 1)
3	CHN	atomization energy	CHN (0, 0), H (0, 1)
4	CO2	atomization energy	CO2 (0, 0)
5	F2	atomization energy	F2 (0, 0)
6	C2H2	atomization energy	C2H2 (0, 0), H (0, 1)
7	CO	atomization energy	CO (0, 0)
8	HLi	atomization energy	HLi (0, 0), H (0, 1), Li (0, 1)
9	Na2	atomization energy	Na2 (0, 0)
10	NO	atomization energy	NO (0, 1)
11	CH	atomization energy	CH (0, 1), H (0, 1)

Index	Point	Kind	Species (name, charge, spin)
12	HO	atomization energy	HO (0, 1), H (0, 1)
13	NO2	atomization energy	NO2 (0, 1)
14	HN	atomization energy	HN (0, 2), H (0, 1)
15	O3	atomization energy	O3 (0, 0)
16	N2O	atomization energy	N2O (0, 0)
17	CH3	atomization energy	CH3 (0, 1), H (0, 1)
18	CH2	atomization energy	CH2 (0, 2), H (0, 1)
19	H2O	atomization energy	H2O (0, 0), H (0, 1)
20	H3N	atomization energy	H3N (0, 0), H (0, 1)
21	OH+N2_to_H+N2O	reaction energy	HO (0, 1), N2 (0, 0), H (0, 1), N2O (0, 0)
22	OH+CH3_to_O+CH4	reaction energy	HO (0, 1), CH3 (0, 1), O (0, 2), CH4 (0, 0), H (0, 1)
23	HF+F_to_H+F2	reaction energy	HF (0, 0), F (0, 1), H (0, 1), F2 (0, 0)
24	Li_IP	ionization potential	Li (0, 1), Li+ (1, 0)
25	C_IP	ionization potential	C (0, 2), C+ (1, 1)

2.2 The descriptor distributions

For every unique species of the pool (30 with the anchors and the cations) one self-consistent PBE calculation at def2-SVP on a level-1 quadrature grid gives, at every grid point g with quadrature weight w_g , the three descriptors

$$\rho^{1/3}, \quad s = \frac{|\nabla\rho|}{2(3\pi^2)^{1/3}\rho^{4/3}}, \quad \alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}} \text{ (clipped to } [0, 100]),$$

with $\tau_W = |\nabla\rho|^2/8\rho$ and $\tau_{\text{unif}} = \frac{3}{10}(3\pi^2)^{2/3}\rho^{5/3}$ as in Sec. 1.3. A point’s descriptor sample is the concatenation of its species’ grid points, so a reaction contributes in proportion to the number of grid points of its species. The reference distribution is the concatenation over all 26 points. For each descriptor $x \in \{\rho^{1/3}, s, \alpha\}$ the reference sample is binned into 200 bins of equal width between its 0.1 and 99.9 percentiles, weighted by the quadrature weights, and every candidate subset is binned on the same edges:

$$h_x^{(b)} = \sum_{g: x_g \in \text{bin } b} w_g, \quad P_x^{(b)} = \frac{h_x^{(b)}}{\sum_{b'} h_x^{(b')}}.$$

The normalization to a probability mass function removes the bin width and the sample size, so the comparison is between the shapes of the weighted distributions.

Two properties of these histograms follow from the linear edges and the percentile range and are stated here because they determine what the metric measures. The ranges are set by the low-density tail of the grids, where s and α grow without bound: the s edges run from 8×10^{-9} to 2.0×10^{10} and the α edges from 0 to 2.6×10^{10} , while the $\rho^{1/3}$ edges run from 10^{-10} to 9.04. With 200 bins of equal width, the quadrature weights (which grow into the tail) place 98.1 percent of the reference mass in the first $\rho^{1/3}$ bin, 88.9 percent in the first s bin and 97.2 percent in the first α bin; 99 percent of the mass lies within the first 2, 126 and 12 bins. What the first bin holds differs between the descriptors. For s and α it is the whole chemically relevant range ($s = 5$ and $\alpha = 10$ both fall in bin 0, whose widths are 10^8), so the metric compares those two distributions over the remaining 11 and 3 percent of the weight, which sits in the low-density tails of the molecular grids. For $\rho^{1/3}$ the first bin is the vacuum tail itself ($\rho < 9.2 \times 10^{-5}$ per bohr³), and the remaining 1.9 percent of the weight is the valence and core region ($\rho = 0.01$ falls in bin 4, $\rho = 1$ in bin 22). Samples outside the edges enter no histogram: 1.92 percent of the pool weight for s and 0.20 percent for α . The s marginal carries most of the divergence (Sec. 2.4).

2.3 The metric and the search

The distance between a candidate subset S and the pool is the Jensen-Shannon divergence (Lin, IEEE Trans. Inf. Theory 37, 145 (1991)) summed over the three marginals with equal weights:

$$\text{JSD}(S) = \sum_{x \in \{\rho^{1/3}, s, \alpha\}} \frac{1}{2} [\text{KL}(P_x \parallel M_x) + \text{KL}(Q_x \parallel M_x)], \quad M_x = \frac{1}{2}(P_x + Q_x),$$

$$\text{KL}(P \parallel Q) = \sum_b P^{(b)} \ln \frac{P^{(b)}}{Q^{(b)}},$$

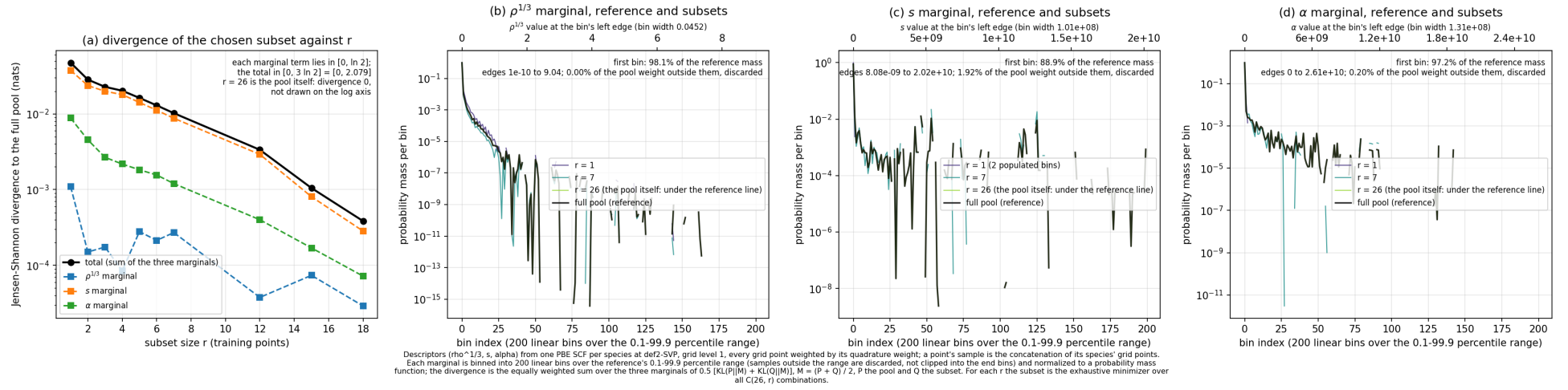
with P_x the reference and Q_x the subset mass function of descriptor x , probabilities floored at 10^{-12} inside the logarithm, in nats. Each marginal term lies in $[0, \ln 2]$ and the sum in $[0, 3 \ln 2]$; a subset whose grid points all fall outside a marginal's range is assigned an infinite divergence and never selected. For each subset size $r \in \{1, \dots, 7, 12, 15, 18, 26\}$ the subset is the exhaustive minimizer of $\text{JSD}(S)$ over all $\binom{26}{r}$ combinations (9.7 million at $r = 12$), computed with the pre-binned per-point histograms; the L2 distance between the same mass functions was run as a second metric and is recorded in the ledger but not used by any v7 run. The subsets are nested only where the minimizer happens to nest: the 7-molecule subset does not contain the 6-molecule one.

2.4 The chosen subsets

r	Points (ledger index order)	AE / BH76 / IP	JSD to the pool	$\rho^{1/3}$ term	s term	α term	s share
1	H2O	1 / 0 / 0	0.0475	0.00109	0.03764	0.00881	79%
2	HLi, OH+N2_to_H+N2O	1 / 1 / 0	0.0284	0.00015	0.02372	0.00455	83%
3	OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	0 / 2 / 1	0.0228	0.00017	0.01995	0.00267	88%
4	CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	1 / 2 / 1	0.0203	0.00008	0.01801	0.00218	89%
5	HLi, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	1 / 2 / 2	0.0163	0.00028	0.01418	0.00183	87%
6	HLi, CH, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	2 / 2 / 2	0.0130	0.00021	0.01120	0.00156	86%
7	CHN, CO2, HLi, CH, OH+N2_to_H+N2O, Li_IP, C_IP	4 / 1 / 2	0.0102	0.00027	0.00877	0.00120	86%
12	CO2, C2H2, CO, HLi, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	7 / 3 / 2	0.0034	0.00004	0.00292	0.00040	87%
15	FLi, CHN, CO2, C2H2, CO, HLi, NO, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	10 / 3 / 2	0.0010	0.00007	0.00081	0.00017	77%
18	N2, FLi, CHN, CO2, C2H2, CO, HLi, Na2, NO, CH, NO2, N2O, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	13 / 3 / 2	0.0004	0.00003	0.00028	0.00007	74%
26	H2, N2, FLi, CHN, CO2, F2, C2H2, CO, HLi, Na2, NO, CH, HO, NO2, HN, O3, N2O, CH3, CH2, H2O, H3N, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	21 / 3 / 2	0.0000	0.00000	0.00000	0.00000	-

Fig. 2.1 Left: the Jensen-Shannon divergence of the chosen subset to the full pool against the subset size, the total and its three marginal terms (log scale; the bound $3\ln 2 = 2.08$ lies far above the axis). The divergence falls from 0.0475 at one molecule to 0.0102 at seven, 0.0034 at twelve, 0.0010 at fifteen and 0.0004 at eighteen, and to zero at the full pool; the s term is 74 to 89 percent of the total at every size (0.0376 of 0.0475 at one molecule, 0.00028 of 0.00038 at eighteen). Right, three panels: the mass functions of the reference (black) and of the subsets at $r = 1, 7$ and 26 over the bin index, log scale, with the first-bin mass printed in each panel.

Training-subset selection: Jensen-Shannon divergence to the full DFS pool



3. Pre-training (functional cloning)

Sources: `xcquinox/pipeline/pretrain.py`, `xcquinox/pipeline/pretrain_data_gen.py`, the `pretrain:` and `fidelity:` blocks of the resolved configurations, `pretrain/<arch>/pretrain_metadata.json` and `pretrain/<arch>/fidelity_certificate.json` of each run.

3.1 Set, targets and objective

Each network is fitted to reproduce its parent functional’s enhancement factor pointwise on the parent’s own self-consistent densities: PBE for the GGA rung, SCAN for the meta-GGA rung. The set is the DFS pre-training inventory (8 free atoms and 23 G2/97 molecules at the GGA level; 22 molecules at the meta-GGA level, which drops H2) plus every single-atom species of the BH76 and W4-11 pools at its production charge and spin (the 12 neutral elements and the F- and Cl- anions), 39 systems for the GGA groups and 38 for the meta-GGA group (`n_systems` of the certificates). The targets are stored per grid point as

$$t_x = F_x^{\text{parent}} - 1 = \frac{\epsilon_x^{\text{parent}}}{\epsilon_x^{\text{LDA}}} - 1, \quad t_c = F_c^{\text{parent}} - 1 = \frac{\epsilon_c^{\text{parent}}}{\epsilon_c^{\text{PW92}}} - 1,$$

from spin-resolved libxc evaluations, the exchange rows posed per spin channel at the doubled density ($2n_\sigma, 4\sigma_{\sigma\sigma}$), the footing the production exchange evaluates (`exchange_footing: spin_channel`). A synthetic (r_s, s, α) mesh (the DFS SI’s (s, α) grid extended to three dimensions) is added as a regularizer at a fixed 30 percent share of the integration weight (`mesh_fraction: 0.3`); its rows belong to no system.

The objective per network is the integration-weighted pointwise residual plus a per-system energy term (`_PretrainLoss`):

$$\begin{aligned} \mathcal{L}_{\text{pre}} = & \frac{\sum_i w_i (F_i^{\text{NN}} - 1 - t_i)^2}{\sum_i w_i} \\ & + w_E \frac{1}{N_{\text{sys}}} \sum_s \left(\sum_{i \in s} w_i^{\text{grid}} n_i \epsilon_i^{\text{LDA}} F_i^{\text{NN}} - E_{xc,s}^{\text{parent}} \right)^2, \\ & w_i = |n_i \epsilon_i^{\text{LDA}}| w_i^{\text{grid}}, \quad w_E = 0.1. \end{aligned}$$

The energy term is present because the pointwise residual alone does not bound a system’s energy: in the earlier campaign the architecture with the lowest exchange residual carried the largest atomization-energy offset from its parent. Schedule: Adam, 20000 steps, learning rate 10^{-3} decaying linearly to 10^{-5} from the midpoint, gradient clip 1.0, a 20 percent held-out validation slice scored every 50 steps with patience 300 checks (`pretrain: block`).

3.2 The fidelity certificate

After the fit, the saved networks are re-evaluated on every system of the set at the production basis and grid and compared with the parent: the per-system exchange-correlation energy error $\Delta E_{xc,s} = E_{xc,s}^{\text{NN}} - E_{xc,s}^{\text{parent}}$, and, for every molecule, the atomization-energy error $\Delta AE = \Delta E_{xc,\text{mol}} - \sum_A \Delta E_{xc,A}$ against the parent’s own atomization energy. The gate (`fidelity: block`, two-tier since 2026-09-02) is

$$\max_{\text{atoms}} |\Delta E_{xc,A}| \leq 1.0 \text{ mHa}, \quad \text{mean}_{\text{mol}} |\Delta AE| \leq 1.0 \text{ kcal/mol}, \quad \max_{\text{mol}} |\Delta AE| \leq 2.0 \text{ kcal/mol},$$

and a failing certificate makes the pre-training task exit without success, which holds the group’s preflight and training array. The certificate is what gives the training-stage comparison against PBE (or SCAN) its meaning: every trained cell starts from a network that is its parent to within these tolerances.

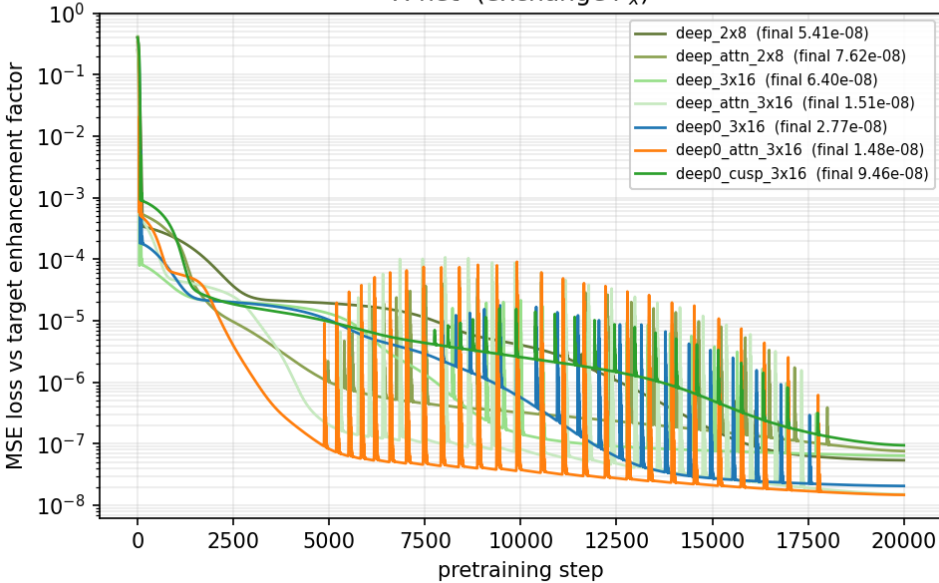
3.3 Pre-training loss curves

Fig. 3.1 The pointwise term of the objective per step taken, for the exchange network (left) and the correlation network (right), one curve per architecture, log scale over the 20000 steps; the final value of each curve is printed in the legend. The regular spikes are bursts of 5 to 14 consecutive steps at 6 to 12 times the local level, recurring every 420 to 590 steps from step 4000 to 8000 onward and ending near step 17500; they do not sit on the 50-step validation grid, and no snapshot is reloaded during a fit (the best network is kept aside while training continues from the current weights), so their cause lies in the optimization itself and is not established here. The first figure holds the seven certified GGA clones of the size and families runs; the second the four meta-GGA clones whose fits completed (their certificates failed, so they are not in the merged view and are drawn from their run; the fifth fit, `deep0_mgga_attn_3x16`, was cancelled before it wrote a network pair).

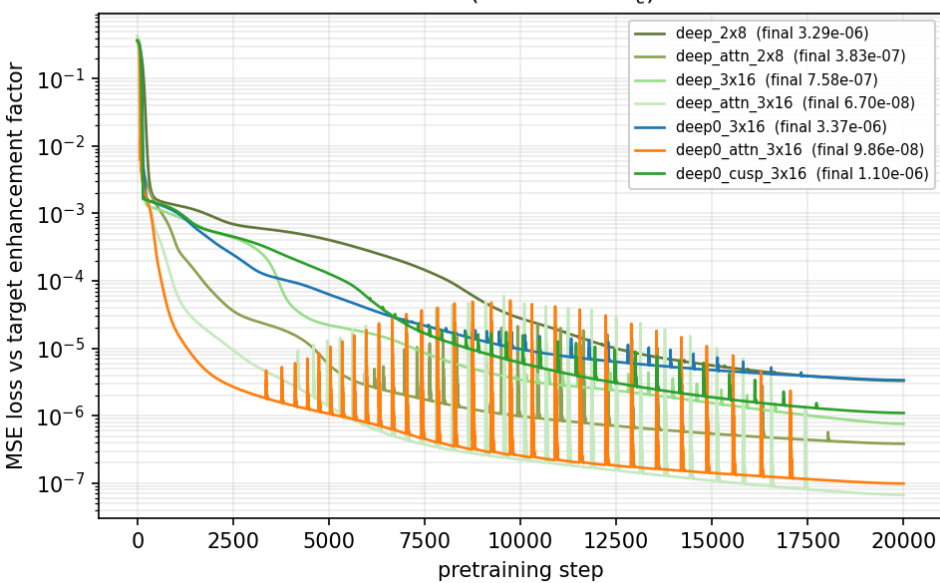
The exchange curves of the four Glorot-start architectures reach 5.4×10^{-8} (`deep_2x8`), 7.6×10^{-8} (`deep_attn_2x8`), 6.4×10^{-8} (`deep_3x16`) and 1.5×10^{-8} (`deep_attn_3x16`), the correlation curves 3.3×10^{-6} , 3.8×10^{-7} , 7.6×10^{-7} and 6.7×10^{-8} in the same order; `deep_attn_3x16` descends earliest and ends lowest on both networks, and the `deep_2x8` correlation network ends highest of the seven. Of the zero-init architectures `deep0_attn_3x16` is lowest on both networks (1.5×10^{-8} , 1.0×10^{-7}), `deep0_3x16` and `deep0_cusp_3x16` within a factor 3.5 of each other (2.8×10^{-8} and 9.5×10^{-8} ; 3.4×10^{-6} and 1.1×10^{-6}). The four meta-GGA clones stop at 3.7×10^{-6} (`deep0_cusp_mgga_3x16`), 3.8×10^{-6} (`deep0_rung35ms_mgga_3x16`), 5.6×10^{-6} (`deep0_mgga_3x16`) and 5.9×10^{-6} (`deep0_rung35_mgga_3x16`) in exchange and at 5.2×10^{-6} , 1.2×10^{-5} , 1.6×10^{-5} and 1.05×10^{-5} in correlation, 40 to 400 times above the GGA clones in exchange and 1.5 to 240 times in correlation after the same 20000 steps, with smooth trajectories and no divergence.

Pretraining loss curves -- 7 archs, 20000 steps
run_20260908T153908Z

X-net (exchange F_x)

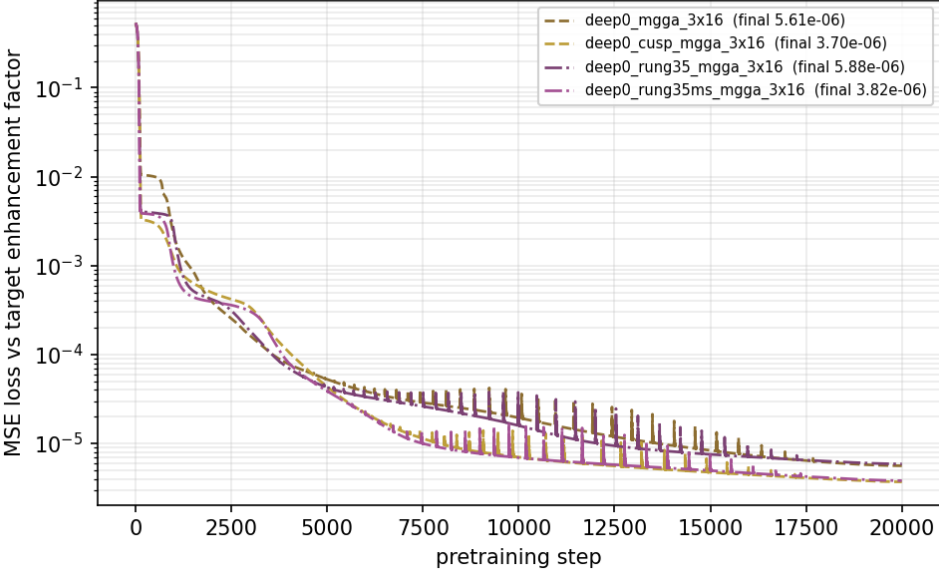


C-net (correlation F_c)

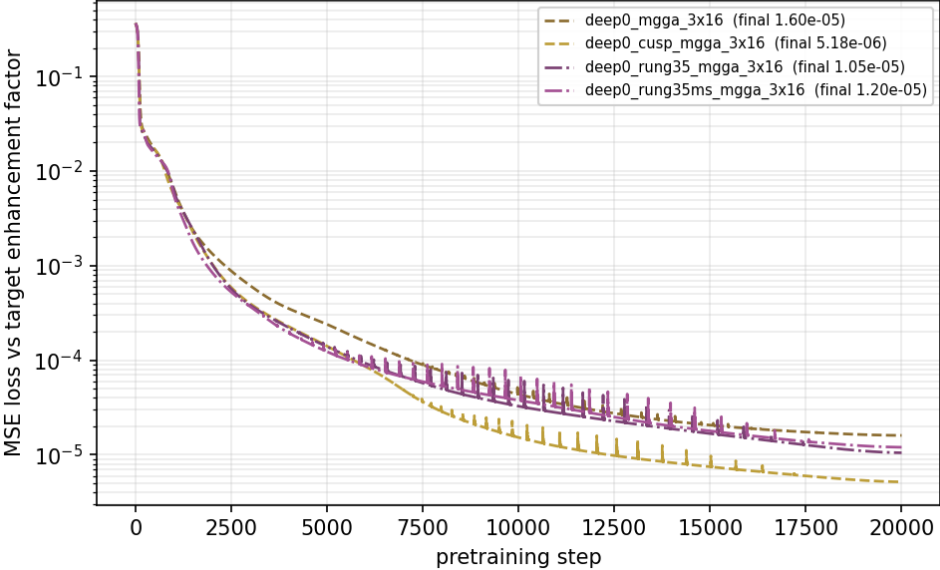


Pretraining loss curves -- 4 archs, 20000 steps
run_20260902T145250Z

X-net (exchange F_x)



C-net (correlation F_c)



3.4 Certificate summary

The certificate-summary figure (Fig. 3.2 of the full report) is not repeated here; its numbers are in the table of Sec. 3.6.

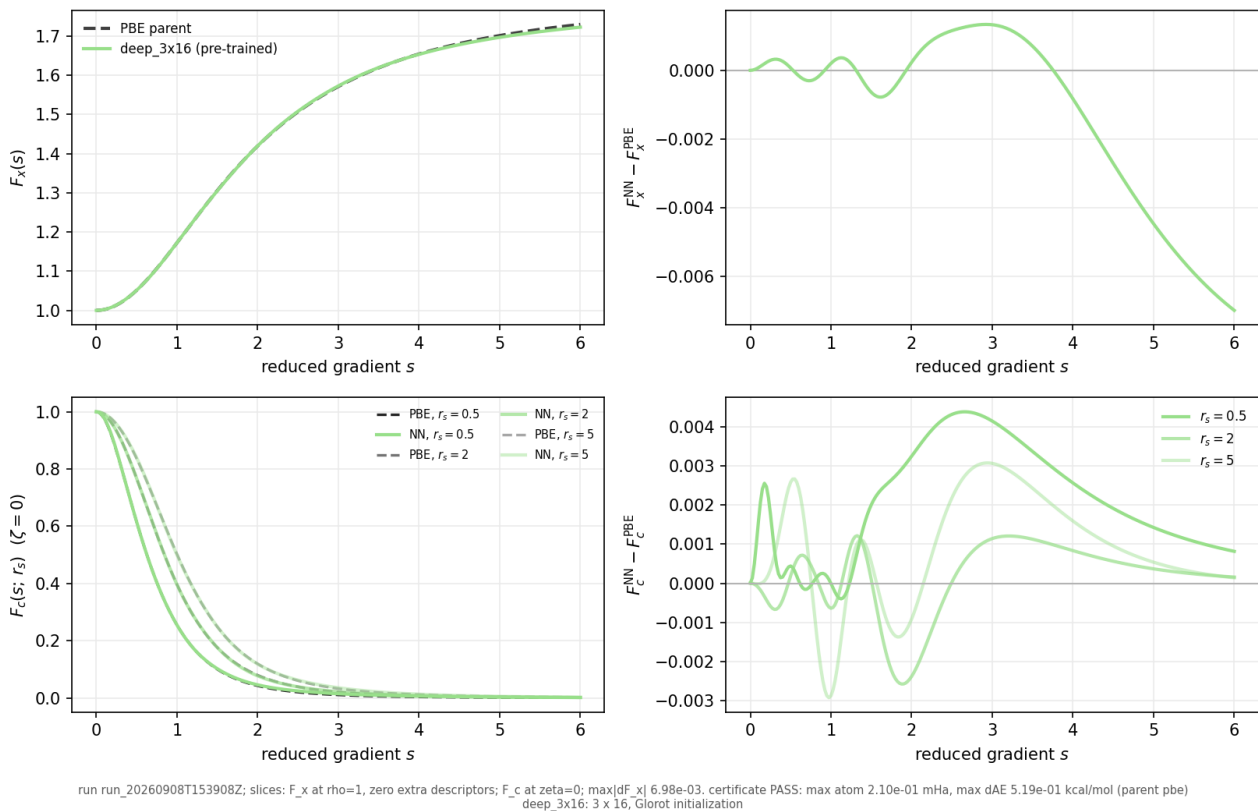
3.5 Pre-trained enhancement factors

Fig. 3.3 For one architecture, the pre-trained network against its parent along two slices of descriptor space. Top left: the exchange enhancement factor $F_x(s)$ at density $n = 1$ and zero extra descriptors, the parent dashed, the network solid, from $s = 0$ to 6. Top right: the difference $F_x^{\text{NN}} - F_x^{\text{parent}}$ on the same slice, which is where the fit is visible at all. Bottom left: the correlation factor $F_c(s; r_s)$ at $\zeta = 0$ for $r_s = 0.5, 2$ and 5, parent dashed and network solid; bottom right: the three differences. For the meta-GGA clones the layout is three columns: $F_x(s; \alpha)$ at $\alpha = 0$ and $\alpha = 1$ (the SCAN Fig. 1 convention, α the exact iso-orbital indicator of the slice) and F_c at the same two α values, each with its difference panel. The footer of each figure names the run, the slices, the largest $|\Delta F_x|$ on the slice and the certificate verdict.

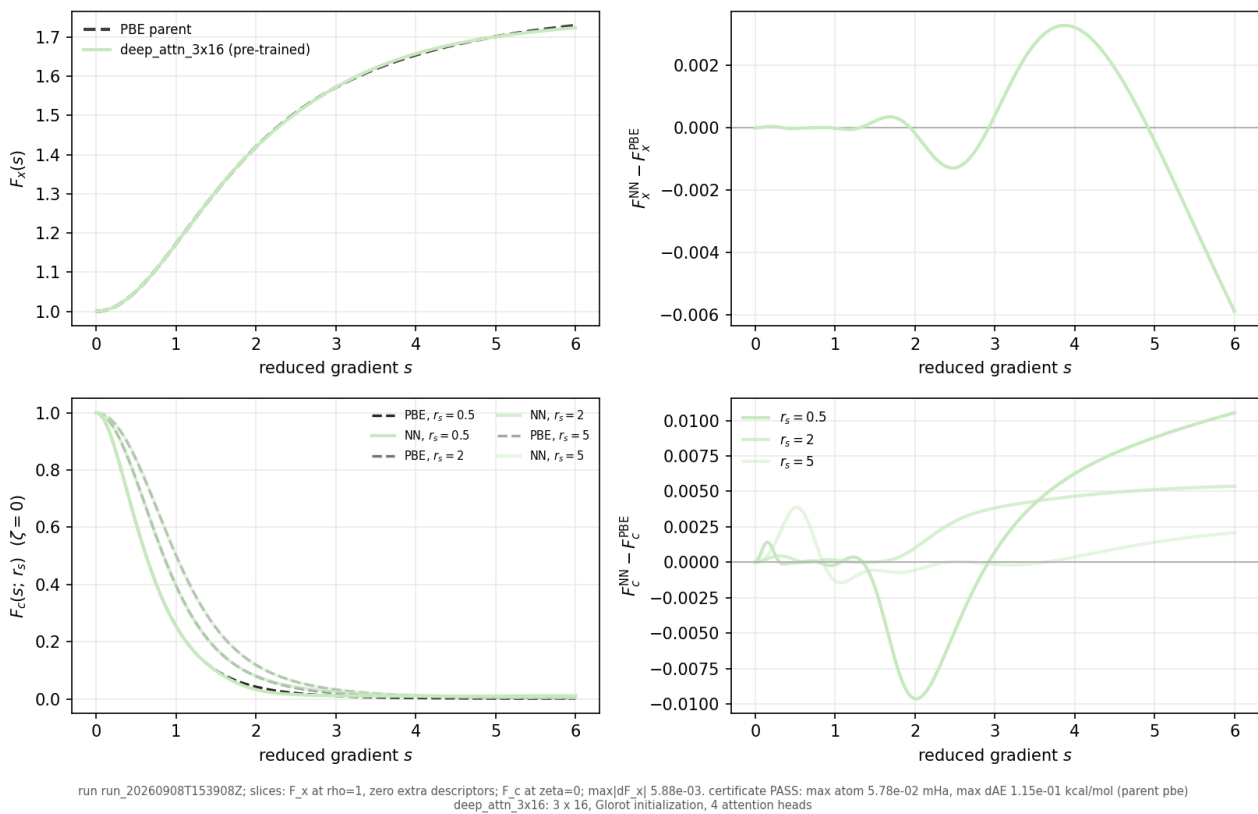
deep_3x16: $|\Delta F_x| \leq 7 \times 10^{-3}$ over $s \leq 6$, a smooth negative deviation growing beyond $s = 4$ (the region the pre-training set barely samples), $|\Delta F_c| \leq 4.5 \times 10^{-3}$ with oscillations below $s = 1.5$ at every r_s . deep_attn_3x16: $|\Delta F_x| \leq 6 \times 10^{-3}$, $|\Delta F_c| \leq 10^{-2}$ with one dip at $s = 2$, $r_s = 0.5$. deep0_3x16: $|\Delta F_x| \leq 2.6 \times 10^{-3}$ (the tightest exchange clone of the three shown), $|\Delta F_c| \leq 1.2 \times 10^{-2}$ at $r_s = 0.5$ and 5. The 2x8 pair (the fourth and fifth figures): deep_2x8 has $|\Delta F_x| \leq 4.9 \times 10^{-3}$, a positive bump of +0.0024 at $s = 2$ and a negative deviation beyond $s = 2.5$ with its minimum -0.0049 at $s = 3.6$, and $|\Delta F_c| \leq 1.2 \times 10^{-2}$ ($r_s = 0.5$, a bump peaking at $s = 1.75$; 2.8×10^{-3} at $r_s = 2$, 8.5×10^{-3} at $r_s = 5$, with oscillations below $s = 1.5$); deep_attn_2x8 has $|\Delta F_x| \leq 1.5 \times 10^{-2}$, the deviation growing negative beyond $s = 4$ to -0.015 at $s = 6$ (the largest exchange miss of the seven GGA clones, in the region the set barely samples), and $|\Delta F_c| \leq 5.7 \times 10^{-3}$ (a peak at $s = 0.65$, $r_s = 5$; 4.6×10^{-3} at $r_s = 0.5$, 2.3×10^{-3} at $r_s = 2$).

The meta-GGA clones against SCAN (the last four figures). deep0_cusp_mgga_3x16: F_x at $\alpha = 0$ runs 0.005 below SCAN up to $s = 1$ and 0.01 to 0.027 above it for $s > 2.5$ (the maximum at $s = 4.4$), and F_x at $\alpha = 1$ 0.004 to 0.013 below it between $s = 1$ and 4; F_c at $\alpha = 1$, $r_s = 0.5$ has a spike of +0.04 near $s = 0.2$, a dip of -0.016 at $s = 0.8$ and sits 0.006 to 0.033 below SCAN from $s = 2$ onward; at $\alpha = 0$ the correlation differences stay within ± 0.02 . deep0_mgga_3x16 is the tightest of the four on the slices ($|\Delta F_x| \leq 1.0 \times 10^{-2}$: at $\alpha = 0$ -0.003 below $s = 1$, $+0.003$ at $s = 1.3$ and 3.7 and a fall to -0.010 at $s = 6$; at $\alpha = 1$ a dip of -0.008 at $s = 0.8$; F_c within ± 0.006 at $\alpha = 0$ and, at $\alpha = 1$, spikes of $+0.024$ at $s = 0.2$ for $r_s = 0.5$ and -0.023 at $s = 0.7$ for $r_s = 5$) and still fails on the atoms by 4.1 mHa (N): the certificate is decided on the production densities, not on these slices. deep0_rung35_mgga_3x16: $|\Delta F_x| \leq 5.0 \times 10^{-2}$, the exchange running from -0.028 at $s = 0$ through zero near $s = 1.5$ to $+0.050$ at $s = 6$ at $\alpha = 0$ and from -0.016 at $s = 1$ to $+0.034$ at $s = 6$ at $\alpha = 1$; F_c at $\alpha = 0$ sits above SCAN at every s for $r_s = 0.5$ (by $+0.061$ at $s = 0.1$, $+0.002$ at $s = 0.8$) and dips to -0.035 and -0.038 near $s = 0.9$ at $r_s = 2$ and 5, and at $\alpha = 1$ has a peak of $+0.21$ at $s = 0.6$ ($r_s = 0.5$) and -0.07 at $s = 0.75$ ($r_s = 5$). deep0_rung35ms_mgga_3x16: $|\Delta F_x| \leq 3.3 \times 10^{-2}$ (from -0.033 at $s = 0$ to -0.007 at $s = 1$ at $\alpha = 0$, then rising to $+0.029$ at $s = 6$; within $+0.01$ at $\alpha = 1$) and the largest correlation misses of any clone: -0.32 to -0.39 at small s at $\alpha = 0$ for every r_s (the minima at $s = 0$, 0 and 1.3 for $r_s = 0.5, 2$ and 5), and at $\alpha = 1$ $+0.15$ at $s = 0.9$ ($r_s = 0.5$) and -0.18 at $s = 0.85$ ($r_s = 5$). These are systematic, same-sign offsets over wide ranges of s , which is the shape of error that does not cancel between a molecule and its atoms (Sec. 3.6).

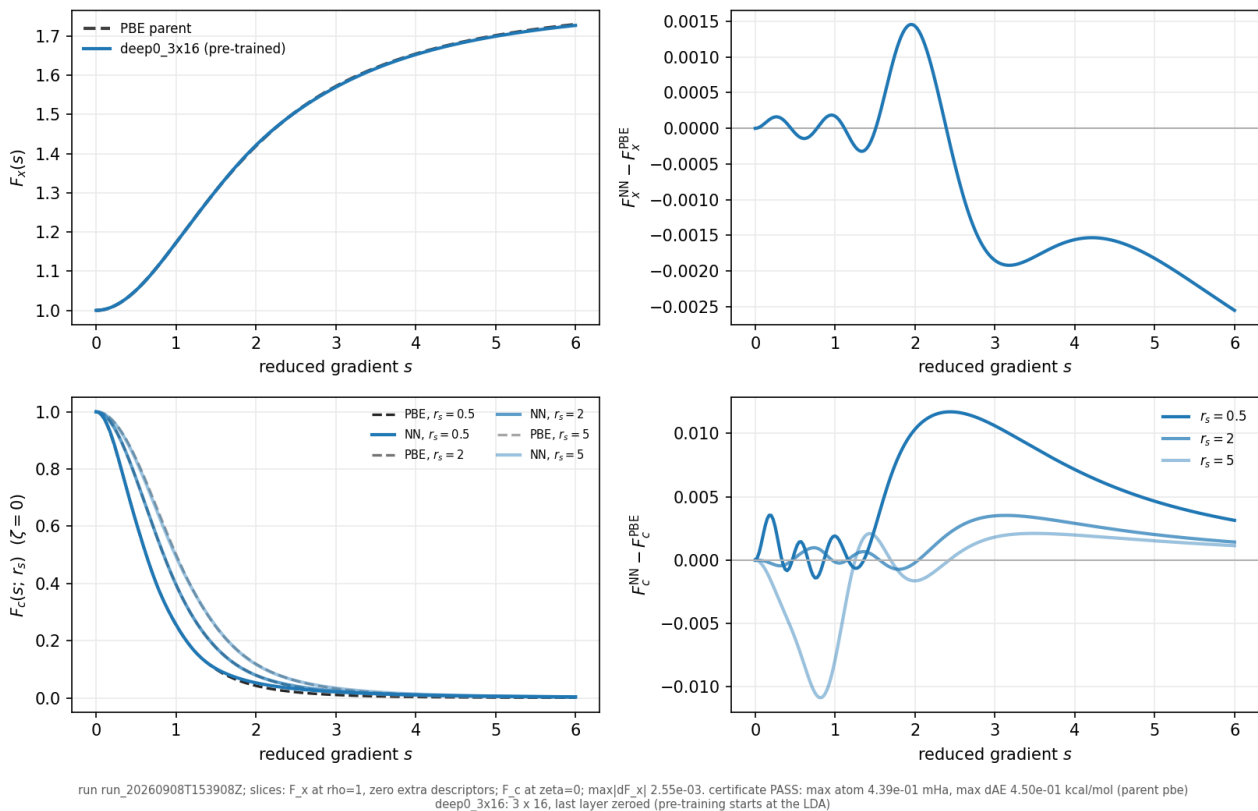
deep_3x16: pre-trained network against the PBE parent



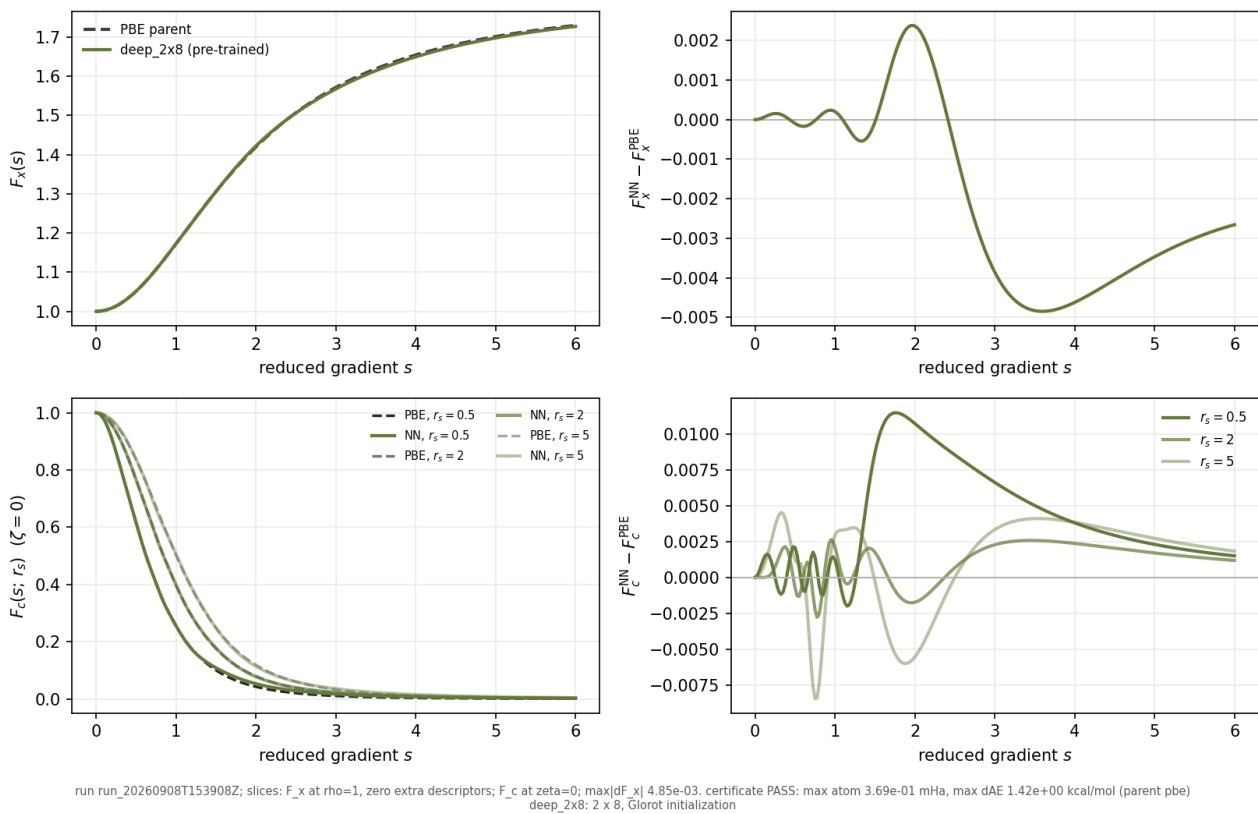
deep_attn_3x16: pre-trained network against the PBE parent



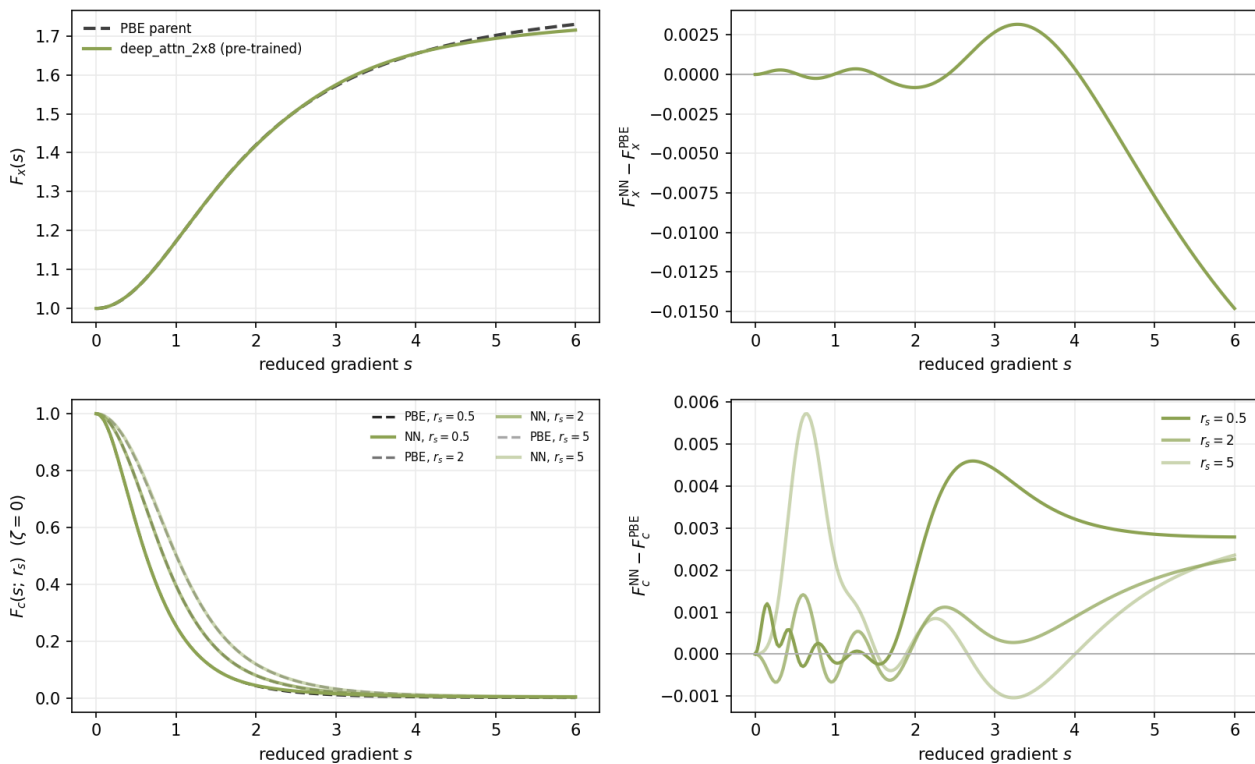
deep0_3x16: pre-trained network against the PBE parent



deep_2x8: pre-trained network against the PBE parent

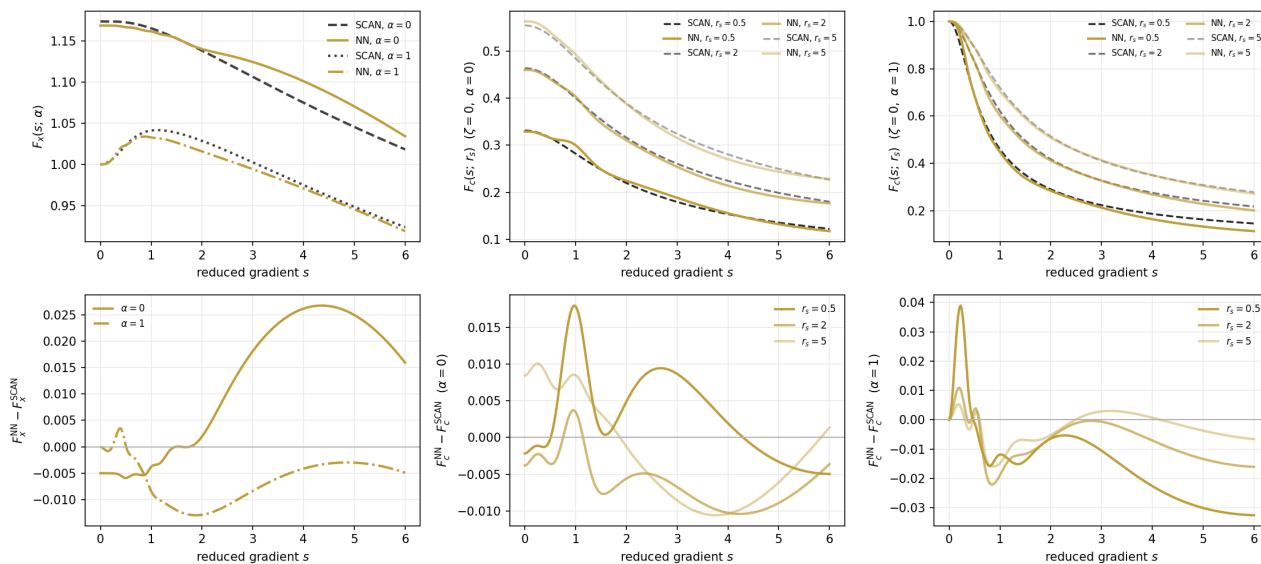


deep_attn_2x8: pre-trained network against the PBE parent



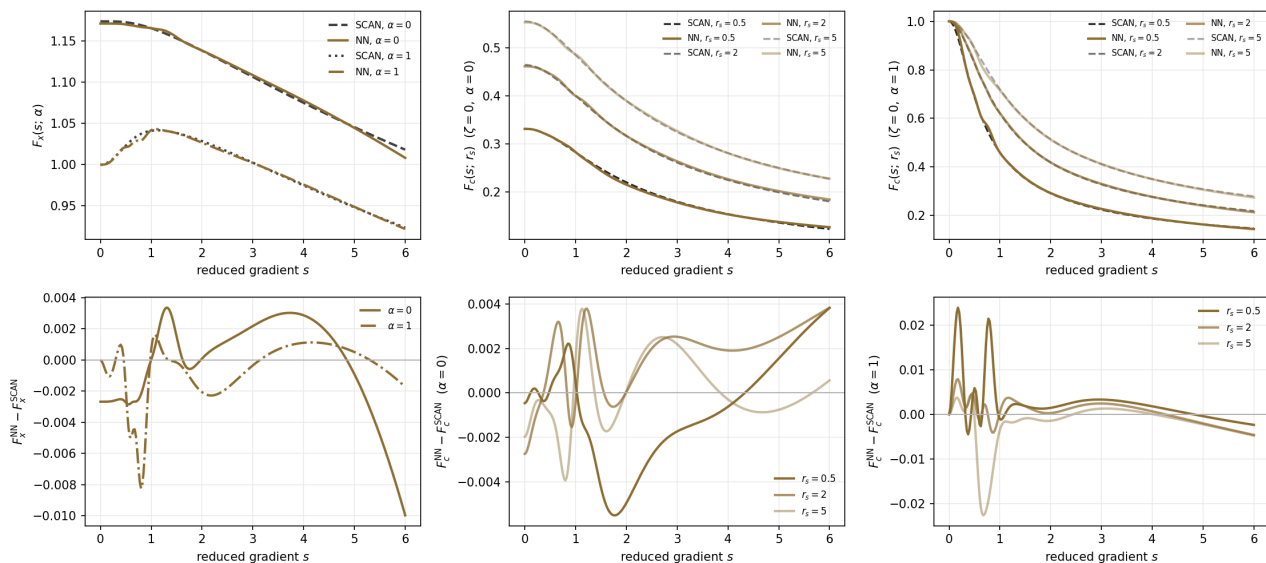
run run_20260908T153908Z; slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$; $\max|dF_x|$ 1.48e-02. certificate PASS: max atom 2.01e-01 mHa, max dAE 5.73e-01 kcal/mol (parent pbe)
deep_attn_2x8: 2 x 8, Glorot initialization, 2 attention heads

deep0_cusp_mgga_3x16: pre-trained network against the SCAN parent



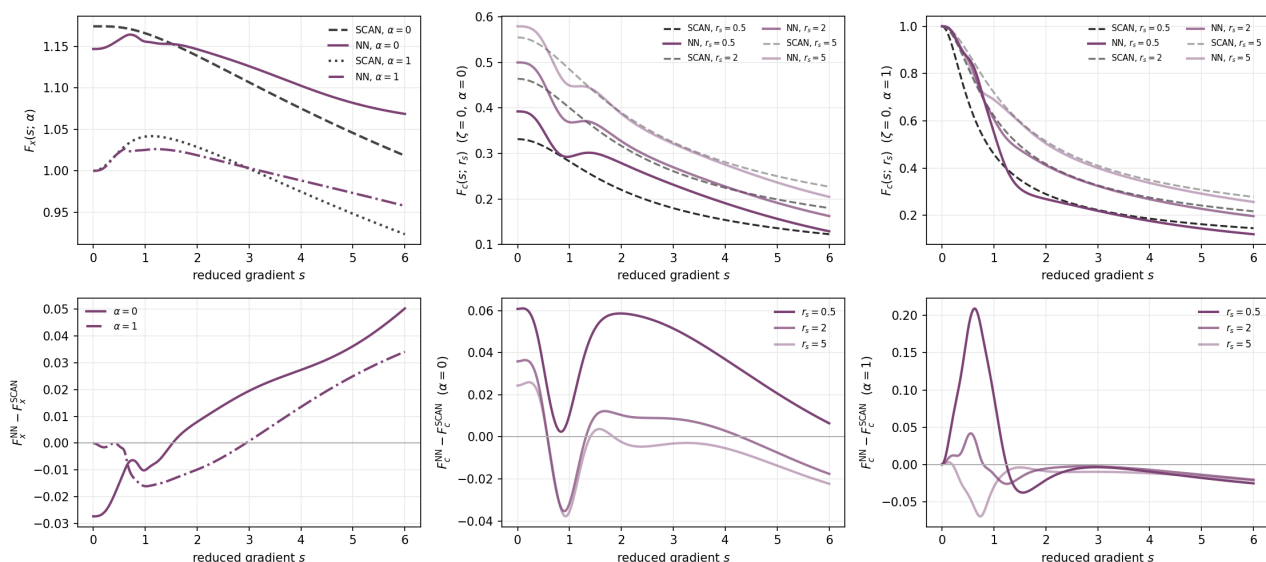
run run_20260902T145250Z; parent SCAN; slices: F_x at $\rho=1$ at the exact iso-orbital indicator alpha in $\{0, 1\}$, zero other descriptors; F_c at $\zeta=0$, same alpha slices; $\max|dF_x|$ 2.67e-02. certificate FAIL: max atom 3.13e+00 mHa, max dAE 7.26e+00 kcal/mol (parent scan)
deep0_cusp_mgga_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), cusp descriptors, meta-GGA (SCAN parent)

deep0_mgga_3x16: pre-trained network against the SCAN parent

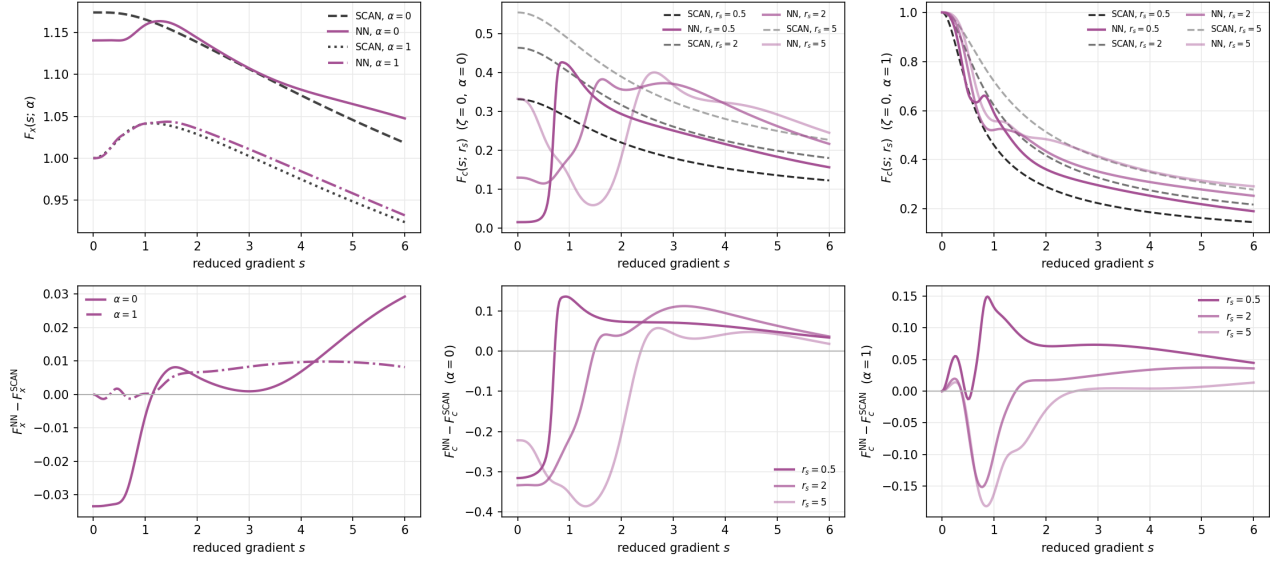


run run_20260902T145250Z; parent SCAN; slices: F_x at $\rho=1$ at the exact iso-orbital indicator alpha in $\{0, 1\}$, zero other descriptors; F_c at $\zeta=0$, same alpha slices; max|df_x| 9.97e-03; certificate FAIL: max atom 4.11e+00 mHa, max dAE 5.60e+00 kcal/mol (parent scan)
deep0_mgga_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), meta-GGA (SCAN parent)

deep0_rung35_mgga_3x16: pre-trained network against the SCAN parent



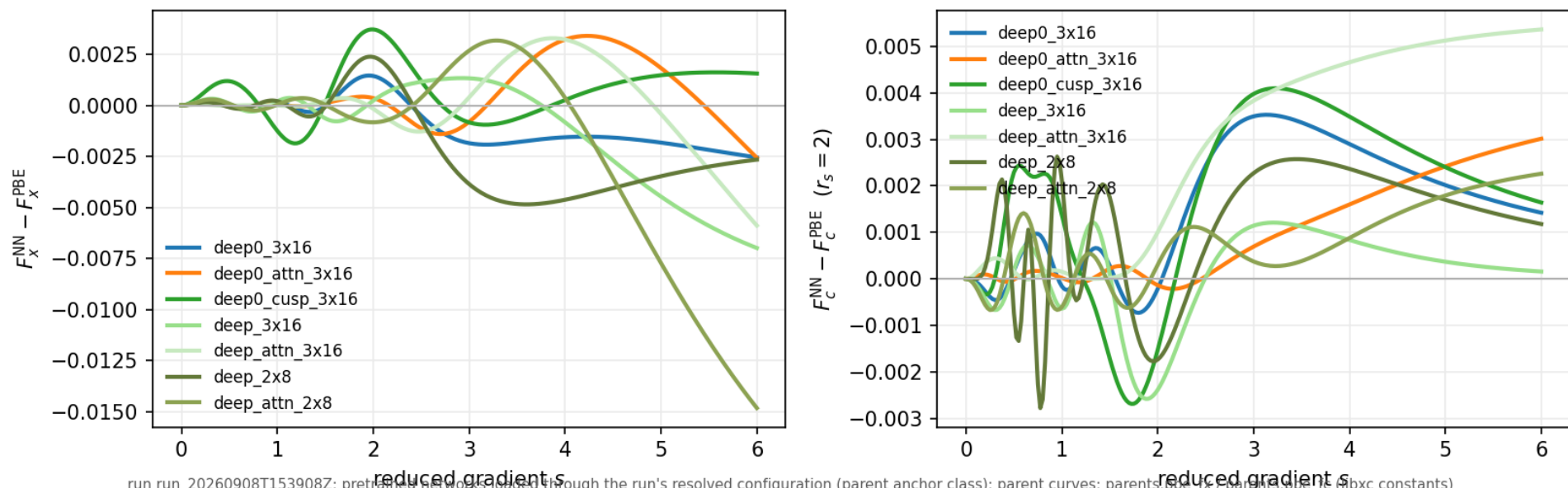
run run_20260902T145250Z; parent SCAN; slices: F_x at $\rho=1$ at the exact iso-orbital indicator alpha in $\{0, 1\}$, zero other descriptors; F_c at $\zeta=0$, same alpha slices; max|df_x| 5.03e-02; certificate FAIL: max atom 3.00e+00 mHa, max dAE 6.30e+00 kcal/mol (parent scan)
deep0_rung35_mgga_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), cusp descriptors, rung-3.5 occupancy, meta-GGA (SCAN parent)



run run_20260902T1452502; parent SCAN; slices: F_x at $\rho=1$ at the exact iso-orbital indicator alpha in $\{0, 1\}$, zero other descriptors; F_c at $\zeta=0$, same alpha slices; max(df_x) 3.34e-02, certificate FAIL: max atom 2.10e+00 mHa, max dAE 3.29e+00 kcal/mol (parent scan)
 deep0_rung35ms_mgga_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), cusp descriptors, rung-3.5 multishell occupancy, meta-GGA (SCAN parent)

Fig. 3.4 The exchange differences (left) and the correlation differences at $r_s = 2$ (right) of every certified GGA clone on one pair of axes. The four Glorot-start exchange clones agree with PBE to 1×10^{-3} below $s = 1.5$ and to 2.5×10^{-3} below $s = 2.5$ (deep_2x8 +0.0024 at $s = 2$) and fan out beyond it (deep_attn_3x16 to +0.003 at $s = 3.9$, deep_attn_2x8 to +0.003 at $s = 3.3$, deep_2x8 to -0.0049 at $s = 3.6$; at $s = 6$ deep_3x16 -0.007, deep_attn_3x16 -0.006, deep_2x8 -0.0027 and deep_attn_2x8 -0.015, the largest miss on the slice); their correlation clones stay within 3×10^{-3} below $s = 3$, deep_attn_3x16 drifting to $+5 \times 10^{-3}$ by $s = 6$ and the 2x8 pair to $+1.2 \times 10^{-3}$ and $+2.3 \times 10^{-3}$. The three zero-init clones stay within 4.2×10^{-3} everywhere on both slices (deep0_attn_3x16 exchange +0.0034 at $s = 4.2$; deep0_cusp_3x16 correlation +0.0041 at $s = 3.2$).

Pre-trained networks against the PBE parent: differences, all architectures



run run_20260908T153908Z; pretrained networks loaded through the run's resolved configuration (parent anchor class); parent curves: parents_pbe_1x (parent pbe-1e (mbxc constants))
 deep0_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA); deep0_attn_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), 4 attention heads; deep0_cusp_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), cusp descriptors; deep_3x16: 3 x 16, Glorot initialization; deep_attn_3x16: 3 x 16, Glorot initialization, 4 attention heads; deep_2x8: 2 x 8, Glorot initialization; deep_attn_2x8: 2 x 8, Glorot initialization, 2 attention heads

The `deep0_attn_3x16` and `deep0_cusp_3x16` clones are drawn in the same directories (`pretrain_fx_fc_<arch>.png`).

3.6 Certificate results

Read from `pretrain/<arch>/fidelity_certificate.json` and `pretrain_metadata.json`; the pointwise losses are the saved network's, in units of the squared enhancement-factor residual.

Every GGA clone passes on all three criteria; the two with a species above 1 kcal/mol (`deep_2x8`: C3H8 and C4H6; `deep0_cusp_3x16`: AlCl3) pass on the mean and under the 2.0 kcal/mol backstop. The attention variants are the tightest clones by every measure (`deep_attn_3x16`: max atom 0.058 mHa, mean 0.031 kcal/mol; `deep0_attn_3x16`: 0.123 mHa, 0.042 kcal/mol), at the price of the longest fits (about 28 h against 3.8 h for their plain twins, `duration_seconds`); the 2x8 pair fits in 1.6 h (`deep_2x8`) and 3.6 h (`deep_attn_2x8`), and `deep_2x8` is the loosest GGA clone (mean 0.343 kcal/mol, max 1.421).

The four meta-GGA clones whose 20000-step fits completed all fail the certificate, every one on the free atoms (the largest atom error 2.10 to 4.11 mHa against the 1.0 mHa tolerance: `deep0_mgga_3x16` 4.11 on N, `deep0_cusp_mgga_3x16` 3.13 on O, `deep0_rung35_mgga_3x16` 3.00 on O, `deep0_rung35ms_mgga_3x16` 2.10 on Be) and three of the four on the mean atomization error as well (1.47 to 2.06 kcal/mol against 1.0); `deep0_rung35ms_mgga_3x16` is the one clone under the 1.0 kcal/mol mean (0.710) and fails on the atoms and, with AlCl3 at 3.29 kcal/mol, on the 2.0 kcal/mol backstop. Their pointwise residuals (3.7e-6 to 5.9e-6 in exchange, 5.2e-6 to 1.6e-5 in correlation) are one to two orders of magnitude above the GGA clones'. `deep0_cusp_mgga_3x16` misses the most on the atomizations: the mean 2.06 kcal/mol, the largest 7.26 (C4H6; C3H8 7.19, O2 4.77, SiCH6 3.52, CH4 3.47), 14 of 22 molecules above 1 kcal/mol, every one of its eight largest errors of the same sign; `deep0_mgga_3x16` (AlCl3 5.60, O2 4.34, N2 3.61, 11 of 22 above 1) and `deep0_rung35_mgga_3x16` (O2 6.30, AlCl3 4.84, Na2 3.71, 11 of 22) carry errors of both signs among their largest. Fig. 3.3 of the full report shows where the clones miss SCAN: same-sign offsets over wide ranges of s (the $\alpha = 0$ exchange slice of `deep0_cusp_mgga_3x16` 0.015 to 0.027 above SCAN for $s > 2$, the rung-3.5 clones' correlation slices off by 0.06 to 0.39 below $s = 1$), which do not cancel between a molecule and its atoms; `deep0_mgga_3x16` sits within 0.01 of SCAN on every slice and fails on the atoms all the same, so the slices are not where its miss lives. The v6 meta-GGA certificates, all PASS, were fitted in the anchored model class (the network learns a correction on top of SCAN); these are the first unanchored meta-GGA clones the two-tier gate has judged, and the rung rather than one architecture is what fails. The fifth fit, `deep0_mgga_attn_3x16`, was cancelled part-way (its directory holds the exchange network and the validation-best pair, no correlation network and no curves). The meta-GGA training array is held; the follow-up is in Sec. 6.

Run	Architecture	Parent	Steps	Pointwise loss X / C	max (mHa)	atom	$ \Delta E_{xc} $	mean (kcal/mol)	$ \Delta AE $	max (kcal/mol)	$ \Delta AE $	Species kcal/mol	over	1	Verdict
size	deep_2x8	PBE	20000	5.41e-08 / 3.29e-06	0.369			0.343		1.421		C3H8, C4H6			PASS
size	deep_attn_2x8	PBE	20000	7.62e-08 / 3.83e-07	0.201			0.131		0.573		none			PASS
size	deep_3x16	PBE	20000	6.40e-08 / 7.58e-07	0.210			0.162		0.519		none			PASS
size	deep_attn_3x16	PBE	20000	1.51e-08 / 6.70e-08	0.058			0.031		0.115		none			PASS
families	deep0_3x16	PBE	20000	2.77e-08 / 3.37e-06	0.439			0.114		0.450		none			PASS
families	deep0_attn_3x16	PBE	20000	1.48e-08 / 9.86e-08	0.123			0.042		0.149		none			PASS
families	deep0_cusp_3x16	PBE	20000	9.46e-08 / 1.10e-06	0.439			0.188		1.071		AlCl3			PASS
meta-GGA	deep0_mgga_3x16	SCAN	20000	5.61e-06 / 1.60e-05	4.106			1.473		5.605		AlCl3, C3H8, C4H6, CH4, Cl2, H2O, HCN, HCl, LiF, N2, O2			FAIL
meta-GGA	deep0_cusp_mgga_3x16	SCAN	20000	3.70e-06 / 5.18e-06	3.132			2.062		7.257		AlCl3, C2H2, C3H8, C4H6, CH2, CH4, CO, CO2, F2, H2O, HCN, O2, PH3, SiCH6			FAIL
meta-GGA	deep0_rung35_mgga_3x16	SCAN	20000	5.88e-06 / 1.05e-05	3.001			1.780		6.303		AlCl3, C2H2, C3H8, C4H6, CO2, Cl2, F2, H2O, HCN, Na2, O2			FAIL
meta-GGA	deep0_rung35ms_mgga_3x16	SCAN	20000	3.82e-06 / 1.20e-05	2.096			0.710		3.294		AlCl3, CO2, H2O, O2			FAIL
meta-GGA	deep0_mgga_attn_3x16	SCAN	cancelled	-	-			-		-		-			none

4. Optimization, in-sample

Sources: `xcquinox/pipeline/losses.py` (L5_gradnorm_vxc_step7, the channel terms), `xcquinox/pipeline/train.py` (the per-molecule loop, the optimizer), `xcquinox/pipeline/oneshot.py` (the SCF tail window), `xcquinox/pipeline/solver_manual.py` (the differentiable SCF), the hyperparams: and solvers: blocks of the resolved configurations, and per cell `train_metadata.json`, `losses.npy`, `aux_log.pkl`, `eval/per_molecule.json`.

4.1 Training protocol

Each cell trains the certified clone of its architecture on a subset of the 26-point DFS pool. The per-molecule update scheme takes one optimizer step per training group (a reaction, or a molecule with its density and potential references) per epoch, so an epoch is one pass over the groups. The loss of a group is the fixed weighted sum

$$\mathcal{L} = 1 \cdot \mathcal{L}_{AE} + 1 \cdot \mathcal{L}_{\text{rxn}} + 1 \cdot \mathcal{L}_{IP} + 1 \cdot \mathcal{L}_{v_{xc}} + 20 \cdot \mathcal{L}_{\rho},$$

the Letter’s $\lambda_{RE} = 1$, $\lambda_n = 20$ structure (effective_channel_weights; the GradNorm balancer is configured but dormant under this scheme, and the vxc_weight 0.01 and density_weight 0.1 pre-scales are forced to 1). The channels, with E^{NN} the self-consistent total energy of the 3-cycle SCF (Sec. 4.2) and $t = 0, \dots, N - 1$ the cycle index:

- Reaction energies (`ae_as_reactions: true` routes the W4-11 atomizations here as molecule-to-atoms reactions with the network’s own atoms, beside the BH76 barriers):

$$\mathcal{L}_{\text{rxn}} = \frac{1}{|T|} \sum_{t \in T} w_t^2 r_t^2, \quad r_t = \sum_i c_i E_{i,t}^{\text{NN}} - E^{\text{ref}}, \quad w_t = \left(\frac{t}{N-1} \right)^2,$$

T the kept SCF tail (the last $\min(N, 10)$ cycles; at $N = 3$ all three, $w_t^2 = 0, 1/16, 1$), the DFS convergence-tail scoring (DFS Eqs. 15 and 16) that penalizes a non-converging SCF instead of one arbitrary final cycle. Absolute Ha² (`loss_metric: absolute`). References: W4-11 atomization energies and the W2-F12 BH76 barrier heights of GMTKN55.

- Atom anchors (the residue of the AE channel once the atomizations are reactions): the free atoms of the subset are held near their exact non-relativistic energies E_Z^{anchor} (Chakravorty et al., PRA 47, 3649 (1993)), $\mathcal{L}_{AE} = 0.01 \cdot \text{mean}_Z (E_Z^{\text{NN}} - E_Z^{\text{anchor}})^2 / (E_Z^{\text{anchor}})^2$.
- Ionization potentials, $\mathcal{L}_{IP} = \text{mean} (E_+^{\text{NN}} - E_0^{\text{NN}} - IP^{\text{ref}})^2$, present in the form but empty on every v7 subset (no IP13 point is selected).
- Exchange-correlation potential, on the reference density (frozen features): $\mathcal{L}_{v_{xc}} = \|V_{xc}^{\text{NN}} - V_{xc}^{\text{ref}}\|_F^2 / n_{\text{AO}}^2$, summed over spin channels for open shells.
- Density, the final mixed density of the SCF against the CCSD reference density on the same grid, per electron squared (the Letter’s Eq. 17 normalization; `density_per_electron: true`):

$$\mathcal{L}_{\rho} = \text{mean}_{\text{mol}} \frac{\sum_g w_g (n_g^{\text{NN}} - n_g^{\text{ref}})^2}{N_e^2}, \quad N_e = \sum_g w_g n_g^{\text{ref}}.$$

Optimizer: AdamW, learning rate 10^{-3} held for the first half of the run and decayed linearly to 10^{-5} over the second (`lr_decay_start: 0.5`), weight decay 10^{-4} , gradient clip 1.0. 200 epochs (`n_steps`), a validation reaction-energy MAE on the 35-reaction validation slice every 25 epochs, early stop after 5 checks without a 0.1 kcal/mol improvement; three checkpoints are kept (final, lowest training loss, lowest validation MAE), and the figures of Secs. 4 and 5 read the validation-best one. Over the 61 cells of the size and families runs the runs took 150 to 200 epochs (early-stopped in 34) and 1.2 to 34.9 h of wall each (`train_metadata.json`).

The two arms of the density program train the same deep_3x16 clone on the same subsets under 25 SCF cycles (`solver: full_25`) for 100 configured epochs with a validation check every 10. The 25-cycle arm keeps everything else of the protocol above (the weights 1, 1, 1, 1, 20, the

learning-rate schedule, the early stop); its three cells ran 70 to 90 epochs, every one early-stopped, in 21.6 to 56.5 h. The dpyscf-parity arm trains under the Letter’s recipe as recorded in its `train_metadata.json`: channel weights 1 (anchors), 0.01 (reactions), 0.01 (ionization), 0 (potential) and 20 (density), Adam with the plateau schedule from 10^{-4} (factor 0.1 after 10 checks without improvement, never triggered in either cell, so the rate stayed at 10^{-4} throughout) and weight decay 10^{-6} ; its two cells ran the full 100 epochs, not early-stopped, in 25.1 and 47.7 h. The parity cells therefore carry no potential term in their training loss, and their reaction term is weighted one hundredth of the size run’s; the tables of Sec. 4.6 of the full report print the weighted channels as trained.

4.2 The differentiable SCF

The energies and densities above are self-consistent: the manual backend runs a fixed $N = 3$ cycles from the converged PBE density (solver: `full_3`), diagonalizing the Kohn-Sham matrix built from the network’s V_{xc} each cycle, with the DFS decaying linear mixing $D_{t+1} = (1 - a_t)D_t + a_t D_t^{\text{out}}$, $a_t = 0.3^t + 0.3$, a freeze once $|\Delta E| < 10^{-6}$ Ha, and gradients through every cycle. The density scored by $\mathcal{L}_\rho$ and by the density metrics is the last mixed density; three cycles do not converge the SCF for a functional far from a fixed point, which is the subject of the two arms submitted on 2026-09-08 (25 cycles; dpyscf parity) and of the converged-SCF re-evaluation, of which one cell-pass exists (Sec. 6). The density record of 2026-09-07 (the HISTORY entry `f0c7b7865`) records the 21-item parity table against dpyscf.

4.3 Training losses

Fig. 4.1 Per-channel training losses, one figure per architecture (eight figures: the six GGA architectures and the two tagged arms), eight panels. Every loss panel is the weighted per-epoch mean over the epoch’s group updates read from the cell’s `aux_log.pkl`, on a log axis against the epoch, one curve per subset size (the color bar: dark one molecule, yellow twenty-six). Panel 1 is the total loss, the quantity the optimizer steps; panels 2 to 6 are the five channels of Sec. 4.1 at their trained weights, and they add up to panel 1 at every epoch. Panel 7 is the validation reaction-energy MAE at each check (every 25 epochs in the size and families runs, every 10 in the arms), the star marking the validation-best epoch whose checkpoint every later figure reads. The footer states the weights and their source; the parity arm’s figure adds the learning-rate panel of its plateau schedule and draws its zero-weighted potential channel dotted as the unweighted residual.

deep_3x16 (the first figure; the per-epoch means from the suite’s collector): the total falls from $4.7\text{e-}5$ (one molecule) and $1.6\text{e-}4$ to $3.2\text{e-}4$ (the rest) at the first epoch to a plateau reached by epoch 25 in every cell, the plateau ordered by subset size from $3.7\text{e-}5$ (one molecule) to $1.0\text{e-}4$ (eighteen), because the potential and density channels grow with the number of supervised molecules. The reaction channel is the one that falls by orders of magnitude: from $1.1\text{e-}4$ to $2.5\text{e-}4$ at the first epoch ($1.1\text{e-}6$ in the one-molecule cell) to $4.6\text{e-}9$ (one molecule), $2.1\text{e-}6$ to $8.0\text{e-}6$ (two to six) and $1.6\text{e-}5$ to $3.2\text{e-}5$ (seven to twenty-six) at the end. The anchors channel is $3\text{e-}9$ to $7\text{e-}7$ throughout and drifts upward in the large-subset cells (the free atoms are released as the molecules pull the functional). The ionization channel (present from three molecules, where Li^+ enters the subset) peaks at $4.6\text{e-}6$ to $1.7\text{e-}5$ within the first 40 epochs and falls to $1.2\text{e-}6$ to $3.8\text{e-}6$. The potential channel is the one that rises: from $1.0\text{e-}5$ to $1.4\text{e-}5$ at the start to $1.3\text{e-}5$ to $1.7\text{e-}5$ (two to six molecules) and from $1.7\text{e-}5$ to $3.5\text{e-}5$ to $2.4\text{e-}5$ to $4.3\text{e-}5$ (seven to twenty-six) at the end, the rise steepest in the first 25 epochs, while the one-molecule cell falls from $2.8\text{e-}5$ to $2.7\text{e-}5$. The density channel drops from $1.8\text{e-}5$ to $6.9\text{e-}5$ at the first epoch to $9.6\text{e-}6$ (one molecule) and $1.8\text{e-}5$ to $2.6\text{e-}5$ (the rest) at the end, most of the fall inside the first 10 epochs in the cells from four molecules upward (the two- and three-molecule cells first rise to $3.3\text{e-}5$ and $2.7\text{e-}5$ by epoch 10 and then fall). The validation panel separates the cells into two families: the two-to-six-molecule cells start at 12 to 19 kcal/mol and rise to 15 to 33 by the last check (the star at epoch 25 for two, three, five and six; at 200 for four, whose MAE falls back from 20 to 14.6), while the one-molecule cell and the cells from seven upward hold 6.9 to 10.4 kcal/mol across every check.

deep_attn_3x16 (the second figure): the same shapes with a reaction channel that falls further ($5.1\text{e-}7$ to $2.1\text{e-}6$ at two to six molecules, $8.2\text{e-}7$ at seven), a density channel that ends lower (6.0e-

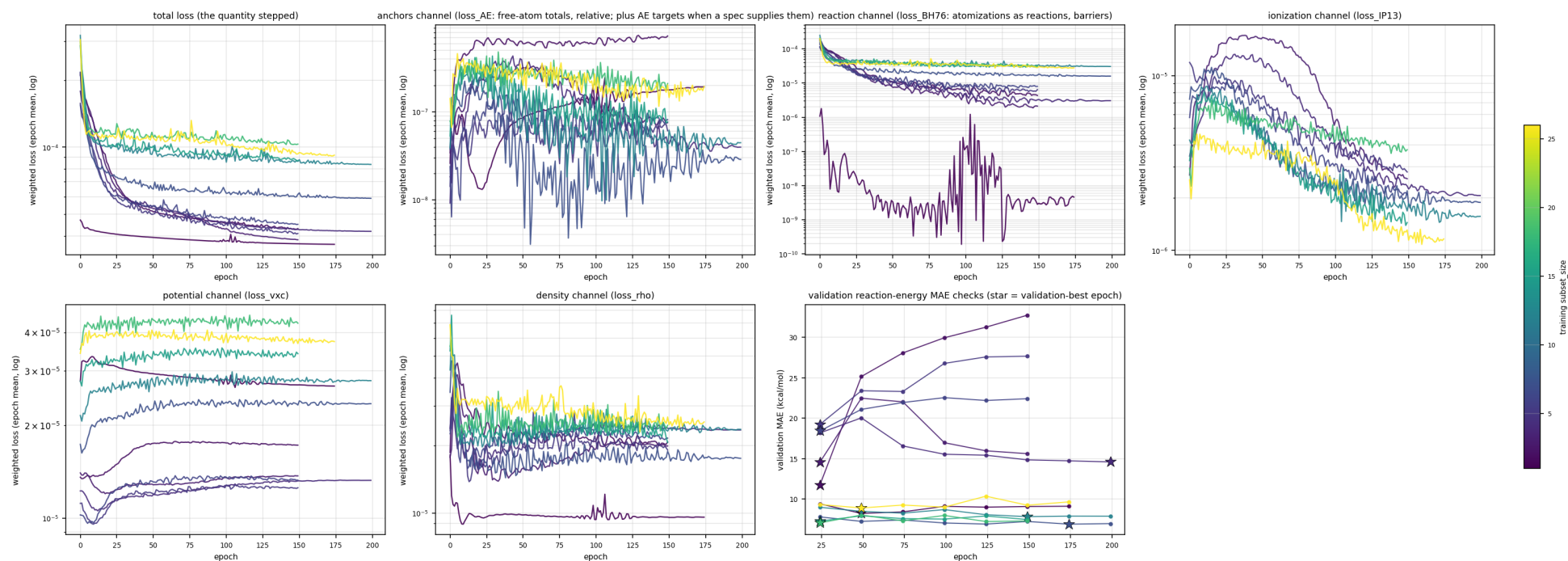
6 to $1.3\text{e-}5$ up to seven molecules, $1.7\text{e-}5$ to $2.1\text{e-}5$ from twelve), and a validation panel where the two-to-six-molecule cells fall rather than rise: from 22 to 37 kcal/mol at the first check to 7.2 to 13.4 at the last in four of the five (the three-molecule cell reaches 13.3 at epoch 75 and climbs back to 21.2), with the validation-best epochs at 75 to 200. `deep0_3x16` (the third figure) reproduces the `deep_3x16` picture, validation included (two molecules: 11.2 to 32.4 kcal/mol; seven to eighteen: 6.2 to 9.2).

`deep_2x8` (the fourth figure): the same plateau structure, with the reaction channel at $1.1\text{e-}4$ to $2.6\text{e-}4$ at the first epoch and $4.8\text{e-}6$ to $9.0\text{e-}6$ (two to six molecules) and $1.3\text{e-}5$ to $3.3\text{e-}5$ (seven to twenty-six) at the end, the one-molecule cell's reaction term falling from $6.4\text{e-}7$ to $1.6\text{e-}8$; the potential channel rises in every cell from two molecules upward ($1.0\text{e-}5$ to $3.6\text{e-}5$ at the start, $1.2\text{e-}5$ to $4.1\text{e-}5$ at the end) and falls in the one-molecule cell ($2.8\text{e-}5$ to $2.4\text{e-}5$); the validation panel repeats the `deep_3x16` split, the two-to-six-molecule cells rising from 7.8 to 9.4 kcal/mol at the first check to 10.8 to 26.0 by the last (validation-best at epoch 25 in all five), the one-molecule cell and the cells from seven upward at 7.1 to 10.2 throughout. `deep_attn_2x8` (the fifth figure): the anchors and reaction channels oscillate by an order of magnitude from epoch to epoch in every cell (the anchors term ranges over three orders of magnitude within one cell, $4\text{e-}10$ to $1.4\text{e-}7$ at three molecules), the reaction channel of the one-molecule cell falling to $7\text{e-}10$; the potential channel of the cells from seven molecules upward rises slowly ($1.7\text{e-}5$ to $3.8\text{e-}5$ at the start, $2.3\text{e-}5$ to $4.3\text{e-}5$ at the end); the validation panel splits like `deep_attn_3x16`'s, the two-to-six-molecule cells starting at 23.6 to 25.2 kcal/mol and either falling (three, four and six molecules: to 18.9, 12.9 and 19.7) or staying (two and five: 29.2, 29.8), the cells from seven upward at 6.8 to 10.4.

The two arms (the sixth and seventh figures, three and two curves): the 25-cycle cells fall from $2.8\text{e-}4$ to $4.1\text{e-}4$ to $7.2\text{e-}5$ (seven molecules) and $1.0\text{e-}4$ (twelve and fifteen) in total, their reaction channels from $2.2\text{e-}4$ to $3.4\text{e-}4$ to $2.6\text{e-}5$ to $4.5\text{e-}5$, their potential channels rising from $1.7\text{e-}5$ to $2.8\text{e-}5$ to $2.4\text{e-}5$ to $3.5\text{e-}5$, their validation MAE 7.7 to 10.2 kcal/mol over checks every 10 epochs with the best at epoch 30, 30 and 40 and the early stop at 80, 70 and 90. The parity cells' total, reaction and density channels fall throughout the 100 epochs under the flat $1\text{e-}4$ learning rate (total $2.5\text{e-}5$ to $1.4\text{e-}5$ and $3.6\text{e-}5$ to $1.6\text{e-}5$; the reaction term, weighted 0.01, from $2.7\text{e-}6$ and $4.3\text{e-}6$ to $1.2\text{e-}6$ and $1.3\text{e-}6$; density $2.2\text{e-}5$ to $1.3\text{e-}5$ and $3.1\text{e-}5$ to $1.5\text{e-}5$), the unweighted potential residual rises ($1.8\text{e-}5$ to $2.3\text{e-}5$, $2.2\text{e-}5$ to $3.0\text{e-}5$), and their validation MAE falls or holds at every check but the last of the twelve-molecule cell, from 11.6 and 10.7 to 7.1 and 7.5 kcal/mol at the last check (the best 7.1 at epoch 100 and 7.4 at epoch 80).

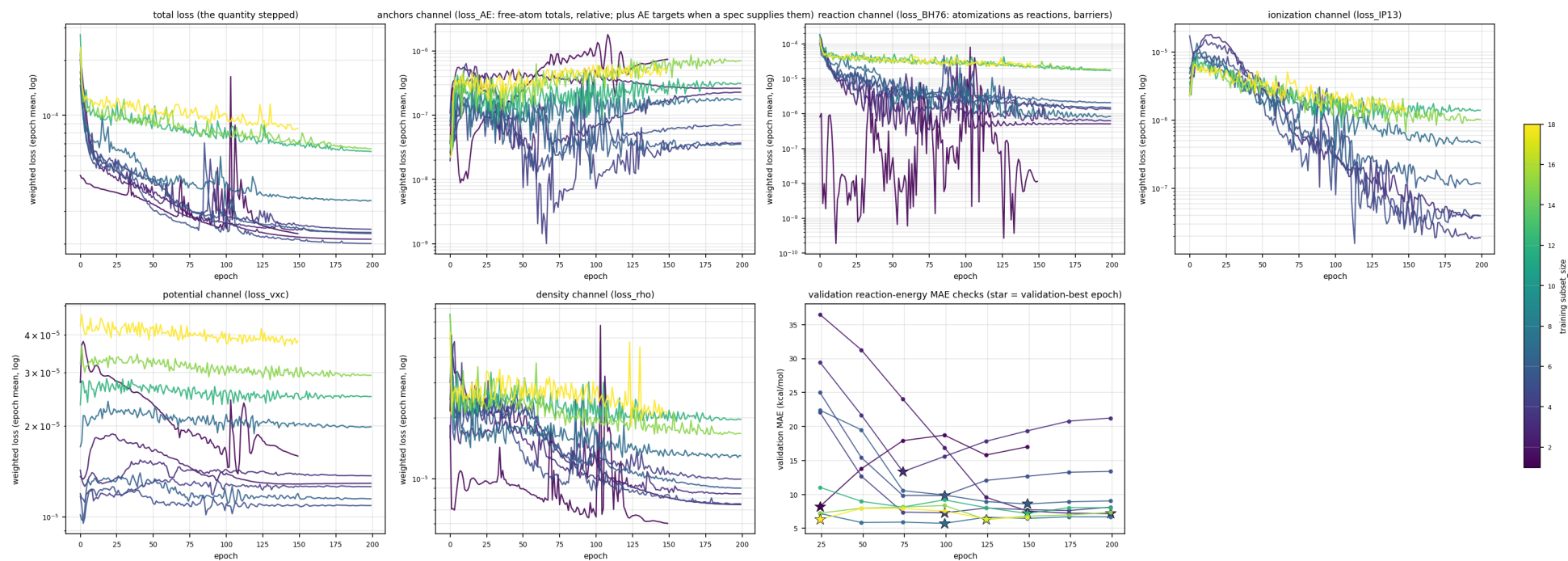
The first and last five-epoch means of every channel and every validation check are in the tables of Sec. 4.6 of the full report.

Per-channel training losses -- deep_3x16 (weighted epoch means; panels sum to the total) · run_20260908T153908Z · val-best



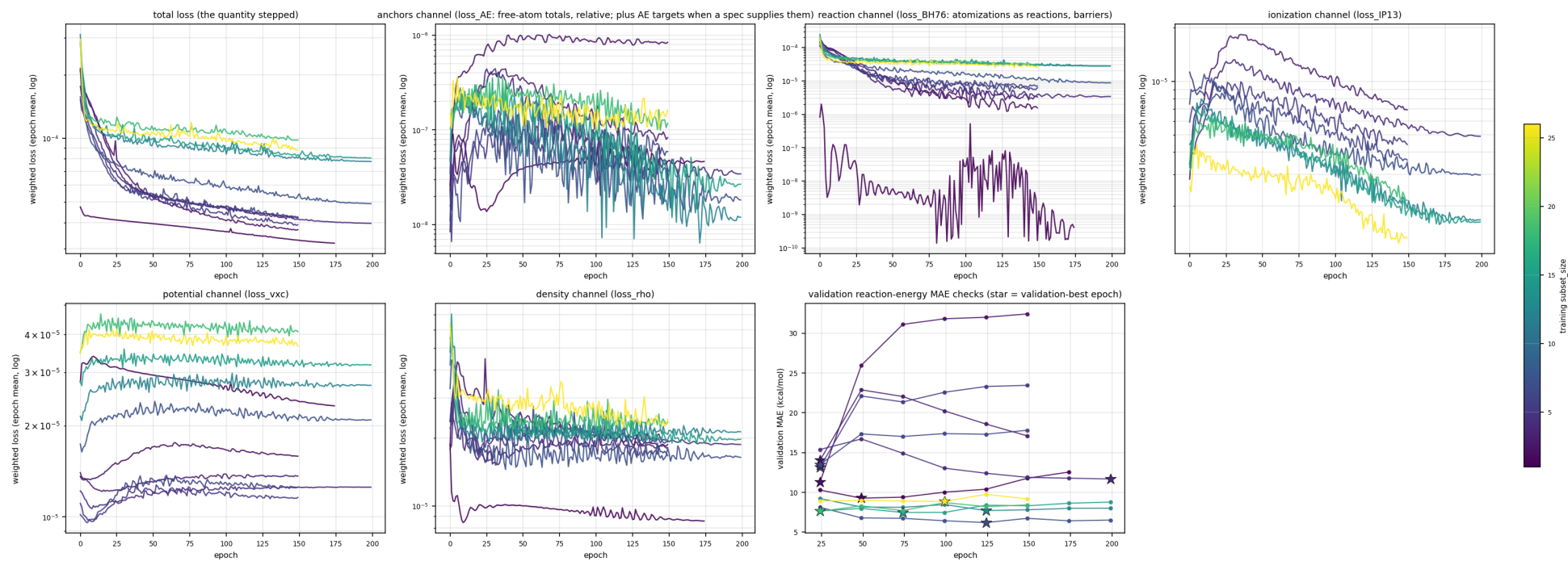
Held-out reactions: 8/29 archs (deep_2x8, deep_attn_2x8, deep_3x16, deep_3x16 (25 cycles), deep_3x16 (dpysc parity), deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep_attn_3x16 (weighted epoch means; panels sum to the total) · run_20260908T153908Z · val-best



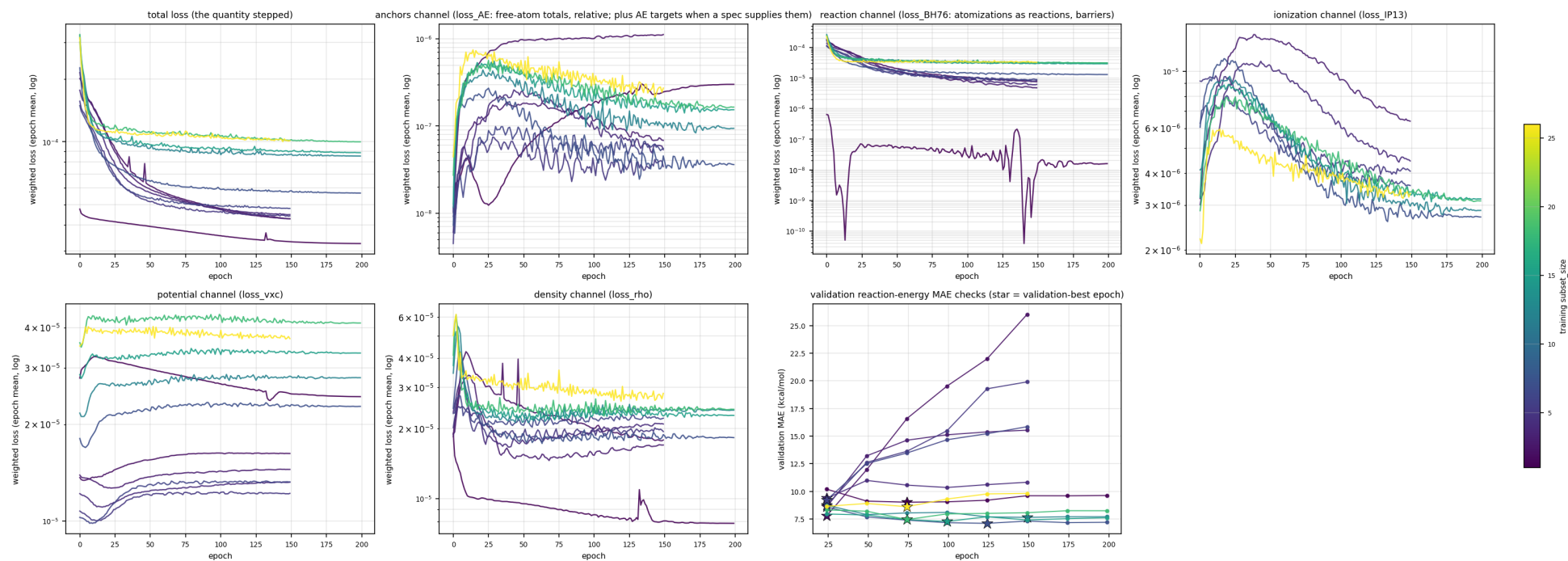
Held-out reactions: 8/29 archs (deep_2x8, deep_attn_2x8, deep_3x16, deep_3x16 [25 cycles], deep_3x16 [dpysc parity], deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep0_3x16 (weighted epoch means; panels sum to the total) · run_20260908T153908Z · val-best



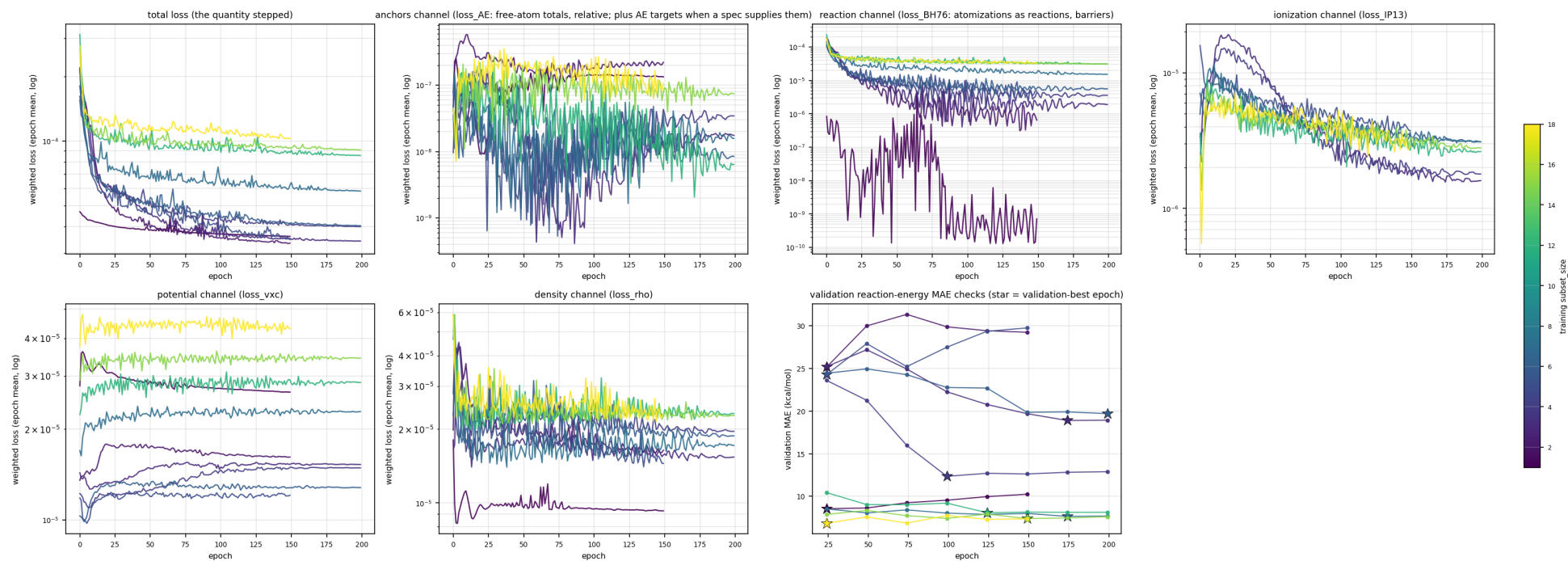
Held-out reactions: 8/29 archs (deep_2x8, deep_attn_2x8, deep_3x16, deep_3x16 [25 cycles], deep_3x16 [dpyscf parity], deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep_2x8 (weighted epoch means; panels sum to the total) · run_20260908T153908Z · val-best



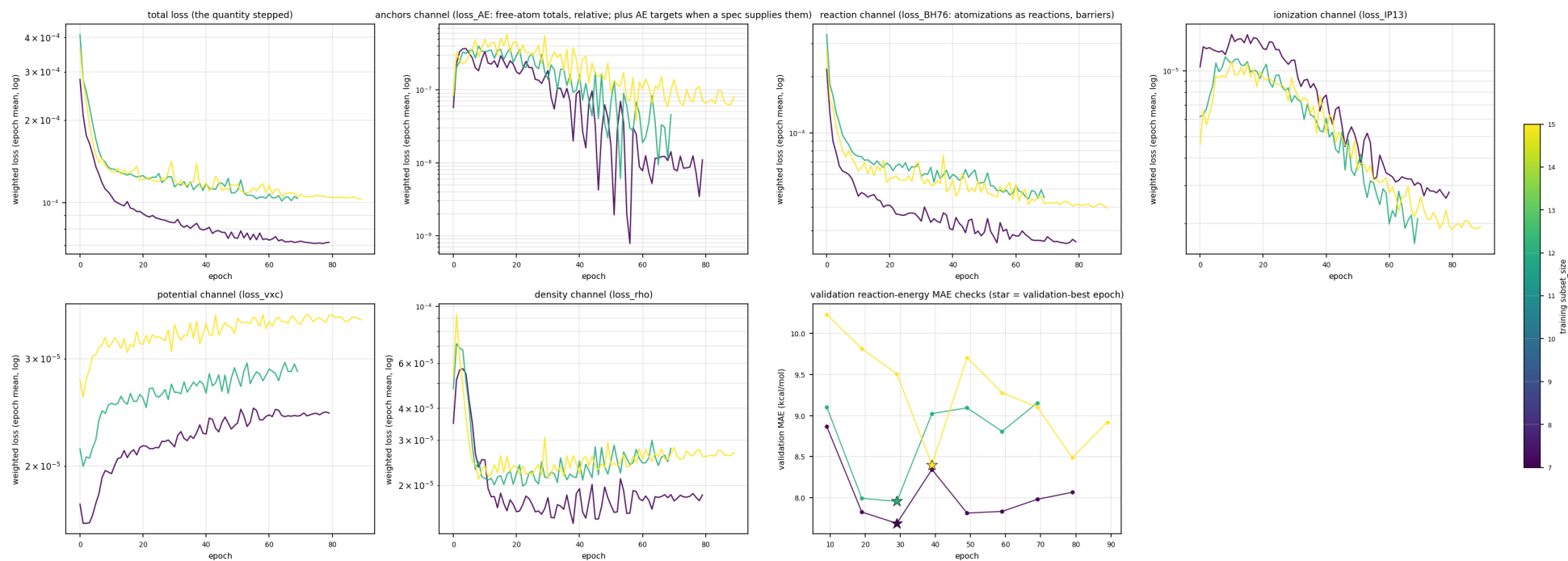
Held-out reactions: 8/29 archs (deep_2x8, deep_attn_2x8, deep_3x16, deep_3x16 [25 cycles], deep_3x16 [dpysc parity], deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep_attn_2x8 (weighted epoch means; panels sum to the total) · run_20260908T153908Z · val-best



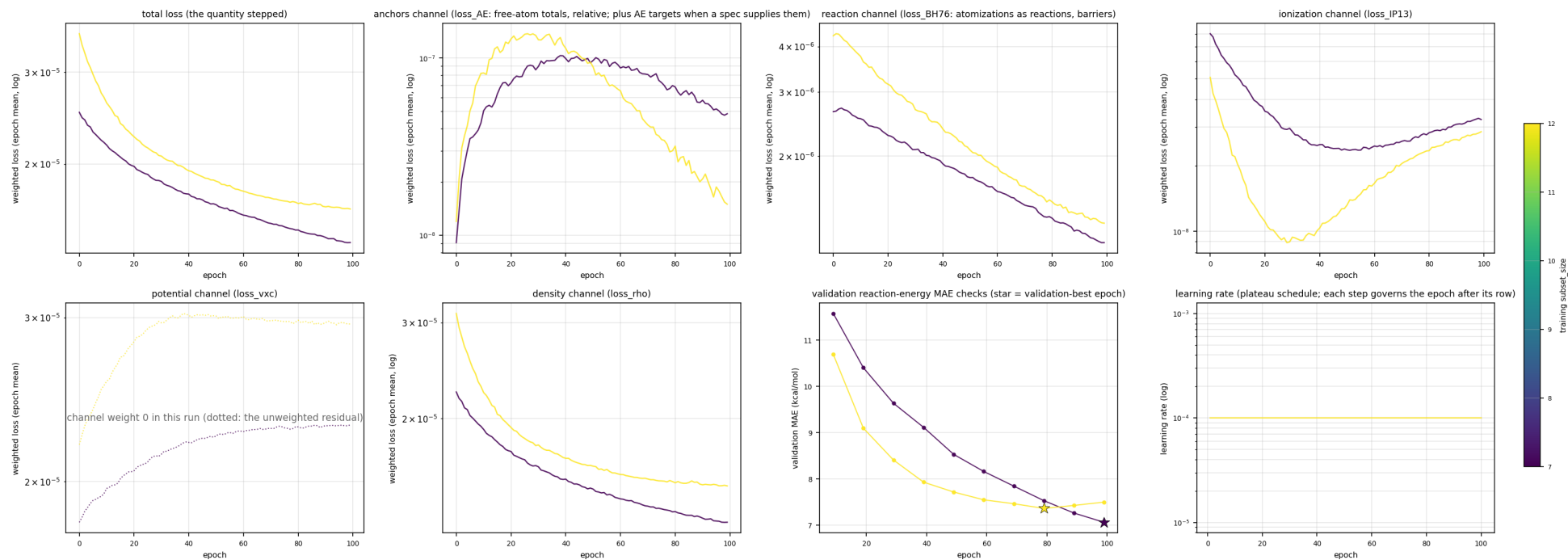
Held-out reactions: 8/29 archs (deep_2x8, deep_attn_2x8, deep_3x16, deep_3x16 [25 cycles], deep_3x16 [dpyscf parity], deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep_3x16 [25 cycles] (weighted epoch means; panels sum to the total) · run_20260908T153908Z · val-best



Held-out reactions: 8/29 archs (deep_2x8, deep_attn_2x8, deep_3x16, deep_3x16 [25 cycles], deep_3x16 [dpysc parity], deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 1, loss_IP13 1, loss_vxc 1, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

Per-channel training losses -- deep_3x16 [dpyscf parity] (weighted epoch means; panels sum to the total) · run_20260908T153908Z · val-best



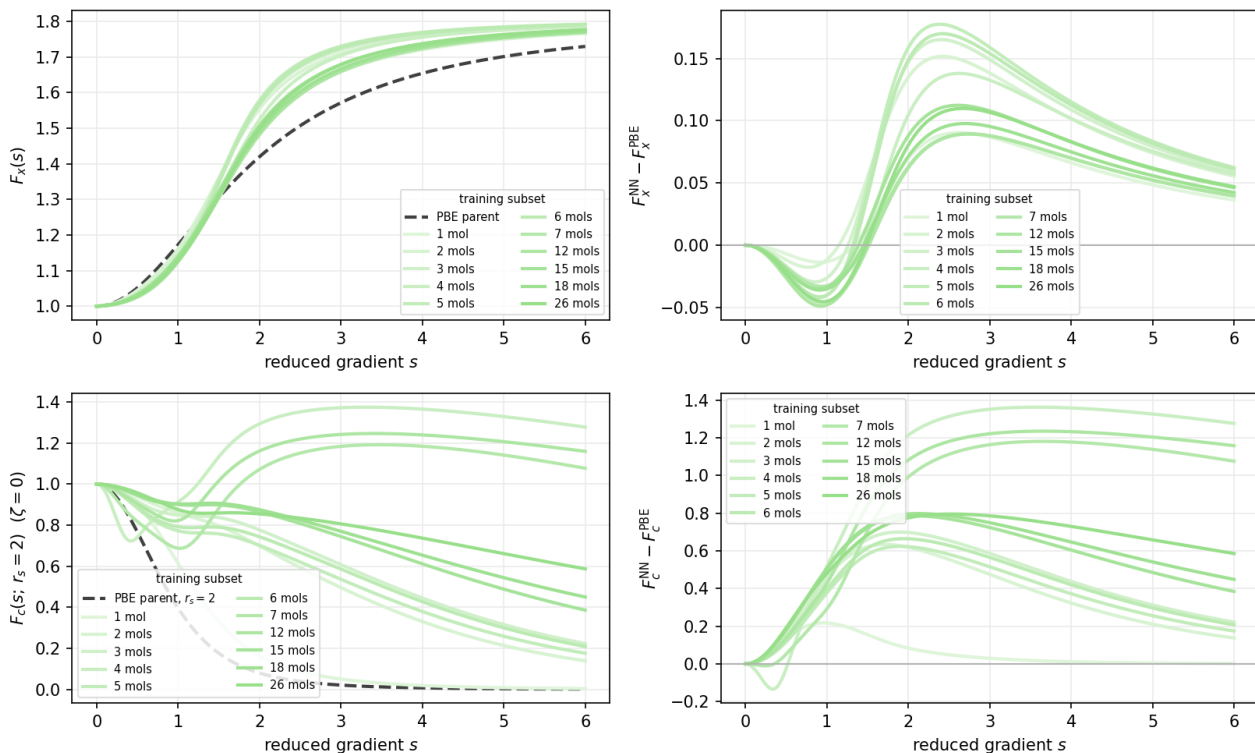
Held-out reactions: 8/29 archs (deep_2x8, deep_attn_2x8, deep_3x16, deep_3x16 [25 cycles], deep_3x16 [dpyscf parity], deep_attn_3x16, deep0_3x16, deep0_attn_3x16). NOT TRAINED in this run: deep0_cusp_3x16.
Channel weights as trained: loss_AE 1, loss_BH76 0.01, loss_IP13 0.01, loss_vxc 0, loss_rho 20 (source: train_metadata); every panel is the weighted epoch mean over the epoch's group updates of aux_log.pkl, so the five channel panels add up to the total.

4.4 Trained enhancement factors

Fig. 4.3 The trained networks against PBE, one figure per architecture, four panels, one curve per subset size at the validation-best checkpoint. Top left: $F_x(s)$ at $n = 1$ and zero extra descriptors, PBE dashed. Top right: $F_x^{\text{NN}} - F_x^{\text{PBE}}$ on the same slice. Bottom left: $F_c(s; r_s = 2)$ at $\zeta = 0$, PBE dashed. Bottom right: the correlation difference. The pre-trained curves of Fig. 3.3 are the starting point of every curve here, and the footer gives the largest $|\Delta F_x|$ over the slice.

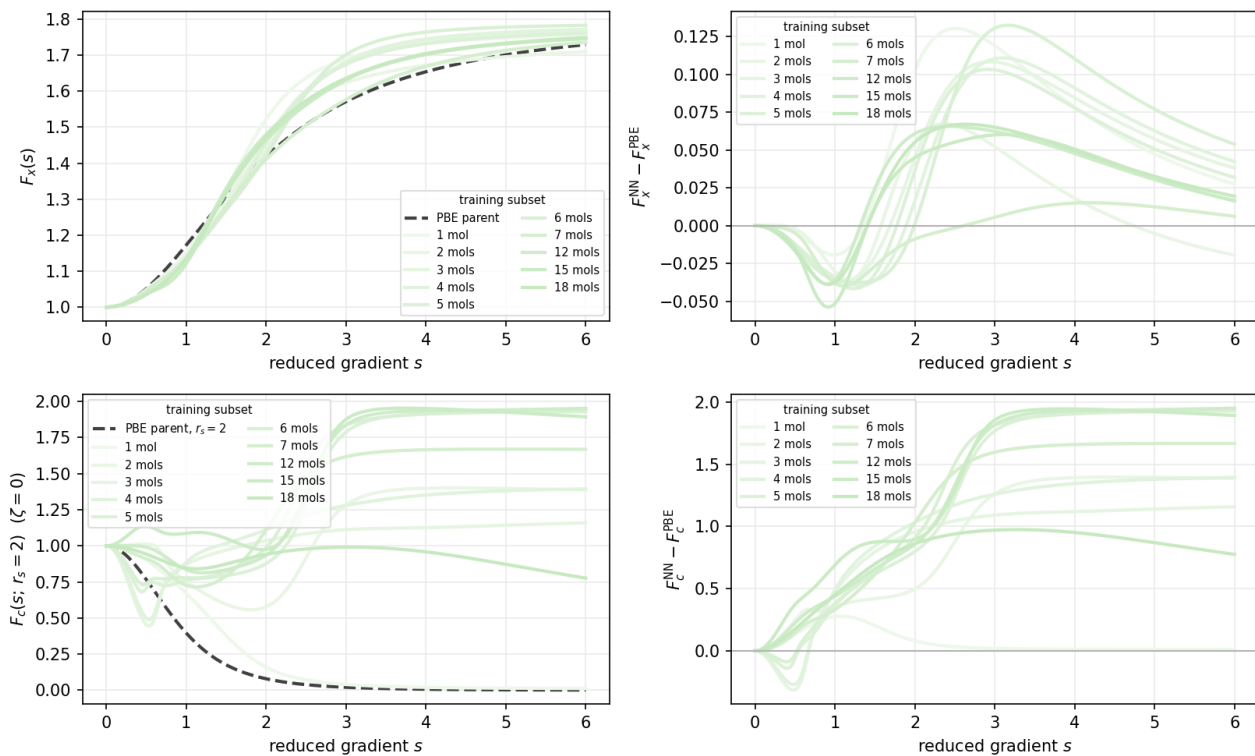
Exchange, all six architectures and both arms (the numbers from `trained_fx_fc_curves.csv`): every trained network lowers F_x below PBE between $s \approx 0.3$ and 1.0 to 1.7 (by 0.002 to 0.06 at the dip, $s = 0.55$ to 1.33) and raises it above PBE beyond, with the largest $|\Delta F_x|$ over the slice at $s = 1.9$ to 3.4 in every cell but `deep_attn_3x16` at seven molecules, whose largest deviation is its dip (0.04 at $s = 1.1$): `deep_3x16` 0.09 to 0.18 (the two-to-six-molecule cells 0.14 to 0.18 , the rest 0.09 to 0.11), `deep_attn_3x16` 0.04 to 0.13 , `deep0_3x16` 0.05 to 0.14 , `deep0_attn_3x16` 0.08 to 0.19 , `deep_2x8` 0.06 to 0.08 (the smallest corrections of the campaign, at $s = 2.2$ to 3.0), `deep_attn_2x8` 0.07 to 0.19 (at $s = 1.9$ to 2.4 , the largest at two molecules), the 25-cycle arm 0.11 to 0.13 and the parity arm 0.11 and 0.13 (against 0.09 to 0.11 for the size group’s `deep_3x16` at the same subsets). At $s = 6$ the difference has decayed to -0.05 to $+0.06$ over the `3x16` cells (below PBE in the two attention cells at two molecules, within $+0.02$ of it in `deep_attn_3x16` from seven molecules upward, $+0.02$ to $+0.06$ in the other `3x16` cells) and to $+0.02$ to $+0.04$ in `deep_2x8`, while `deep_attn_2x8` ends within ± 0.035 of PBE there. The shape is the same at every subset size; the one-molecule cell moves least. Correlation at $r_s = 2$: PBE’s F_c falls from 1 at $s = 0$ to 0.02 at $s = 3$; up to $s = 2$ the trained F_c runs from 0.10 (the parity cells, which stay at PBE) to 1.29 (`deep_3x16` at four molecules) across the cells, and beyond that the cells separate. `deep_3x16`: the one-molecule cell returns to PBE (a difference of 0.2 at $s = 1$ and 0 beyond 3), the four-molecule cell and the cells at seven and twelve molecules rise to 1.1 to 1.3 at $s = 6$, the two-, three-, five- and six-molecule cells decay to 0.14 to 0.22 and the cells at fifteen to twenty-six to 0.4 to 0.6 . `deep0_3x16`: the one-molecule cell as for `deep_3x16`, the four-molecule cell 1.5 at $s = 6$, the fifteen-molecule cell flat at 0.8 from $s = 2$, the seven-, twelve- and twenty-six-molecule cells 1.0 to 1.2 , the others 0.1 to 0.25 . `deep_attn_3x16`: the one-molecule cell returns to PBE, every other cell rises above 1 from $s = 2$ to 3 onward, to 1.2 (three molecules), 1.4 (two and four), 1.7 (seven), 1.9 (five, six, twelve and fifteen) and back to 0.8 at eighteen, at $s = 6$. `deep0_attn_3x16`: the one-molecule cell returns to PBE, the others end between 0.4 (twelve) and 1.8 (five) at $s = 6$. The `2x8` pair: `deep_2x8`’s one-molecule cell returns to PBE, its two-to-six-molecule cells decay to 0.20 to 0.41 at $s = 6$ and its cells from seven upward rise to 1.02 to 1.27 , the mirror image of `deep_3x16`’s small-subset rise; `deep_attn_2x8` ends at 0.05 to 0.08 (two and five molecules), 0.33 (eighteen) and 0.73 to 1.02 (the rest). The arms: the 25-cycle cells end at 0.17 , 0.43 and 0.65 (seven, twelve, fifteen) against 1.08 , 1.16 and 0.39 for the size group’s cells of the same subsets, and the parity cells keep F_c within 0.07 of PBE at $r_s = 2$ over the whole slice (the largest difference 0.067 at $s = 1.0$; 0.075 at $r_s = 5$), the one trained correlation factor of the campaign that stays at its parent. The correlation difference therefore reaches 0.2 to 1.9 at large s depending on the cell. The corrections are therefore not small perturbations of PBE: the density-gradient dependence of both channels is reshaped in the region $s > 1.5$, which is populated by the valence tails and the bonding regions of molecules, and what Sec. 5 measures is how that reshaping transfers.

deep_3x16: trained networks against the PBE parent



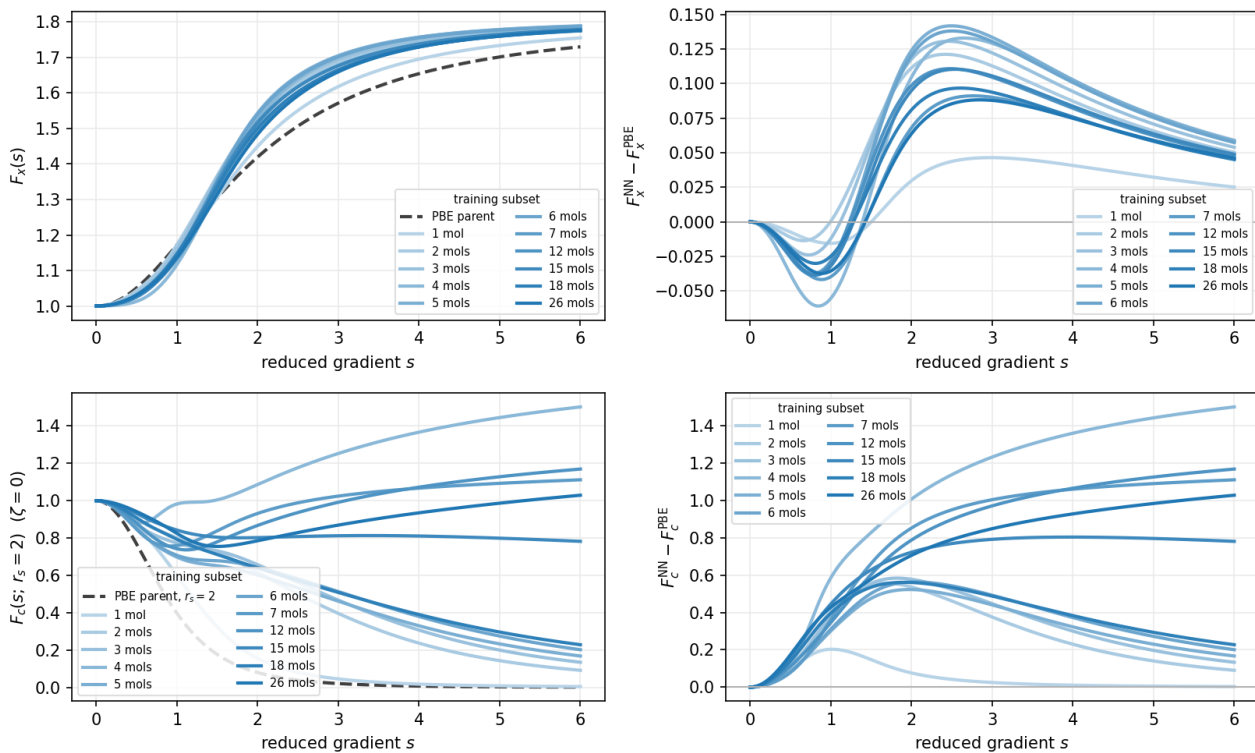
run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 11 completed cell(s); max|dF_x| 1.78e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_3x16: 3 x 16, Glorot initialization

deep_attn_3x16: trained networks against the PBE parent



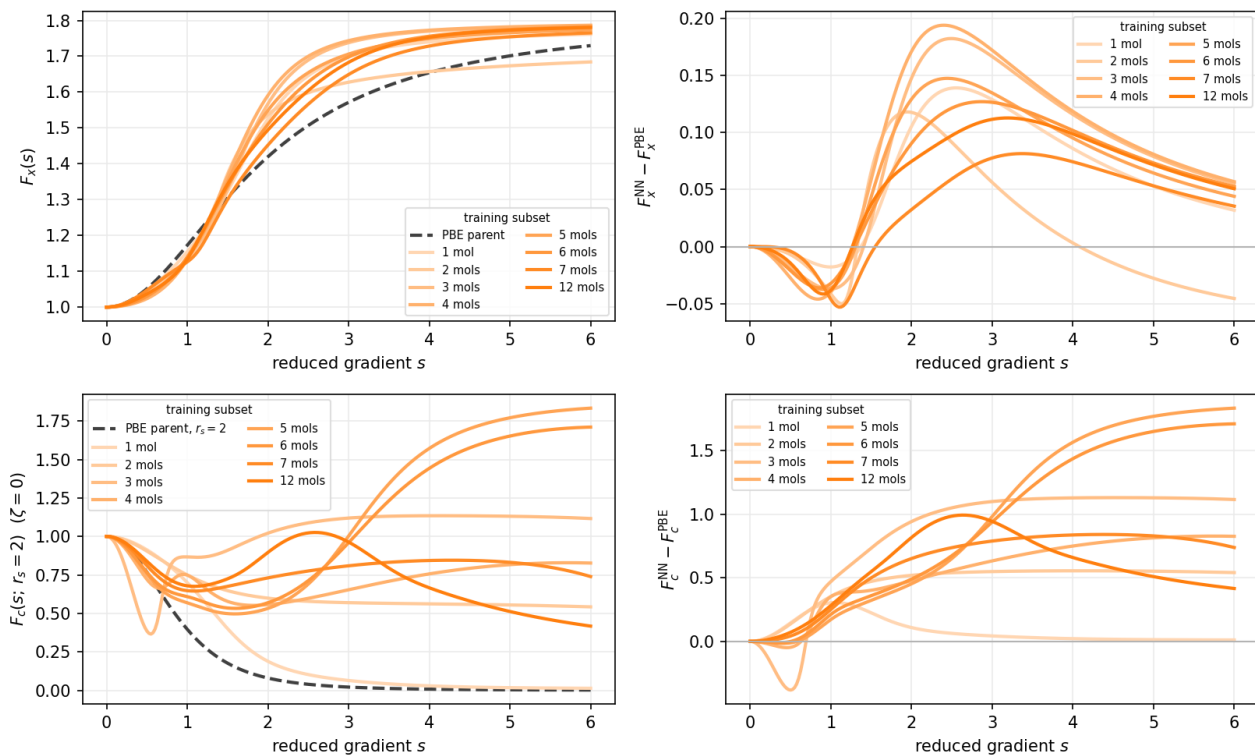
run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 10 completed cell(s); max|dF_x| 1.32e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_attn_3x16: 3 x 16, Glorot initialization, 4 attention heads

deep0_3x16: trained networks against the PBE parent



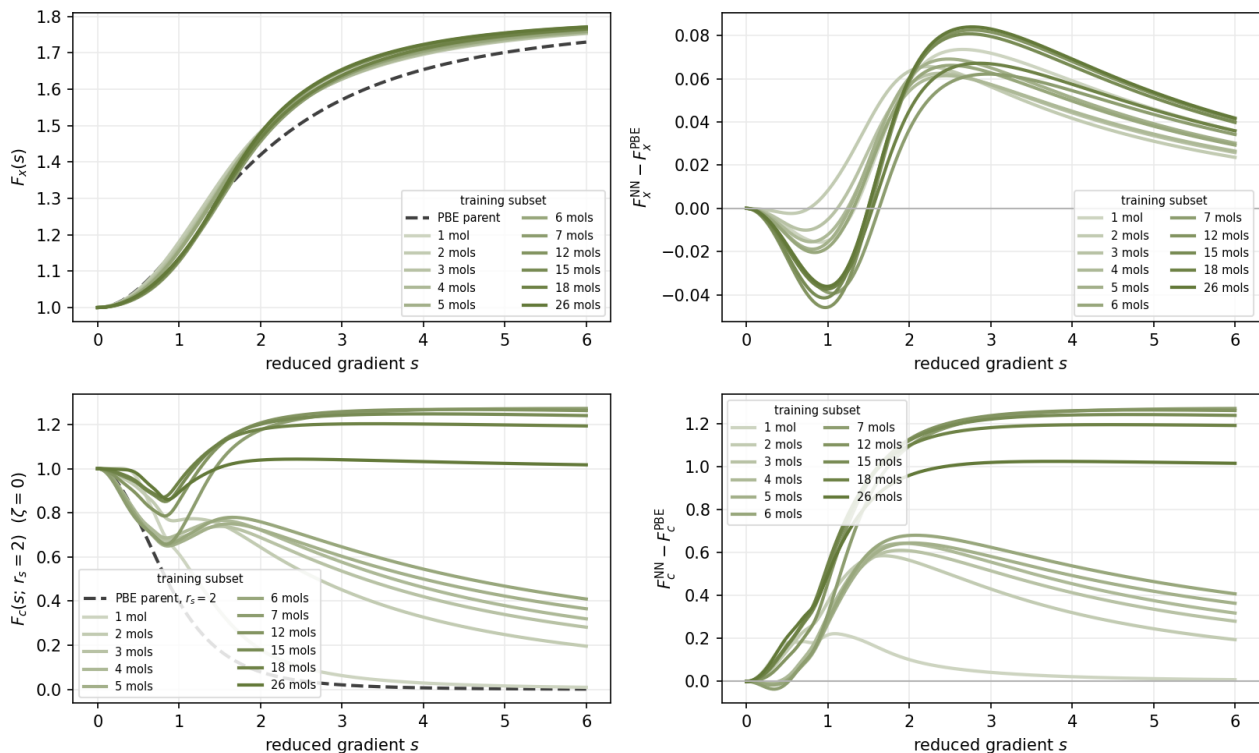
run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 11 completed cell(s); $\max|dF_x|$ 1.42e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep0_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA)

deep0_attn_3x16: trained networks against the PBE parent



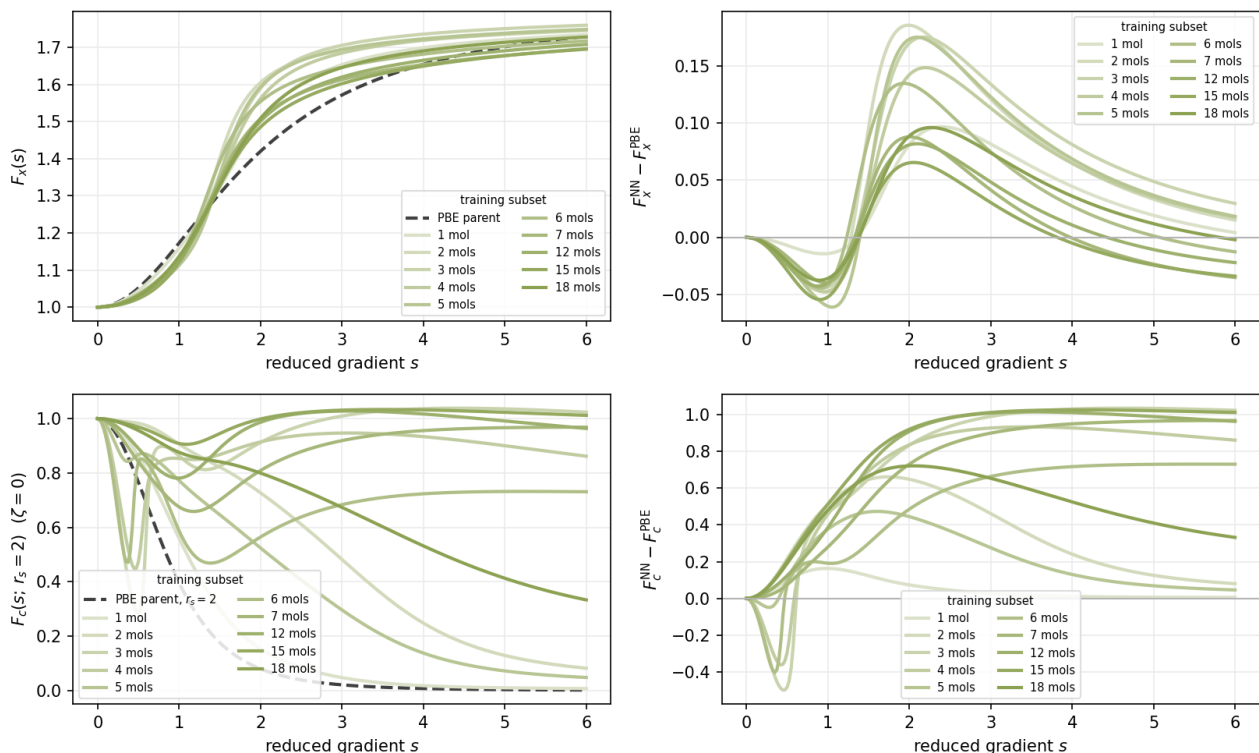
run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at $\rho=1$, zero extra descriptors; F_c at $\zeta=0$, $r_s=2$; 8 completed cell(s); $\max|dF_x|$ 1.94e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep0_attn_3x16: 3 x 16, last layer zeroed (pre-training starts at the LDA), 4 attention heads

deep_2x8: trained networks against the PBE parent



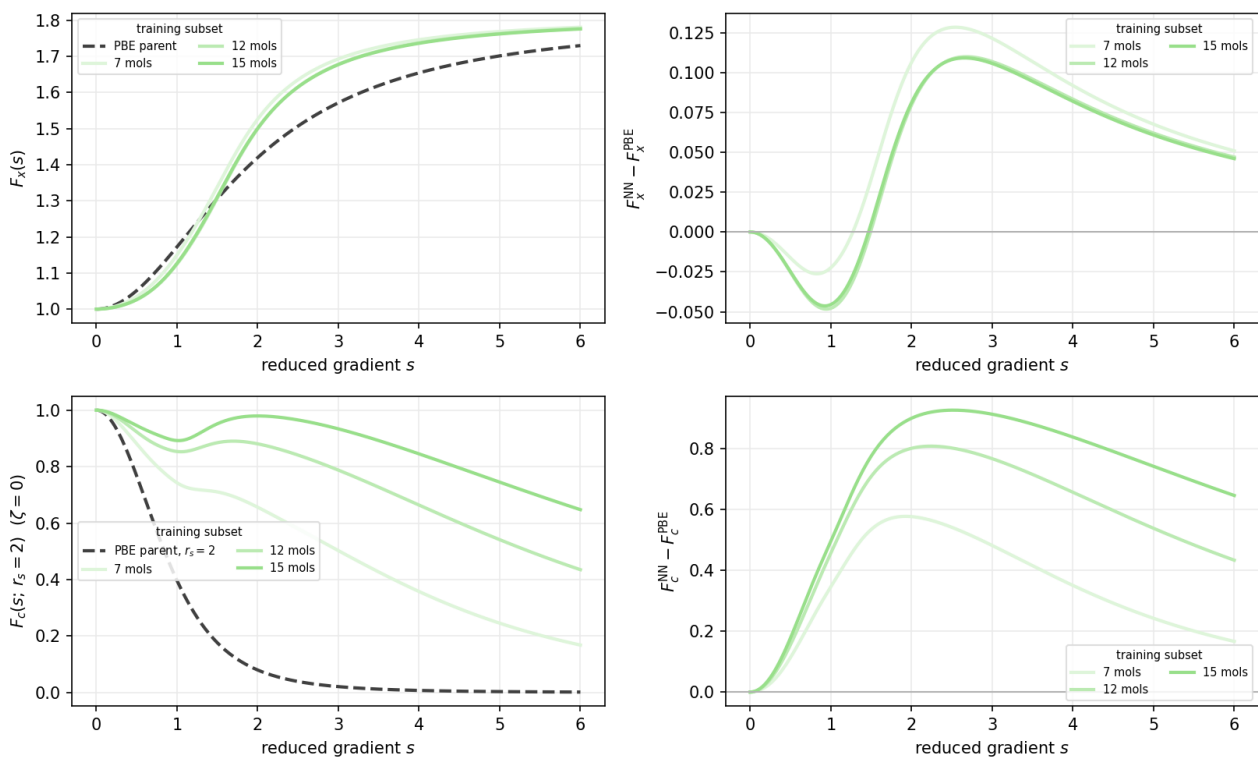
run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_X at $\rho=1$, zero extra descriptors; F_C at $\zeta=0$, $r_s=2$; 11 completed cell(s); max|dF_X| 8.40e-02; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_2x8: 2 x 8, Glorot initialization

deep_attn_2x8: trained networks against the PBE parent



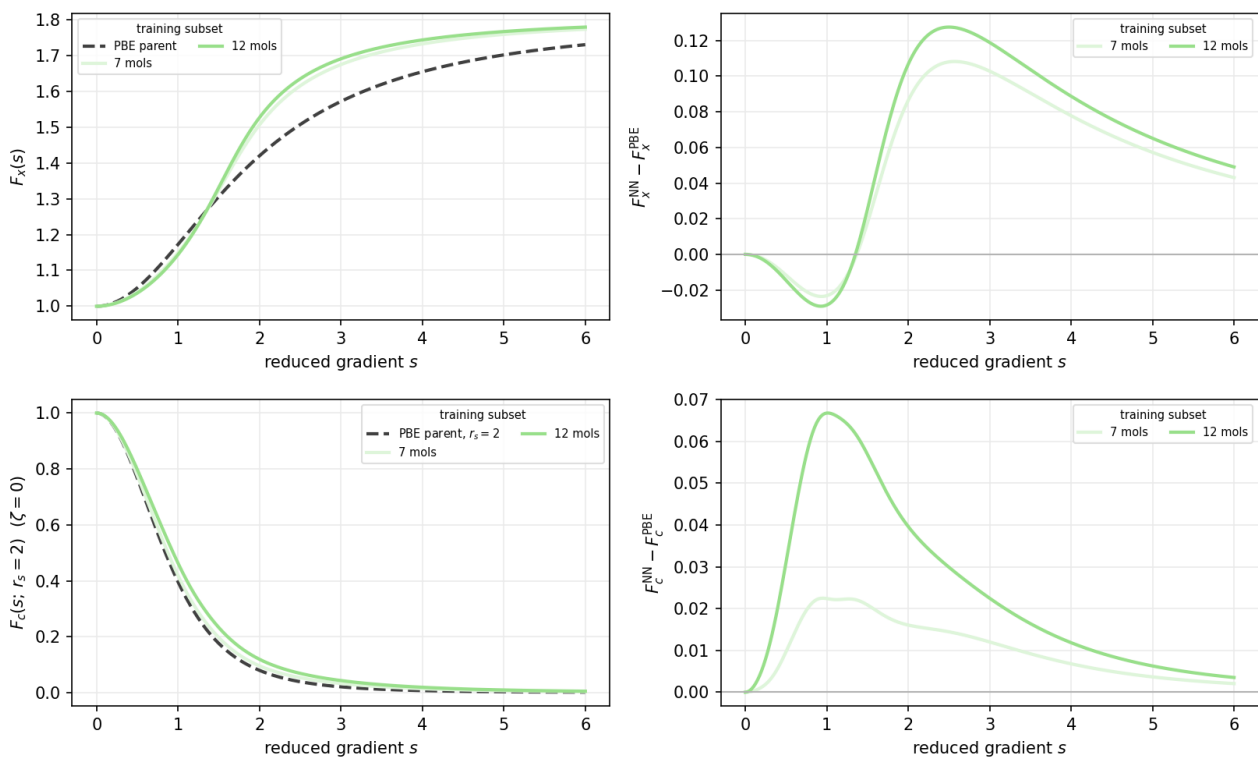
run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_X at $\rho=1$, zero extra descriptors; F_C at $\zeta=0$, $r_s=2$; 10 completed cell(s); max|dF_X| 1.85e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_attn_2x8: 2 x 8, Glorot initialization, 2 attention heads

deep_3x16 [25 cycles]: trained networks against the PBE parent



run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_X at $\rho=1$, zero extra descriptors; F_C at $\zeta=0$, $r_s=2$; 3 completed cell(s); $\max|dF_X|$ 1.28e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_3x16 [25 cycles]: 3 x 16, Glorot initialization; 25 cycles

deep_3x16 [dpyscf parity]: trained networks against the PBE parent



run run_20260908T153908Z; checkpoint channel val_best (model_val_best.eqx); slices: F_X at $\rho=1$, zero extra descriptors; F_C at $\zeta=0$, $r_s=2$; 2 completed cell(s); $\max|dF_X|$ 1.28e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_3x16 [dpyscf parity]: 3 x 16, Glorot initialization; dpyscf parity

4.5 In-sample fit

The grid-RMSE density fit (Fig. 4.4 of the full report) is not repeated here; the per-cell ratios are in its table there.

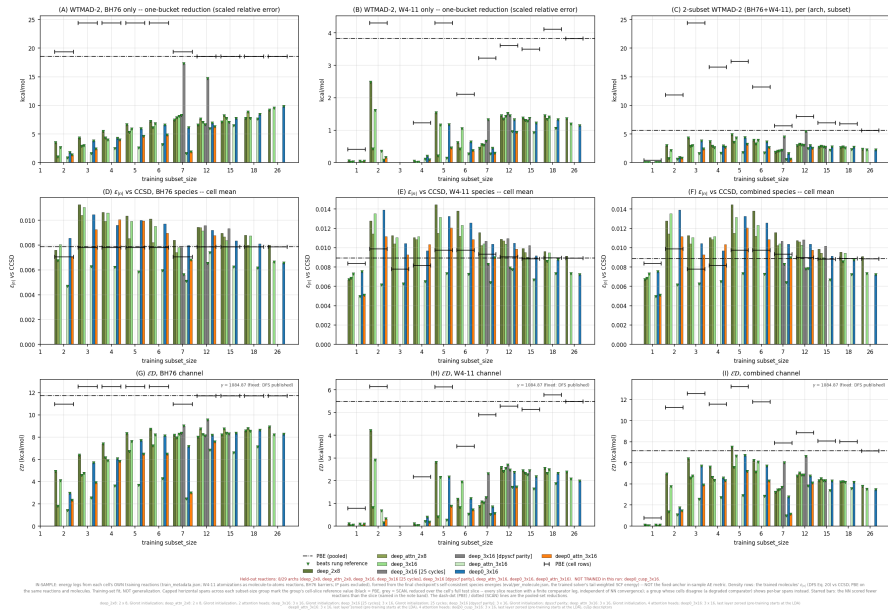
Fig. 4.5 In-sample energy and density in DFS units, the 3x3 layout of Fig. 5.5 on the training set: rows WTMAD-2 / ε_n / $\mathcal{ED}$, columns BH76 / W4-11 / combined, each cell against PBE on its own training reactions and molecules. The energy leg is the training reactions themselves (the W4-11 atomizations as molecule-to-atoms reactions and the BH76 barriers of `train_metadata.json`), each formed from the final checkpoint’s self-consistent species energies in `eval/per_molecule.json` against its reference; PBE on the same reactions from the same file’s PBE energies. The dashed line is PBE on the union of the run’s training reactions (18.6 kcal/mol for the BH76 barriers, 3.83 for the W4-11 atomizations, 5.67 combined, the same union for every run since the runs train the same subsets); the capped spans and the verdict triangles use each cell’s own reactions. A cell whose subset has no reaction of a column (the one-molecule cell in the BH76 column; the three-molecule cell, whose subset is H₂O, Li⁺ and a barrier, in the W4-11 column) has no bar there.

Panel A (BH76 energy): `deep_3x16`’s barrier residual grows with the subset from 2.7 kcal/mol at two molecules to 9.5 at twenty-six, against 18.6 for PBE on the union; `deep_attn_3x16` 0.8 to 7.6, `deep0_3x16` 1.8 to 9.8, `deep0_attn_3x16` 1.4 to 6.3, `deep_2x8` 3.5 to 9.2, `deep_attn_2x8` 1.0 to 8.8; the 25-cycle arm 6.7 to 8.3 beside the size group’s 7.0 to 8.1 at the same subsets, and the parity arm 17.3 and 14.7 (its reaction term trained at weight 0.01). Panel B (W4-11): `deep_3x16` 0.02 (one molecule, H₂O fitted to 0.02 kcal/mol) to 1.6, `deep_attn_3x16` 0.05 to 1.1, `deep0_3x16` 0.00 to 1.3, `deep0_attn_3x16` 0.05 to 0.9, `deep_2x8` 0.05 to 2.5, `deep_attn_2x8` 0.01 to 1.4, the arms 0.7 to 1.5, against PBE’s 3.83. Panel C: the two-subset combination, 0.00 to 5.4 against 5.67 on the union of the training reactions of every cell, the parity arm (4.5 and 5.4) the closest to PBE. Panels D to F (density): the trained molecules’ ε_n is 0.0073 (one molecule) to 0.0135 (two) for `deep_3x16`, 0.0049 to 0.0078 for `deep_attn_3x16`, 0.0072 to 0.0139 for `deep0_3x16`, 0.0051 to 0.0120 for `deep0_attn_3x16`, 0.0067 to 0.0145 for `deep_2x8`, 0.0068 to 0.0115 for `deep_attn_2x8`, 0.0102 to 0.0108 for the 25-cycle arm and 0.0078 and 0.0083 for the parity arm, against PBE’s 0.0089 on the union of the trained molecules; the `deep_3x16` cells at two to eighteen molecules sit above PBE on this per-electron L1 while their grid RMSE (Fig. 4.4 of the full report) sits below it: the two norms rank the small-subset fits differently. Panels G to I: every one of the 66 cells’ $\mathcal{ED}$ is below PBE’s on its own slice (0.00 to 7.5 against 0.79 to 13.3), the energy leg deciding the combination at every subset.

In-sample per-channel story (DF5 units): WTMAD-2 | ϵ_H | ϵ_H (BH76, W4-11, combined) -- training fit - run_20260908T153908Z - val-best

$\epsilon_{H_{\text{val}}} = 25.5\% \cdot \epsilon_{H_{\text{val}}} = \frac{1}{N} \sum_{i=1}^N \epsilon_{H_{\text{val}}}^i$ (Eq. 20 per species). ONE gamma shared by all channels, the letters published 2024.87 kspiral (near the non-axial RL) stamped in each paper -- $\epsilon_{H_{\text{val}}}$ compare across channels, $\epsilon_{H_{\text{val}}}$ of PRE vs L PRE. Density row = adjusted $\epsilon_{H_{\text{val}}}$ near the trained molecule. The dashed PRE (pink) line reduce the UNDO of the cell's training reaction's molecules, not any one cell's the clipped gene and the dash results use each cell's own reaction. CC&M reference architectures carry no dash; mark here its CC&M counterpart as the training reactions). Final checkpoint only (final) has no val best variant: identical in the final step and val best direction, only the title's checkpoint stamp differs.

Example final checkpoint: each left own training reaction the left's color: BH76: 1 - W4-11: 2, alternates as reaction and barrier. P: pink (withhold: reaction images, final best), density: 22 (trained species in CC&M left column excluded).



4.6 Results per cell

In-sample per cell in DFS units (insample_by_pool_3x3_eps.csv of the family_val_best set, identical in the final-step set; combined leg; E = 2-subset WTMAD-2 over the cell's own training reactions, kcal/mol; eps over its training molecules; ED under gamma = 1084.87 kcal/mol; the PBE columns are the union of the run's training reactions / molecules for E and eps and the cell's own slice for ED; n = training reactions and trained species with a density).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_2x8	1	1	1	0.05	5.67	0.00670	0.00888	0.11	0.79	-0.68	yes
deep_2x8	2	2	3	3.00	5.67	0.01278	0.00888	4.93	11.27	-6.35	yes
deep_2x8	3	2	3	4.32	5.67	0.01126	0.00888	6.39	12.57	-6.18	yes
deep_2x8	4	3	4	3.65	5.67	0.01107	0.00888	5.60	11.58	-5.98	yes
deep_2x8	5	3	4	4.92	5.67	0.01447	0.00888	7.50	13.25	-5.76	yes
deep_2x8	6	4	5	3.93	5.67	0.01381	0.00888	6.22	11.78	-5.56	yes
deep_2x8	7	5	6	1.84	5.67	0.01157	0.00888	3.22	7.89	-4.68	yes
deep_2x8	12	10	11	2.98	5.67	0.01075	0.00888	4.75	8.86	-4.11	yes
deep_2x8	15	13	14	2.70	5.67	0.00986	0.00888	4.31	8.08	-3.77	yes
deep_2x8	18	16	16	2.63	5.67	0.00956	0.00888	4.19	8.01	-3.82	yes
deep_2x8	26	24	22	2.33	5.67	0.00909	0.00888	3.77	7.14	-3.37	yes
deep_attn_2x8	1	1	1	0.01	5.67	0.00683	0.00888	0.02	0.79	-0.77	yes
deep_attn_2x8	2	2	3	0.72	5.67	0.01144	0.00888	1.37	11.27	-9.91	yes
deep_attn_2x8	3	2	3	2.86	5.67	0.01041	0.00888	4.56	12.57	-8.01	yes
deep_attn_2x8	4	3	4	2.87	5.67	0.01086	0.00888	4.62	11.58	-6.96	yes
deep_attn_2x8	5	3	4	3.59	5.67	0.01150	0.00888	5.58	13.25	-7.68	yes
deep_attn_2x8	6	4	5	3.25	5.67	0.01122	0.00888	5.13	11.78	-6.65	yes
deep_attn_2x8	7	5	6	2.02	5.67	0.01023	0.00888	3.41	7.89	-4.48	yes
deep_attn_2x8	12	10	11	3.23	5.67	0.01061	0.00888	5.05	8.86	-3.81	yes
deep_attn_2x8	15	13	14	2.89	5.67	0.00946	0.00888	4.51	8.08	-3.56	yes
deep_attn_2x8	18	16	16	2.74	5.67	0.00855	0.00888	4.23	8.01	-3.78	yes
deep_3x16	1	1	1	0.02	5.67	0.00728	0.00888	0.05	0.79	-0.74	yes
deep_3x16	2	2	3	2.14	5.67	0.01353	0.00888	3.73	11.27	-7.55	yes
deep_3x16	3	2	3	2.96	5.67	0.01106	0.00888	4.75	12.57	-7.82	yes
deep_3x16	4	3	4	2.63	5.67	0.01118	0.00888	4.33	11.58	-7.25	yes
deep_3x16	5	3	4	4.31	5.67	0.01315	0.00888	6.63	13.25	-6.63	yes
deep_3x16	6	4	5	3.92	5.67	0.01234	0.00888	6.06	11.78	-5.72	yes
deep_3x16	7	5	6	2.04	5.67	0.01042	0.00888	3.46	7.89	-4.44	yes
deep_3x16	12	10	11	3.11	5.67	0.01026	0.00888	4.86	8.86	-4.00	yes
deep_3x16	15	13	14	2.78	5.67	0.00900	0.00888	4.32	8.08	-3.76	yes
deep_3x16	18	16	16	2.60	5.67	0.00944	0.00888	4.14	8.01	-3.86	yes
deep_3x16	26	24	22	2.23	5.67	0.00731	0.00888	3.49	7.14	-3.66	yes
deep_3x16 [25 cycles]	7	5	6	2.18	5.67	0.01070	0.00888	3.67	7.89	-4.22	yes
deep_3x16 [25 cycles]	12	10	11	3.06	5.67	0.01081	0.00888	4.86	8.86	-4.00	yes
deep_3x16 [25 cycles]	15	13	14	2.68	5.67	0.01016	0.00888	4.31	8.08	-3.77	yes
deep_3x16 [dpyscf parity]	7	5	6	4.52	5.67	0.00829	0.00888	6.01	7.89	-1.88	yes
deep_3x16 [dpyscf parity]	12	10	11	5.42	5.67	0.00780	0.00888	6.61	8.86	-2.25	yes
deep_attn_3x16	1	1	1	0.05	5.67	0.00494	0.00888	0.09	0.79	-0.70	yes
deep_attn_3x16	2	2	3	0.59	5.67	0.00617	0.00888	1.08	11.27	-10.20	yes
deep_attn_3x16	3	2	3	1.57	5.67	0.00626	0.00888	2.55	12.57	-10.02	yes
deep_attn_3x16	4	3	4	1.68	5.67	0.00648	0.00888	2.71	11.58	-8.87	yes
deep_attn_3x16	5	3	4	1.77	5.67	0.00732	0.00888	2.89	13.25	-10.36	yes
deep_attn_3x16	6	4	5	1.72	5.67	0.00724	0.00888	2.83	11.78	-8.96	yes
deep_attn_3x16	7	5	6	0.53	5.67	0.00640	0.00888	0.98	7.89	-6.92	yes
deep_attn_3x16	12	10	11	2.45	5.67	0.00784	0.00888	3.80	8.86	-5.05	yes
deep_attn_3x16	15	13	14	2.20	5.67	0.00667	0.00888	3.37	8.08	-4.70	yes
deep_attn_3x16	18	16	16	2.29	5.67	0.00722	0.00888	3.54	8.01	-4.47	yes
deep0_3x16	1	1	1	0.00	5.67	0.00753	0.00888	0.00	0.79	-0.79	yes
deep0_3x16	2	2	3	0.91	5.67	0.01392	0.00888	1.72	11.27	-9.56	yes
deep0_3x16	3	2	3	3.81	5.67	0.01046	0.00888	5.71	12.57	-6.86	yes
deep0_3x16	4	3	4	2.92	5.67	0.00969	0.00888	4.57	11.58	-7.01	yes
deep0_3x16	5	3	4	4.36	5.67	0.01327	0.00888	6.69	13.25	-6.56	yes
deep0_3x16	6	4	5	3.61	5.67	0.01257	0.00888	5.71	11.78	-6.08	yes
deep0_3x16	7	5	6	1.58	5.67	0.01038	0.00888	2.77	7.89	-5.12	yes
deep0_3x16	12	10	11	3.00	5.67	0.01037	0.00888	4.74	8.86	-4.12	yes
deep0_3x16	15	13	14	2.72	5.67	0.00913	0.00888	4.27	8.08	-3.80	yes
deep0_3x16	18	16	16	2.66	5.67	0.00873	0.00888	4.16	8.01	-3.85	yes
deep0_3x16	26	24	22	2.22	5.67	0.00721	0.00888	3.46	7.14	-3.68	yes

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep0_attn_3x16	1	1	1	0.05	5.67	0.00508	0.00888	0.09	0.79	-0.70	yes
deep0_attn_3x16	2	2	3	0.77	5.67	0.01118	0.00888	1.45	11.27	-9.83	yes
deep0_attn_3x16	3	2	3	2.42	5.67	0.00926	0.00888	3.91	12.57	-8.66	yes
deep0_attn_3x16	4	3	4	2.68	5.67	0.01033	0.00888	4.32	11.58	-7.26	yes
deep0_attn_3x16	5	3	4	3.23	5.67	0.01205	0.00888	5.18	13.25	-8.07	yes
deep0_attn_3x16	6	4	5	2.61	5.67	0.01084	0.00888	4.27	11.78	-7.52	yes
deep0_attn_3x16	7	5	6	0.60	5.67	0.00898	0.00888	1.14	7.89	-6.76	yes
deep0_attn_3x16	12	10	11	2.55	5.67	0.00975	0.00888	4.10	8.86	-4.75	yes

5. Optimization, hold-out

Sources: `xcquinox/pipeline/evaluation.py` (the density metrics), `docs/notes/HOLDOUT_SET.md` (the set), `tools/analysis/make_ablation_arch_figure.py` (the reductions, WTMAD-2 and ED), `per_cell_eval_holdout_val_best/per_reaction.json` and `per_molecule.json`, and the CSVs written beside the figures.

5.1 The held-out set and the comparators

The held-out pool is the union of the GMTKN55 BH76 barrier heights and the W4-11 atomization energies. Its test slice, after the removal of the 35-reaction validation slice (which selects the validation-best checkpoint and never enters a reported number), of each cell’s verbatim training reactions, carries 61 BH76 rows (three listed twice by name, four more as permuted-name twins; the energy legs average one term per barrier identity, 54) and 120 W4-11 reactions (the `n_rxn` column of the tables below counts name-deduplicated rows, 58 BH76 at one point), and the density comparison runs over 189 molecular species with CCSD reference densities at the same basis and grid (atoms excluded). PBE is evaluated on exactly the same slice as each cell, so every comparison is like against like; the pooled PBE values are 7.98 (BH76), 13.74 (W4-11) and 11.86 kcal/mol (combined) reaction-energy MAE, and 8.92 kcal/mol 2-subset WTMAD-2. The functionals are evaluated with the same 3-cycle SCF they were trained with (Sec. 4.2); the PBE density is the converged reference calculation’s.

5.2 Metrics

Energy. Per-reaction absolute error $|E_{\text{rxn}}^{\text{NN}} - E_{\text{rxn}}^{\text{ref}}|$, reduced either as the plain MAE over a pool or as the two-bucket WTMAD-2 of GMTKN55 (Goerigk et al., PCCP 19, 32184 (2017)),

$$\text{WTMAD-2} = \frac{56.84}{N} \sum_{i \in \{\text{BH76}, \text{W4-11}\}} N_i \frac{\text{MAD}_i}{|\Delta E_{\text{ref}}|_i},$$

a labeled reweighting of the two present subsets, not the full-GMTKN55 quantity (the figures say so). When one pool is reported alone the same expression collapses to $56.84 \text{MAD}_i / |\Delta E_{\text{ref}}|_i$, the pool’s mean absolute deviation rescaled by its mean absolute reference energy: the BH76 column of the 3x3 figures is a scaled relative error, which is why it reads three times the plain MAE (a mean barrier of about 19 kcal/mol) while the W4-11 column reads a fifth of it (a mean atomization energy of about 350 kcal/mol). **Density.** Per species, the grid-weighted RMSE $\sqrt{\sum_g w_g (n_g - n_g^{\text{ref}})^2 / \sum_g w_g}$ (the self-calibrated figures) and the DFS Eq. 20 per-electron error

$$\varepsilon_n = \frac{1}{N_e} \int |n - n^{\text{ref}}| d^3r, \quad N_e = \int n^{\text{ref}} d^3r$$

(the DFS-units figures), each reduced to a cell mean over species with case twins collapsed. **Combined.** The DFS Eq. 21 harmonic combination

$$\mathcal{ED} = \frac{2}{1/E + 1/(\gamma D)},$$

with E the WTMAD-2 energy leg and D the density leg. In DFS units $\gamma = 1084.87$ kcal/mol, the Letter’s published slope, is dimensionally valid on ε_n and is used as is; in the RMSE units the figures self-calibrate $\gamma = E_{\text{PBE}}/D_{\text{PBE}}$ from the pooled PBE anchors, so that $\mathcal{ED}_{\text{PBE}} = E_{\text{PBE}}$ by construction and the NN against PBE ranking is independent of the density scale. A cell “beats PBE” when its $\mathcal{ED}$ is below the PBE $\mathcal{ED}$ of the same slice. The harmonic mean is dominated by its smaller leg: with $\gamma\varepsilon_n$ between 9.6 and 13.6 kcal/mol for every functional here and E between 5.9 and 16.2 kcal/mol, a change of 1 kcal/mol in E moves $\mathcal{ED}$ by 0.4 to 0.9 kcal/mol (the derivative $2(\gamma D)^2/(E + \gamma D)^2$ over the 64 cells’ own legs), while the observed spread of the density leg (0.0088 to 0.0125 per electron, 9.6 to 13.6 kcal/mol after γ) moves it by at most 2.7 kcal/mol, at the largest energy leg, and by 1.3 at $E = 8$.

5.3 Held-out energy

The energy figures of the full report (Figs. 5.1 to 5.3) are not repeated here; their readings are in Sec. 5.8.

5.4 Held-out density

The grid-RMSE density figures of the full report (Fig. 5.4 of the full report) are not repeated here; their readings are in Sec. 5.8.

5.5 The per-pool 3x3 in DFS units

Fig. 5.5 The held-out result in the Letter’s units, nine panels: the rows are the energy leg (WTMAD-2), the density leg (ε_n , Eq. 20) and the combination ($\mathcal{ED}$ under the one shared $\gamma = 1084.87$ kcal/mol); the columns are the BH76 barriers, the W4-11 atomizations and both pooled. In every panel the abscissa is the subset size, the bars are the architectures (the greens deep_2x8, deep_attn_2x8, deep_3x16 and deep_attn_3x16, the greys the two arms, the blue deep0_3x16, the orange deep0_attn_3x16, in the display order), the dashed line is PBE on the pooled slice and the black cap over each subset size is PBE on that cell’s own slice; a triangle on a bar marks a cell below its cap. The first figure of each pair is the full set, the second the tail-excluded one. The panel values are in the four tables of Sec. 5.7.

Row 1 (energy). Panel A, BH76: PBE’s scaled relative error is 22.1 kcal/mol; deep_3x16 reads 21.3 at one molecule, 23.6 to 36.6 at two to six (the peak at five), 16.3 at seven, 16.7 to 20.0 from twelve to twenty-six; deep_attn_3x16 22.8 at one, then 20.6, 15.3, 14.6, 13.3, 14.0 and 15.5. Panel B, W4-11: PBE 2.54; deep_3x16 1.41 at one, 2.95 to 3.54 at two to six, 1.70 to 2.01 from seven; deep_attn_3x16 1.17, 1.80, 3.46, 2.06, 2.63, 2.38, 1.31. Panel C, the two-subset WTMAD-2 of Fig. 5.1 of the full report.

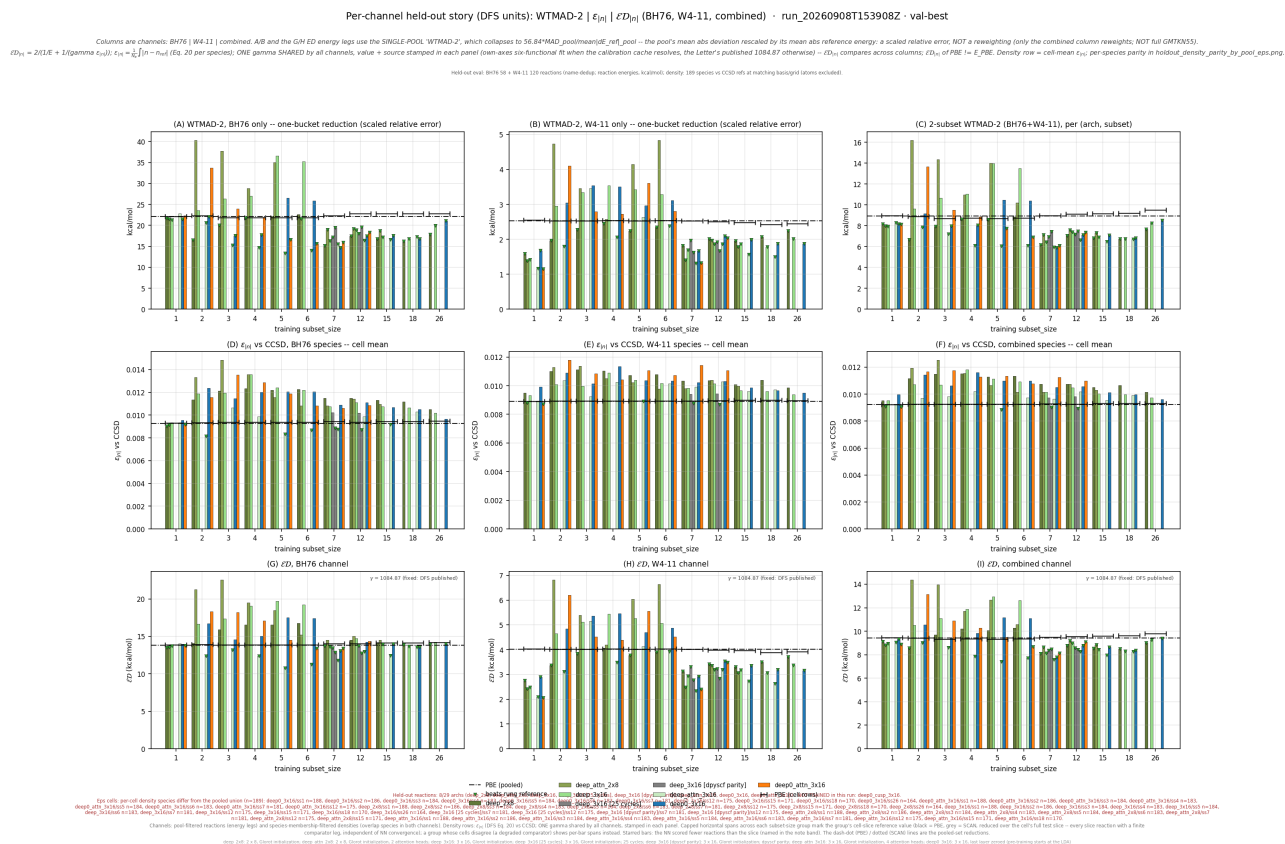
Row 2 (density). PBE’s ε_n is 0.00928 (BH76 species), 0.00891 (W4-11) and 0.00923 (combined) per electron. Panel D: deep_3x16 0.0093 at one, 0.0119 to 0.0136 at two to six, 0.0102 to 0.0111 from seven; deep_attn_3x16 0.0093, 0.0081, 0.0107, 0.0099, 0.0083, 0.0087, 0.0087, so the attention twin is below PBE at two, five, six and seven. Panel E: every deep_3x16 cell is above PBE (0.0093 to 0.0109), and so is every deep_attn_3x16 cell, at one and five molecules by less than 0.4 percent (0.00894 and 0.00892 against 0.00891). Panel F, the combination of both.

Row 3 (the combination). Panel G, BH76: PBE 13.8; deep_3x16 13.7 at one, 16.7 to 19.7 at two to six, 13.6 to 14.7 from seven; deep_attn_3x16 14.0 at one and 10.8 to 13.2 from two to seven. Panel H, W4-11: PBE 4.02; deep_3x16 2.48 at one, 4.65 to 5.45 at two to six, 2.93 to 3.36 from seven; deep_attn_3x16 2.09, 3.10, 5.14, 3.48, 4.13, 3.92, 2.33. Panel I, combined, the verdict column of the tables.

deep0_3x16, the same panels: BH76 energy 21.5, 22.0, 17.5, 17.8, 26.6, 25.9, 14.6, 17.5, 17.6, 16.8, 21.1 at one to twenty-six; W4-11 energy 1.68, 3.05, 3.54, 3.50, 2.97, 3.12, 1.67, 2.10, 1.99, 1.88, 1.87; combined ε_n 0.0096 to 0.0116, above PBE in every cell; combined $\mathcal{ED}$ 9.32, 10.54, 9.44, 9.83, 11.16, 11.10, 7.77, 8.78, 8.63, 8.37, 9.38 against the cell caps 9.46, 9.44, 9.32, 9.34, 9.32, 9.34, 9.49, 9.56, 9.60, 9.62, 9.80. The deep_attn_3x16 cells at twelve, fifteen and eighteen molecules: BH76 energy 16.45, 16.56 and 17.24, W4-11 1.87, 1.56 and 1.50, combined ε_n 0.0102, 0.0096 and 0.0099, combined $\mathcal{ED}$ 8.27, 7.97 and 8.27 against 9.56, 9.60 and 9.62. deep0_attn_3x16 over its eight cells: BH76 energy 22.38, 33.73, 23.92, 21.92, 16.56, 15.62, 15.91 and 18.34, W4-11 1.15, 4.10, 2.80, 2.72, 3.61, 2.80, 1.33 and 2.04, combined ε_n 0.0091 to 0.0118 (above PBE in every cell but the first), combined $\mathcal{ED}$ 8.89, 13.14, 10.91, 10.28, 9.49, 8.68, 8.12 and 9.09 against 9.46, 9.44, 9.32, 9.34, 9.32, 9.34, 9.49 and 9.56.

The 2x8 pair. deep_2x8: BH76 energy 21.68, 16.51, 20.02, 21.72, 22.15, 22.66, 15.19, 17.43, 16.82, 16.17 and 17.88 at one to twenty-six; W4-11 1.59, 1.97, 2.28, 2.45, 2.25, 2.35, 1.82, 2.02, 1.95, 2.07 and 2.24, below PBE’s 2.54 at every size; combined ε_n 0.0095 at one, 0.0112 to 0.0115 at two to six and 0.0102 to 0.0108 from seven; combined $\mathcal{ED}$ 9.13, 8.58, 9.69, 10.20, 10.09, 10.29, 8.08, 8.78, 8.53, 8.51 and 9.03 against the caps above, below PBE at one, two and from seven on. deep_attn_2x8: BH76 energy 21.52, 40.33, 37.72, 28.82, 35.08, 21.57, 18.97, 19.18 and 18.70 at one to fifteen; W4-11 1.38, 4.73, 3.46, 2.58, 4.15, 4.84, 1.40, 1.97 and 1.79; combined ε_n 0.0091 at one, 0.0101 to 0.0125 elsewhere; combined $\mathcal{ED}$ 8.82, 14.39, 13.97, 11.70, 12.66,

10.58, 8.65, 9.17 and 8.84, below PBE at one, seven, twelve and fifteen. The two arms at seven and twelve: BH76 energy 17.15 and 18.05 (25 cycles), 19.50 and 19.53 (dpyscf parity); W4-11 1.97 and 1.92, 1.62 and 1.67; combined ε_n 0.0097 and 0.0098 (25 cycles), 0.0090 and 0.0089 (dpyscf parity, the two cells of the campaign below PBE's 0.0092 on the full slice at seven or more molecules); combined $\mathcal{E}\mathcal{D}$ 8.36 and 8.57, 8.46 and 8.44, against 9.49 and 9.56, every one below PBE.



5.7 Results per cell

The two tables below (the per-leg tables are in the full report) are read from `holdout_by_pool_3x3_eps.csv` of the `family_val_best` set (the tail-excluded one from the `_excl_tail` sibling): `E` is the WTMAD-2 of the leg in kcal/mol (the one-pool legs collapse to the scaled relative error of Sec. 5.2), `eps` the per-electron density error of the leg's species, `ED` the combination under $\gamma = 1084.87$ kcal/mol; the `E PBE` and `eps PBE` columns are the leg's pooled PBE values (the same in every row), `ED PBE (cell)` is PBE's ED on the cell's own slice, and "beats" whether the network's ED is below it. `n rxn` and `n species` are the reactions and species of the cell's slice. Every table carries the 64 evaluated cells in the display order of the architectures (a tagged architecture after its untagged one) and then the subset size; the run of a cell is that of its architecture in the scope table (the report's tables of Sec. 4.6 carry it as a column). The pooled PBE columns of the tail-excluded table are its own file's, computed without the excluded species (eps PBE 0.00909 against 0.00923).

Combined leg (both pools).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_2x8	1	177	188	8.18	8.92	0.00952	0.00923	9.13	9.46	-0.33	yes
deep_2x8	2	177	186	6.65	8.92	0.01116	0.00923	8.58	9.44	-0.85	yes
deep_2x8	3	176	184	7.92	8.92	0.01148	0.00923	9.69	9.32	+0.37	no
deep_2x8	4	175	183	8.62	8.92	0.01152	0.00923	10.20	9.34	+0.86	no
deep_2x8	5	176	184	8.58	8.92	0.01128	0.00923	10.09	9.32	+0.77	no
deep_2x8	6	175	183	8.85	8.92	0.01134	0.00923	10.29	9.34	+0.95	no
deep_2x8	7	175	181	6.18	8.92	0.01076	0.00923	8.08	9.49	-1.41	yes
deep_2x8	12	169	175	7.03	8.92	0.01076	0.00923	8.78	9.56	-0.78	yes
deep_2x8	15	168	171	6.82	8.92	0.01050	0.00923	8.53	9.60	-1.07	yes
deep_2x8	18	166	170	6.74	8.92	0.01064	0.00923	8.51	9.62	-1.11	yes
deep_2x8	26	159	164	7.65	8.92	0.01015	0.00923	9.03	9.80	-0.77	yes
deep_attn_2x8	1	177	188	7.98	8.92	0.00909	0.00923	8.82	9.46	-0.64	yes
deep_attn_2x8	2	177	186	16.19	8.92	0.01193	0.00923	14.39	9.44	+4.95	no
deep_attn_2x8	3	176	184	14.36	8.92	0.01254	0.00923	13.97	9.32	+4.65	no
deep_attn_2x8	4	175	183	10.97	8.92	0.01156	0.00923	11.70	9.34	+2.36	no
deep_attn_2x8	5	176	184	13.99	8.92	0.01066	0.00923	12.66	9.32	+3.34	no
deep_attn_2x8	6	175	183	10.19	8.92	0.01014	0.00923	10.58	9.34	+1.24	no
deep_attn_2x8	7	175	181	7.12	8.92	0.01016	0.00923	8.65	9.49	-0.83	yes
deep_attn_2x8	12	169	175	7.57	8.92	0.01073	0.00923	9.17	9.56	-0.39	yes
deep_attn_2x8	15	168	171	7.32	8.92	0.01028	0.00923	8.84	9.60	-0.76	yes
deep_3x16	1	177	188	7.92	8.92	0.00951	0.00923	8.96	9.46	-0.50	yes
deep_3x16	2	177	186	9.61	8.92	0.01071	0.00923	10.52	9.44	+1.08	no
deep_3x16	3	176	184	10.67	8.92	0.01068	0.00923	11.11	9.32	+1.79	no
deep_3x16	4	175	183	11.06	8.92	0.01182	0.00923	11.88	9.34	+2.53	no
deep_3x16	5	176	184	13.97	8.92	0.01113	0.00923	12.96	9.32	+3.64	no
deep_3x16	6	175	183	13.51	8.92	0.01091	0.00923	12.62	9.34	+3.28	no
deep_3x16	7	175	181	6.44	8.92	0.01017	0.00923	8.13	9.49	-1.35	yes
deep_3x16	12	169	175	7.41	8.92	0.01047	0.00923	8.97	9.56	-0.59	yes
deep_3x16	15	168	171	6.89	8.92	0.01004	0.00923	8.44	9.60	-1.16	yes
deep_3x16	18	166	170	6.74	8.92	0.00999	0.00923	8.31	9.62	-1.31	yes
deep_3x16	26	159	164	8.23	8.92	0.00973	0.00923	9.25	9.80	-0.55	yes
deep_3x16 [25 cycles]	7	175	181	6.92	8.92	0.00972	0.00923	8.36	9.49	-1.13	yes
deep_3x16 [25 cycles]	12	169	175	7.17	8.92	0.00983	0.00923	8.57	9.56	-0.99	yes
deep_3x16 [dpyscf parity]	7	175	181	7.44	8.92	0.00903	0.00923	8.46	9.49	-1.03	yes
deep_3x16 [dpyscf parity]	12	169	175	7.48	8.92	0.00892	0.00923	8.44	9.56	-1.12	yes
deep_attn_3x16	1	177	188	8.26	8.92	0.00925	0.00923	9.06	9.46	-0.40	yes
deep_attn_3x16	2	177	186	7.87	8.92	0.00972	0.00923	9.01	9.44	-0.43	yes
deep_attn_3x16	3	176	184	7.22	8.92	0.00981	0.00923	8.60	9.32	-0.72	yes
deep_attn_3x16	4	175	183	6.09	8.92	0.01020	0.00923	7.85	9.34	-1.49	yes
deep_attn_3x16	5	176	184	6.03	8.92	0.00883	0.00923	7.40	9.32	-1.92	yes
deep_attn_3x16	6	175	183	6.09	8.92	0.00973	0.00923	7.73	9.34	-1.61	yes
deep_attn_3x16	7	175	181	5.94	8.92	0.00963	0.00923	7.57	9.49	-1.91	yes
deep_attn_3x16	12	169	175	6.61	8.92	0.01019	0.00923	8.27	9.56	-1.29	yes
deep_attn_3x16	15	168	171	6.47	8.92	0.00956	0.00923	7.97	9.60	-1.63	yes
deep_attn_3x16	18	166	170	6.72	8.92	0.00993	0.00923	8.27	9.62	-1.35	yes
deep0_3x16	1	177	188	8.18	8.92	0.00997	0.00923	9.32	9.46	-0.14	yes
deep0_3x16	2	177	186	9.16	8.92	0.01143	0.00923	10.54	9.44	+1.10	no
deep0_3x16	3	176	184	7.99	8.92	0.01062	0.00923	9.44	9.32	+0.12	no
deep0_3x16	4	175	183	8.06	8.92	0.01160	0.00923	9.83	9.34	+0.48	no
deep0_3x16	5	176	184	10.48	8.92	0.01099	0.00923	11.16	9.32	+1.83	no
deep0_3x16	6	175	183	10.40	8.92	0.01097	0.00923	11.10	9.34	+1.76	no
deep0_3x16	7	175	181	5.90	8.92	0.01051	0.00923	7.77	9.49	-1.72	yes
deep0_3x16	12	169	175	7.11	8.92	0.01058	0.00923	8.78	9.56	-0.78	yes
deep0_3x16	15	168	171	7.10	8.92	0.01013	0.00923	8.63	9.60	-0.97	yes
deep0_3x16	18	166	170	6.82	8.92	0.00996	0.00923	8.37	9.62	-1.26	yes
deep0_3x16	26	159	164	8.52	8.92	0.00961	0.00923	9.38	9.80	-0.42	yes

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep0_attn_3x16	1	177	188	8.11	8.92	0.00906	0.00923	8.89	9.46	-0.57	yes
deep0_attn_3x16	2	177	186	13.64	8.92	0.01168	0.00923	13.14	9.44	+3.70	no
deep0_attn_3x16	3	176	184	9.52	8.92	0.01177	0.00923	10.91	9.32	+1.59	no
deep0_attn_3x16	4	175	183	8.87	8.92	0.01127	0.00923	10.28	9.34	+0.93	no
deep0_attn_3x16	5	176	184	7.73	8.92	0.01134	0.00923	9.49	9.32	+0.17	no
deep0_attn_3x16	6	175	183	6.90	8.92	0.01078	0.00923	8.68	9.34	-0.66	yes
deep0_attn_3x16	7	175	181	6.08	8.92	0.01126	0.00923	8.12	9.49	-1.37	yes
deep0_attn_3x16	12	169	175	7.34	8.92	0.01100	0.00923	9.09	9.56	-0.47	yes

Combined leg without the recurring tail (the species named in Sec. 5.8 removed from the density leg).

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep_2x8	1	177	180	8.18	8.92	0.00879	0.00909	8.80	9.39	-0.59	yes
deep_2x8	2	177	178	6.65	8.92	0.01029	0.00909	8.33	9.37	-1.04	yes
deep_2x8	3	176	176	7.92	8.92	0.01048	0.00909	9.34	9.26	+0.08	no
deep_2x8	4	175	175	8.62	8.92	0.01039	0.00909	9.77	9.28	+0.49	no
deep_2x8	5	176	176	8.58	8.92	0.01012	0.00909	9.63	9.26	+0.38	no
deep_2x8	6	175	175	8.85	8.92	0.01016	0.00909	9.81	9.28	+0.54	no
deep_2x8	7	175	173	6.18	8.92	0.00996	0.00909	7.86	9.42	-1.56	yes
deep_2x8	12	169	167	7.03	8.92	0.00973	0.00909	8.44	9.49	-1.05	yes
deep_2x8	15	168	163	6.82	8.92	0.00958	0.00909	8.23	9.53	-1.29	yes
deep_2x8	18	166	162	6.74	8.92	0.00979	0.00909	8.25	9.55	-1.30	yes
deep_2x8	26	159	157	7.65	8.92	0.00918	0.00909	8.66	9.74	-1.09	yes
deep_attn_2x8	1	177	180	7.98	8.92	0.00852	0.00909	8.56	9.39	-0.83	yes
deep_attn_2x8	2	177	178	16.19	8.92	0.01049	0.00909	13.37	9.37	+4.00	no
deep_attn_2x8	3	176	176	14.36	8.92	0.01158	0.00909	13.40	9.26	+4.15	no
deep_attn_2x8	4	175	175	10.97	8.92	0.01077	0.00909	11.32	9.28	+2.04	no
deep_attn_2x8	5	176	176	13.99	8.92	0.00949	0.00909	11.86	9.26	+2.61	no
deep_attn_2x8	6	175	175	10.19	8.92	0.00912	0.00909	10.04	9.28	+0.76	no
deep_attn_2x8	7	175	173	7.12	8.92	0.00930	0.00909	8.35	9.42	-1.07	yes
deep_attn_2x8	12	169	167	7.57	8.92	0.00963	0.00909	8.78	9.49	-0.71	yes
deep_attn_2x8	15	168	163	7.32	8.92	0.00921	0.00909	8.45	9.53	-1.07	yes
deep_3x16	1	177	180	7.92	8.92	0.00886	0.00909	8.68	9.39	-0.71	yes
deep_3x16	2	177	178	9.61	8.92	0.00969	0.00909	10.04	9.37	+0.67	no
deep_3x16	3	176	176	10.67	8.92	0.00959	0.00909	10.53	9.26	+1.28	no
deep_3x16	4	175	175	11.06	8.92	0.01086	0.00909	11.41	9.28	+2.13	no
deep_3x16	5	176	176	13.97	8.92	0.00981	0.00909	12.08	9.26	+2.83	no
deep_3x16	6	175	175	13.51	8.92	0.00964	0.00909	11.79	9.28	+2.51	no
deep_3x16	7	175	173	6.44	8.92	0.00933	0.00909	7.87	9.42	-1.55	yes
deep_3x16	12	169	167	7.41	8.92	0.00947	0.00909	8.61	9.49	-0.88	yes
deep_3x16	15	168	163	6.89	8.92	0.00917	0.00909	8.14	9.53	-1.38	yes
deep_3x16	18	166	162	6.74	8.92	0.00921	0.00909	8.05	9.55	-1.50	yes
deep_3x16	26	159	157	8.23	8.92	0.00876	0.00909	8.82	9.74	-0.92	yes
deep_3x16 [25 cycles]	7	175	173	6.92	8.92	0.00908	0.00909	8.13	9.42	-1.29	yes
deep_3x16 [25 cycles]	12	169	167	7.17	8.92	0.00917	0.00909	8.33	9.49	-1.16	yes
deep_3x16 [dpyscf parity]	7	175	173	7.44	8.92	0.00848	0.00909	8.23	9.42	-1.19	yes
deep_3x16 [dpyscf parity]	12	169	167	7.48	8.92	0.00835	0.00909	8.19	9.49	-1.29	yes
deep_attn_3x16	1	177	180	8.26	8.92	0.00858	0.00909	8.75	9.39	-0.64	yes
deep_attn_3x16	2	177	178	7.87	8.92	0.00846	0.00909	8.47	9.37	-0.90	yes
deep_attn_3x16	3	176	176	7.22	8.92	0.00936	0.00909	8.43	9.26	-0.82	yes
deep_attn_3x16	4	175	175	6.09	8.92	0.00929	0.00909	7.59	9.28	-1.69	yes
deep_attn_3x16	5	176	176	6.03	8.92	0.00785	0.00909	7.06	9.26	-2.19	yes
deep_attn_3x16	6	175	175	6.09	8.92	0.00859	0.00909	7.37	9.28	-1.91	yes
deep_attn_3x16	7	175	173	5.94	8.92	0.00852	0.00909	7.23	9.42	-2.19	yes
deep_attn_3x16	12	169	167	6.61	8.92	0.00895	0.00909	7.87	9.49	-1.62	yes
deep_attn_3x16	15	168	163	6.47	8.92	0.00841	0.00909	7.57	9.53	-1.95	yes
deep_attn_3x16	18	166	162	6.72	8.92	0.00909	0.00909	7.99	9.55	-1.56	yes
deep0_3x16	1	177	180	8.18	8.92	0.00927	0.00909	9.02	9.39	-0.37	yes
deep0_3x16	2	177	178	9.16	8.92	0.01041	0.00909	10.12	9.37	+0.75	no
deep0_3x16	3	176	176	7.99	8.92	0.00966	0.00909	9.07	9.26	-0.19	yes
deep0_3x16	4	175	175	8.06	8.92	0.01062	0.00909	9.49	9.28	+0.21	no
deep0_3x16	5	176	176	10.48	8.92	0.00982	0.00909	10.57	9.26	+1.31	no
deep0_3x16	6	175	175	10.40	8.92	0.00983	0.00909	10.53	9.28	+1.26	no
deep0_3x16	7	175	173	5.90	8.92	0.00974	0.00909	7.57	9.42	-1.85	yes
deep0_3x16	12	169	167	7.11	8.92	0.00956	0.00909	8.44	9.49	-1.05	yes
deep0_3x16	15	168	163	7.10	8.92	0.00908	0.00909	8.26	9.53	-1.27	yes
deep0_3x16	18	166	162	6.82	8.92	0.00913	0.00909	8.08	9.55	-1.47	yes
deep0_3x16	26	159	157	8.52	8.92	0.00847	0.00909	8.84	9.74	-0.90	yes

Architecture	Subset	n rxn	n species	E NN	E PBE (pool)	eps NN	eps PBE (pool)	ED NN	ED PBE (cell)	dED	beats
deep0_attn_3x16	1	177	180	8.11	8.92	0.00844	0.00909	8.60	9.39	-0.79	yes
deep0_attn_3x16	2	177	178	13.64	8.92	0.01047	0.00909	12.40	9.37	+3.03	no
deep0_attn_3x16	3	176	176	9.52	8.92	0.01107	0.00909	10.62	9.26	+1.36	no
deep0_attn_3x16	4	175	175	8.87	8.92	0.01052	0.00909	9.98	9.28	+0.70	no
deep0_attn_3x16	5	176	176	7.73	8.92	0.01050	0.00909	9.21	9.26	-0.05	yes
deep0_attn_3x16	6	175	175	6.90	8.92	0.00990	0.00909	8.40	9.28	-0.87	yes
deep0_attn_3x16	7	175	173	6.08	8.92	0.01037	0.00909	7.89	9.42	-1.53	yes
deep0_attn_3x16	12	169	167	7.34	8.92	0.00974	0.00909	8.66	9.49	-0.83	yes

5.8 Reading

Energy. On the 2-subset WTMAD-2 the deep_3x16 architecture is below PBE (8.92 kcal/mol) at one molecule and at every subset from 7 upward (6.44 to 8.23), and above it at 2 to 6 molecules (9.61 to 13.97); on the plain reaction-energy MAE (Fig. 5.2 of the full report) the same cells read 0.63 and 0.68 to 0.83 of PBE against 1.15 to 1.43. The attention twin is below PBE at every one of its ten cells on WTMAD-2 (5.94 to 8.26), and on the MAE ratio at nine of them (3 molecules: 1.24). deep0_3x16, the same network under the zero-init start, is below PBE on WTMAD-2 at 1, 3, 4 and 7 to 26 (5.90 to 8.52) and above it at 2, 5 and 6 (9.16, 10.48, 10.40); its verdicts agree with deep_3x16's at every size but 3 and 4, where it sits just below PBE (7.99, 8.06) and deep_3x16 above (10.67, 11.06), and on the MAE ratio those two cells are above PBE as well. deep0_attn_3x16 is below PBE at 1 and at 4 to 12 (6.08 to 8.87) and above it at 2 and 3 (13.64, 9.52); on the MAE ratio it is below at 1, 7 and 12 (0.55 to 0.81) and above at 2 to 6 (1.03 to 1.61). The 2x8 pair: deep_2x8 is below PBE in every one of its eleven cells, on WTMAD-2 (6.18 to 8.85) and on the MAE ratio (0.69 to 0.97); deep_attn_2x8 is below at 1, 7, 12 and 15 (7.12 to 7.98; MAE ratio 0.61 to 0.79) and above at 2 to 6 (10.19 to 16.19; 1.08 to 1.87). The arms at seven and twelve molecules read 6.92 and 7.17 (25 cycles) and 7.44 and 7.48 (dpyscf parity) against 6.44 and 7.41 for the size group's deep_3x16, below PBE on both legs. The 2-to-6-molecule rise above PBE is seen in deep_3x16, deep0_3x16, deep_attn_2x8 and, at two and three molecules, deep0_attn_3x16; in deep_2x8 the cells at three to six molecules sit at 7.9 to 8.9, below PBE but above its cells from seven upward (6.2 to 7.7), and deep_attn_3x16 shows no rise (6.03 to 7.87 at those sizes). The subsets at those sizes are not nested and are not small-molecule subsets (Sec. 2.4): the 2-point subset is HLi with the OH + N₂ barrier, the 3-point subset is two barriers with the Li ionization, and the 4-, 5- and 6-point subsets add CH₂, or HLi with the C ionization, or HLi, CH and the C ionization, to the two barriers and the Li ionization; from seven points on, the subsets hold four or more atomization energies. Which property of these subsets produces the rise is not established. From 7 molecules the correction transfers, and the largest subsets do not improve on 7 to 18 (at 26: deep_3x16 8.23, deep0_3x16 8.52, deep_2x8 7.65).

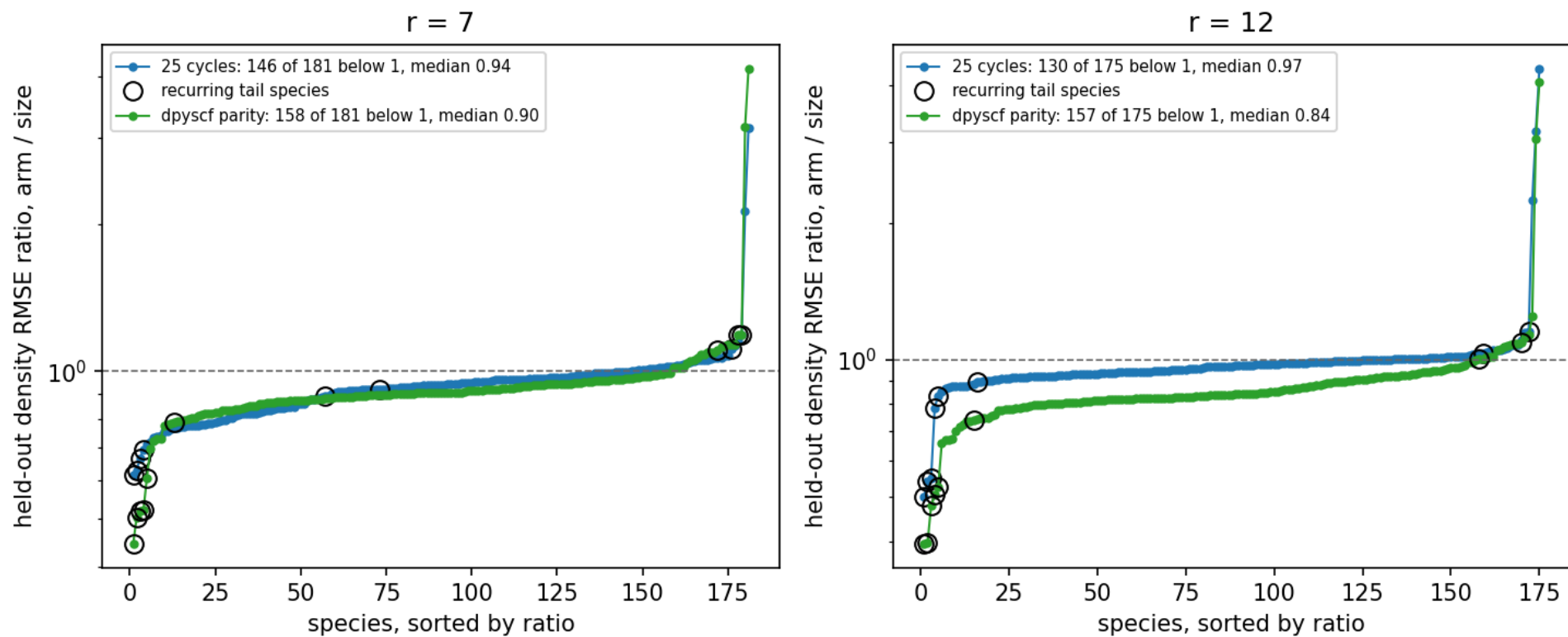
Density. In DFS units the held-out ε_n of the networks sits at or above PBE's (0.00923 per electron on the full slice) in every cell but five (deep_attn_2x8 at 1, 0.00909; deep_attn_3x16 at 5, 0.00883; deep0_attn_3x16 at 1, 0.00906; the parity arm at 7 and 12, 0.00903 and 0.00892), by 0.2 to 36 percent; the per-species parity (Figs. 5.4 and 5.6 of the full report) shows the bulk of species near the diagonal and a tail well above it. The tail is recurring: over the 64 cells, eight species (H₂, RKT06, RKT17, bn, bn3pi, cloo, cn and cs) are above 1.5 x PBE in at least half of the cells carrying them, computed from eval_holdout_val_best/per_molecule.json by the rule of Fig. 5.4 of the full report, the same eight as over the 33 cells of 2026-09-09; bn is the largest in 55 of the 64 cells (hoch3fcomp2 in 5, c2 in 4). Without them the networks' ε_n moves to 0.0079 to 0.0116 against PBE's 0.0091 on the same reduced set: below PBE in deep_attn_3x16 at 1, 2, 5, 6, 7, 12 and 15 (equal to it at 18), in deep_3x16 at 1 and 26, in deep_2x8 and deep_attn_2x8 at 1, in deep0_3x16 at 15 and 26, in deep0_attn_3x16 at 1 and in both parity cells, within 0.5 percent of it in the 25-cycle arm at 7 and in deep0_3x16 at 18, and above it elsewhere; in the deep_3x16 column from 7 molecules upward the ratio is 0.96 to 1.04. The tail species are the multireference and open-shell cases (BN and its 3-Pi state, CN, CS, ClOO, the OH + Cl and the H + CH₄ transition states, and H₂), and they are the species whose three-cycle SCF from the PBE seed is farthest from settled (Sec. 4.2). The 2026-09-07 density record found the comparison asymmetric by protocol (an unconverged network density against the converged PBE one) and could not decide from the pulled data whether the tail belongs to the functionals or to the three-cycle evaluation; the density channel itself trains (Sec. 4.5), and the converged-SCF re-evaluation, of which one cell-pass exists (Sec. 6), is the direct test. The model-free sibling of the tail-excluded view removes the 38 species whose CCSD T1 diagnostic exceeds 0.02 (Sec. 5.4 of the full report, the _excl_t1 figures; six of the eight tail species, all but H₂ and CS, are among them): on the 151 remaining species the pooled PBE ε_n is 0.0089, and the combined verdicts change at deep_2x8 at 3, deep0_3x16 at 3 and deep0_attn_3x16 at 5 molecules (9.17, 8.97 and 9.15 against 9.18, all three moving below PBE), the two cells the tail exclusion moves plus deep_2x8 at 3.

Combined. On $\mathcal{ED}$ in DFS units, against the cell-slice PBE value, 41 of the 64 cells are below PBE: deep_3x16 at 1, 7, 12, 15, 18 and 26 (above it at 2 to 6); deep_attn_3x16 at every one of its ten cells; deep0_3x16 at 1, 7, 12, 15, 18 and 26 (above at 2 to 6, where at 3 and 4 molecules the small energy margin (7.99 and 8.06 against 8.92) is outweighed by the density leg (0.0106 and 0.0116 against 0.0092), so that the two starts of the 3x16 network agree on every verdict);

deep0_attn_3x16 at 1, 6, 7 and 12 (8.89, 8.68, 8.12 and 9.09 against 9.46, 9.34, 9.49 and 9.56; above at 2 to 5); deep_2x8 at 1, 2 and 7 to 26 (above at 3 to 6, 9.69 to 10.29 against 9.32 to 9.34, where its energy leg is within 1.0 kcal/mol of PBE and its density leg 22 to 25 percent above); deep_attn_2x8 at 1, 7, 12 and 15 (above at 2 to 6); and both arms at 7 and 12 (8.36 and 8.57, 8.46 and 8.44, against 9.49 and 9.56). The verdicts are unchanged by removing the tail except deep0_3x16 at 3 molecules (9.07 against 9.26) and deep0_attn_3x16 at 5 (9.21 against 9.26), which then move below PBE. In every cell below PBE the margin comes from the energy leg: on the decomposition plane (Fig. 5.6 of the full report) each such cell lies to the left of the PBE point, and its density leg is below PBE’s only in the five cells named above. The lowest $\mathcal{ED}$ values are deep_attn_3x16 at 4 to 7 molecules (7.40 to 7.85 against 9.32 to 9.49) and deep0_3x16 at 7 (7.77 against 9.49). The tables of Sec. 5.7 print the difference $\mathcal{ED}_{\text{NN}} - \mathcal{ED}_{\text{PBE}}$ per cell, the negative values (the network below PBE) in bold.

Units and legs. The verdicts above are the DFS-units ones (the ε_n leg under the Letter’s γ). Under the self-calibrated grid-RMSE units of Figs. 5.6 and 5.7 of the full report (holdout_ed_combined.csv, WTMAD-2 leg) the verdicts differ from the DFS-units ones at deep_2x8 at 3 to 6 molecules (below PBE there, 8.33 to 8.77 against 8.84 to 8.88), deep_attn_3x16 at 2 (above, 9.30 against 8.93) and deep0_3x16 at 3 (below, 8.01 against 8.84); under the plain-MAE energy leg the verdicts of deep_attn_3x16 at 2, 3, 5 and 6, of deep0_3x16 at 3 and of deep0_attn_3x16 at 6 differ from the WTMAD-2 ones (Sec. 5.6 of the full report); the other 54 cells carry the same verdict on the three legs. Against the DFS-units verdict itself the plain-MAE leg moves eight cells (deep_2x8 at 3 to 6, deep_attn_3x16 at 3, 5 and 6, deep0_attn_3x16 at 6) and the self-calibrated units the six above, deep_2x8 at 3 to 6 under both choices, the same ten cells in all; the deck’s backup table lists that reading. The cells whose verdict depends on the choice of leg or units are the ones within about 10 percent of PBE on at least one leg.

The 25-cycle question. The two arms trained on the same subsets as the size run’s deep_3x16 cells (7 and 12 molecules) answer, species by species, whether a 25-cycle self-consistent field improves the held-out density (arm_vs_size_density.py, Fig. 5.9). Per held-out species the arm’s density RMSE against CCSD divided by the size cell’s on the same species is below one for 146 of 181 species at 7 molecules (median ratio 0.939) and 130 of 175 at 12 (0.970) for the 25-cycle arm, and for 158 of 181 (0.905) and 157 of 175 (0.840) for the parity arm. The gain is concentrated in the recurring tail (bn, bn3pi, cloo, cn, cs, H2, RKT06, RKT17): six of the eight are below one in every panel, the first four ranks of every curve are tail species, and the tail’s median ratio is below the other species’ in every panel (0.79 against 0.94 and 0.57 against 0.91 at 7 molecules, 0.81 against 0.97 and 0.52 against 0.84 at 12), while cloo and RKT17 are worse under both arms at both sizes (ratios 1.00 to 1.19). In the pooled numbers the 25-cycle arm lowers ε_n from 0.01017 to 0.00972 at 7 and from 0.01047 to 0.00983 at 12 (4.3 and 6.2 percent) and stays above PBE’s 0.00923; in the grid RMSE the gain holds at 7 (2.39e-4 to 2.20e-4) and vanishes at 12 (2.59e-4 to 2.62e-4, where o3, c2 and fo2 regress by factors 4.4, 3.2 and 2.2); WTMAD-2 moves from 6.44 to 6.92 at 7 and from 7.41 to 7.17 at 12, and the combined $\mathcal{ED}$ of every cell (8.13 to 8.97) stays under its PBE cap (9.49 at 7, 9.56 at 12). The parity arm reaches ε_n 0.00903 and 0.00892, the only deep_3x16 cells below PBE on the ε_n leg, at a worse barrier leg (BH76 WTMAD-2 19.5 against 16.3 and 18.9). The comparison conflates the training protocol with the evaluation protocol: the arms were evaluated with the 25-cycle budget they were trained with (168 and 173 of the 181 species converged at 7, 161 and 169 of 175 at 12), the size cells with 3 cycles (1 and 0 converged), so the converged-SCF re-evaluation of the same networks (Sec. 6) is the direct functional-against-functional test; the 15-molecule pass of the 25-cycle arm is one of the two set-aside cells, and the 25-cycle arm’s cells at 18 and 26 and the parity arm’s at 15, 18 and 26 are running or queued.



val_best channel; each cell evaluated with its own SCF cycle budget (density species): size cell at $r = 7$: max_cycles 3, 1 of 181 converged; size cell at $r = 12$: max_cycles 3, 0 of 175 converged; 25 cycles at $r = 7$: max_cycles 25, 168 of 181 converged; dpyscf parity at $r = 7$: max_cycles 25, 173 of 181 converged; 25 cycles at $r = 12$: max_cycles 25, 161 of 175 converged; dpyscf parity at $r = 12$: max_cycles 25, 169 of 175 converged.

Fig. 5.9 The 25-cycle arm and the dpyscf-parity arm against the size run’s deep_3x16 cell at 7 (left) and 12 molecules (right): per held-out species the ratio of the arm’s density RMSE to the size cell’s, sorted and drawn against its rank, the ratio on a logarithmic axis, the line at one, the eight recurring tail species as hollow markers, the counts below one and the medians in the legend; the footer states each cell’s SCF cycle budget and converged count over the twin-collapsed held-out species (arm_vs_size_density.png, written by arm_vs_size_density.py from the view’s per-molecule records).

Two cells set aside. The view holds 66 trained cells and 64 evaluated ones. The held-out passes of deep_attn_2x8 at 18 molecules and of the 25-cycle arm at 15 ran while the T1 backfill job was rewriting the benchmark reference files with a key the evaluation’s reference reader did not accept: every reference was refused, the passes wrote their tables from no species (every metric NaN) and exited 0. The two cells’ training logs, checkpoints and in-sample evaluations, which read the training references and not the rewritten files, are intact and appear in the tables of Sec. 4.6 of the full report; their held-out directories are set aside locally, and their re-evaluation with the corrected reader is deferred to a later cluster session (Sec. 6).

6. Pending work and open items

- Cells: deep_attn_3x16 at 26 (its training was stopped by the 48 h wall with its resume checkpoint set on disk; the harness relaunches it from that checkpoint once the array has drained), deep_attn_2x8 at 26, deep0_attn_3x16 at 15, 18 and 26, deep0_cusp_3x16 at every subset, the 25-cycle arm at 18 and 26 and the dpyscf-parity arm at 15, 18 and 26 are running or queued on the cluster. Two evaluated cells are excluded pending their re-evaluation (deep_attn_2x8 at 18 and deep_3x16 [25 cycles] at 15, Sec. 5.8): their held-out passes ran while the T1 backfill was rewriting the benchmark reference files with a key the evaluation’s reference reader did not accept, every reference was refused and the tables were written from no species with the task exiting 0; the reader now accepts the key, a reference the reader cannot read stops the pass instead of dropping species one by one, an evaluation that precomputed no species is refused before any table is written; the corrected code is committed and not yet on the cluster, so every task whose evaluation starts before it is deployed writes the same tables. The deploy, the two cells’ re-run by the repair job that re-evaluates every held-out pass of one cell, and the re-run of any further task the cluster’s logs show refused, are one later cluster session (docs/open_items.md, item 33). The figures and tables are regenerated from the same suite as cells land.
- Meta-GGA group: the four completed certificates read FAIL and the fifth fit was cancelled (Sec. 3.6); the training array is held. The follow-up is in the repository: three wider meta-GGA architectures registered (3x32, 4x16, 4x32), the pre-training probe in configuration mode, and a lever study as one 48 h job awaiting submission (fourteen 10000-step fits at the run’s identity: the energy-term weight at 0.1, 10 and 100 over the four widths under the run’s objective, and the failed run’s integration objective on the 3x16 at 0.1 and 100, each certified by the run’s own certificate). The decision between the widths, the weight and the anchored class follows its tables.
- Converged-SCF re-evaluation: the retroactive job covered one cell-pass (deep_3x16 at 1, the final checkpoint, 214 of 214 species converged, 181 reactions) in about 13 h of its 24 h wall (the time stamps of its preflight directory, 14:00 on 2026-09-12, and of the pass, 03:18 on 2026-09-13); at that rate the campaign’s 66 cells at two checkpoints are about 1700 node-hours, out of reach of the resumable single job. No converged reading is drawn from one cell, and the program’s continuation (an array over (cell, checkpoint) pairs on the cells the density reading needs) is a decision still to be taken. The CCSD T1 diagnostics are on disk (203 species, threshold 0.02), and the _excl_t1 siblings of every density figure, the model-free counterpart of the _excl_tail view, are rendered from them (Sec. 5.8).
- The two training arms of the density program on the deep_3x16 architecture at 7, 12, 15, 18 and 26 molecules: 25 SCF cycles with the current protocol (arm A) and the dpyscf-parity protocol (arm B: no convergence freeze, the executed first-mix index, the randomized atomic/PBE seed mixture, Adam with the plateau schedule and the Letter’s L2 and channel weights). Their cells at 7 and 12 are in the figures (Secs. 4.4 and 5.8); 15 to 26 are running or queued, and arm A’s cell at 15 is one of the two to be re-evaluated.

- In-sample: the suite's in-sample atomization panels stay excluded until that metric uses the network's own atoms; Figs. 4.1 and 4.5 carry the trained quantities instead.